Partial Differential Equatons

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Part I

Four Important Linear PDE

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§1 Fundmental Solution

Definition 1.1. *The function*

$$\Phi(x) := \begin{cases} -\frac{1}{2\pi}\log|x| & (n=2) \\ \frac{1}{n(n-2)\alpha(n)}\frac{1}{|x|^{n-2}} & (n \geq 3) \end{cases}$$

defined for $x \in \mathbb{R}^n - \{0\}$, is the ***fundamental solution of Laplace's equation****.*

Proposition 1.2. *Observe also that we have the estimates*

$$|D\Phi(x)| \lesssim \frac{1}{|x|^{n-1}}, \left|D^2\Phi(x)\right| \lesssim \frac{1}{|x|^n} \quad (x \neq 0)$$

D *for some constant $C > 0$.*

§2 Solving Poisson's equation

Theorem 2.1. *Assume that $f \in C_c^2(\mathbb{R}^n)$. Define u*

$$u(x) := \Phi * f(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\, \mathrm{d}y$$

Then

1. $u \in C^2(\mathbb{R}^n)$
2. $-\Delta u = f$ *in* $\mathbb{R}^n$.

Proof. 1. We have

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\, \mathrm{d}y = \int_{\mathbb{R}^n} \Phi(y) f(x-y)\, \mathrm{d}y$$

hence

$$\frac{u(x+he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \Phi(y) \left[\frac{f(x+he_i-y) - f(x-y)}{h} \right] dy$$

where $h \neq 0$ and $e_i = (0, \ldots, 1, \ldots, 0)$, the 1 in the ith -slot. But

$$\frac{f(x+he_i-y) - f(x-y)}{h} \to f_{x_i}(x-y)$$

uniformly on $\mathbb{R}^n$ as $h \to 0$ since $f \in C_c^2(\mathbb{R}^n)$, and thus

$$u_{x_i}(x) = \int_{\mathbb{R}^n} \Phi(y) f_{x_i}(x-y) dy$$

Similarly

$$u_{x_i x_j}(x) = \int_{\mathbb{R}n} \Phi(y) f_{x_i x_j}(x-y) dy$$

As the expression on the right-hand side is continuous in the variable x, we see $u \in C^2(\mathbb{R}^n)$.

2. Since Φ blows up at 0 , we will need for subsequent calculations to isolate this singularity inside a small ball. So fix $\varepsilon > 0$. Then an integration by parts yields

$$\begin{aligned}
\Delta u(x) &= \int_{B(0,\varepsilon)} \Phi(y) \Delta_x f(x-y) dy + \int_{\mathbb{R}^n - B(0,\varepsilon)} \Phi(y) \Delta_x f(x-y) dy \\
&:= I_\varepsilon - \int_{\mathbb{R}^n - B(0,\varepsilon)} D\Phi(y) \cdot D_y f(x-y) dy + \int_{\partial B(0,\varepsilon)} \Phi(y) \frac{\partial f}{\partial \nu}(x-y) dS(y) \\
&:= I_\varepsilon + K_\varepsilon + L_\varepsilon
\end{aligned}$$

ν denoting the inward pointing unit normal along $\partial B(0,\varepsilon)$. Now We readily check

$$|I_\varepsilon| \leq C \left\| D^2 f \right\|_{L^\infty(\mathbb{R}^n)} \int_{B(0,\varepsilon)} |\Phi(y)| dy = \mathcal{O}(1)$$

and

$$|L_\varepsilon| \le \|Df\|_{L^\infty(\mathbb{R}^n)} \int_{\partial B(0,\varepsilon)} |\Phi(y)| dS(y) = \mathcal{O}(1)$$

3. We continue by integrating by parts once again in the term K_ε, to discover

$$\begin{aligned}
K_\varepsilon &= -\int_{\mathbb{R}^n - B(0,\varepsilon)} D\Phi(y) \cdot D_y f(x-y) dy \\
&= \int_{\mathbb{R}^n - B(0,\varepsilon)} \Delta\Phi(y) f(x-y) dy - \int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y) dS(y) \\
&= -\int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y) dS(y) \\
&= -\int_{\partial B(0,\varepsilon)} \frac{-1}{n\alpha(n)} \frac{y}{|y|^n} \left(-\frac{y}{\varepsilon}\right) f(x-y) dS(y) \\
&= -\int_{\partial B(0,\varepsilon)} \frac{1}{n\alpha(n)\varepsilon^{n-1}} f(x-y) dS(y) \\
&\to -f(x) \quad \text{as } \varepsilon \text{ to } 0
\end{aligned}$$

□

§3 Harmonic Function

§3.1 Mean-value and Smooth properties

Theorem 3.1 (Mean-value formulas)**.** *Consider now open set $U \subset \mathbb{R}^n$ and $u \in C^2(U)$ is harmonic, then*

$$u(x) = \fint_{\partial B(x,r)} u(y)\, \mathrm{d}S = \fint_{B(x,r)} u(y)\, \mathrm{d}y$$

for each ball $B(x,r) \subset U$.

Proof. 1. Set

$$\phi(r) := \fint_{\partial B(x,r)} u(y)\, \mathrm{d}S = \fint_{\partial B(0,1)} u(x+rz)\, \mathrm{d}S(z)$$

Then

$$\begin{aligned}
\phi'(r) &= \fint_{\partial B(0,1)} Du(x+rz) \cdot z\, \mathrm{d}S(z) \\
&= \fint_{\partial B(x,r)} Du(y) \cdot \frac{y-x}{r}\, \mathrm{d}S(y) \\
&= \fint_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\, \mathrm{d}S(y) \\
&= \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\, \mathrm{d}y = 0
\end{aligned}$$

Hence ϕ is constant, and so

$$\phi(r) = \lim_{t \to 0} \phi(t) = u(x)$$

2. Observe next that our employing polar coordinates gives

$$\begin{aligned}\int_{B(x,r)} u(y)dy &= \int_0^r \left(\int_{\partial B(x,s)} u \,\mathrm{d}S\right) \mathrm{d}s \\ &= u(x)\int_0^r n\alpha(n)s^{n-1}\,\mathrm{d}s = \alpha(n)r^n u(x)\end{aligned}$$

□

Remark *We can see that the two Mean-value formulas is equivalent for $u \in C(U)$.*

Theorem 3.2 (Smoothness)**.** *If $u \in C(U)$ satisfies the mean-value property for each ball $B(x,r) \subset U$, then*

$$u \in C^\infty(U)$$

Proof. Let η be a standard mollifier, and recall that η is a radial function. Set $u^\varepsilon := \eta_\varepsilon * u$ in $U_\varepsilon = \{x \in U \mid \mathrm{dist}(x, \partial U) > \varepsilon\}$, then $u^\varepsilon \in C^\infty(U_\varepsilon)$. We will prove u is smooth by demonstrating that in fact $u \equiv u^\varepsilon$ on U_ε. Indeed if $x \in U_\varepsilon$, then

$$\begin{aligned}u^\varepsilon(x) &= \int_U \eta_\varepsilon(x-y)u(y)\,\mathrm{d}y \\ &= \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)u(y)\,\mathrm{d}y \\ &= \int_0^\varepsilon \eta_\varepsilon(r)\left(\int_{\partial B(x,r)} u\,\mathrm{d}S\right)\mathrm{d}r \\ &= u(x)\int_0^\varepsilon \eta_\varepsilon(r)\left(\int_{\partial B(x,r)} \mathrm{d}S\right)\mathrm{d}r \\ &= u(x)\int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\,\mathrm{d}y\end{aligned}$$

Thus $u^\varepsilon \equiv u$ in U_ε, and so $u \in C^\infty(U_\varepsilon)$ for each $\varepsilon > 0$. □

Corollary 3.3 (Converse to mean-value property)**.** *If $u \in C(U)$ satisfies mean-value property for each ball $B(x,r) \subset U$, then u is harmonic.*

Proof. We have $u \in C^\infty(U)$. If $\Delta u \not\equiv 0$, there exists some ball $B(x,r) \subset U$ such that, say, $\Delta u > 0$ within $B(x,r)$. But then for ϕ as above,

$$0 = \phi'(r) = \frac{r}{n} \fint_{B(x,r)} \Delta u(y)dy > 0,$$

a contradiction. □

Corollary 3.4. *The Laplace's equation $\Delta u = 0$ is rotation invariant; that is, if O is an orthogonal $n \times n$ matrix and we define*

$$v(x) := u(Ox)$$

$x \in \mathbb{R}^n$, then $\Delta v = 0$.

Theorem 3.5 (Harnack's inequality)**.** *For each connected open set $V \subset\subset U$, there exists a positive constant C_V, depending only on V, such that*

$$\sup_V u \leq C \inf_V u$$

for all nonnegative harmonic functions u in U.

Proof. Let $r := \frac{1}{4}\,\mathrm{dist}(V, \partial U)$. Choose $x, y \in V, |x-y| \leq r$. Then

$$u(x) = \fint_{B(x,2r)} u dz \geq \frac{1}{\alpha(n) 2^n r^n} \int_{B(y,r)} u dz = \frac{1}{2^n} u(y)$$

Thus $2^n u(y) \geq u(x) \geq \frac{1}{2^n} u(y)$ if $x, y \in V, |x-y| \leq r$. Since V is connected and $\bar{V}$ is compact, we can cover $\bar{V}$ by a chain of finitely many balls $\{B_i\}_{i=1}^N$, each of which has radius $\frac{r}{2}$ and $B_i \cap B_{i-1} \neq \varnothing$ for $i = 2, \ldots, N$. Then

$$u(x) \geq \frac{1}{2^{n(N+1)}} u(y)$$

for all $x, y \in V$. □

§3.2 Maximum Principle

Theorem 3.6 (Strong maximum principle)**.** *Assuming that open set $U \subset\subset \mathbb{R}^n$. Suppose $u \in C^2(U) \cap C(\bar{U})$ is harmonic within U .*

1. *Then*

$$\max_{\bar{U}} u = \max_{\partial U} u$$

2. *Furthermore, if U is connected and there exists a point $x_0 \in U$ such that*

$$u(x_0) = \max_{\bar{U}} u$$

then u is constant within $\bar{U}$.

Replacing u by $-u$, we recover also similar assertions with "min" replacing "max".

Proof. Suppose there exists a point $x_0 \in U$ with $u(x_0) = M := \max_{\bar{U}} u$. Then for $0 < r < \mathrm{dist}(x_0, \partial U)$, the mean-value property asserts

$$M = u(x_0) = \fint_{B(x_0,r)} u dy \leq M$$

As equality holds only if $u(y) = M$ for all $y \in B(x_0, r)$. Hence the set $\{x \in U \mid u(x) = M\}$ is both open and closed in U relatively and thus equals U if U is connected. This proves assertion (ii), from which (i) follows. □

Corollary 3.7. *If U is a bounded region and $u \in C^2(U) \cap C(\bar{U})$ satisfies*

$$\begin{cases} \Delta u = 0 \text{ in } U \\ u = g \text{ on } \partial U \end{cases}$$

where $g \geq 0$ and $g \not\equiv 0$, then u is positive everywhere in U.

§3.3 Local Estimates on Derivatives and Analyticity

Theorem 3.8 (Local Estimates on derivatives)**.** *Assume u is harmonic in $U \subset \mathbb{R}^n$. Then*

$$|D^\alpha u(x_0)| \leq \frac{C_{|\alpha|}}{r^{n+k}} \|u\|_{L^1(B(x_0,r))} \tag{1}$$

for each ball $B(x_0, r) \subset U$ and each multiindex α. Here $|\alpha| = k$

$$C_0 = \frac{1}{\alpha(n)}, C_k = \frac{(2^{n+1}nk)^k}{\alpha(n)} \quad (k = 1, \ldots) \tag{2}$$

$$\left\|D^k u\right\|_{L^\infty} = \mathcal{O}\left(\frac{1}{r^{n+k}}\right)$$

Proof. 1. We establish (1), (2) by induction on k, the case $k = 0$ being immediate from the mean-value formula. For $k = 1$, we note upon differentiating Laplace's equation that u_{x_i} is harmonic. Consequently

$$\begin{aligned} |u_{x_i}(x_0)| &= \left| \fint_{B(x_0,r/2)} u_{x_i} \, \mathrm{d}x \right| \\ &= \left| \frac{2^n}{\alpha(n) r^n} \int_{\partial B(x_0,r/2)} u \nu_i dS \right| \\ &\leq \frac{2n}{r} \|u\|_{L^\infty\left(\partial B\left(x_0, \frac{r}{2}\right)\right)} \end{aligned}$$

Now if $x \in \partial B(x_0, r/2)$, then $B(x, r/2) \subset B(x_0, r) \subset U$, and so

$$|u(x)| \leq \frac{1}{\alpha(n)} \left(\frac{2}{r}\right)^n \|u\|_{L^1(B(x_0,r))}$$

for $k = 0$. Combining the inequalities above, we deduce

$$|D^\alpha u(x_0)| \leq \frac{2^{n+1} n}{\alpha(n)} \frac{1}{r^{n+1}} \|u\|_{L^1(B(x_0,r))}$$

if $|\alpha| = 1$. This verifies (1), (2) for $k = 1$.

2. Assume now $k \geq 2$ and (1), (2) are valid for all balls in U and each multiindex of order less than or equal to $k - 1$. Fix $B(x_0, r) \subset U$ and let α be a multiindex with $|\alpha| = k$. Then

$D^\alpha u = \left(D^\beta u\right)_{x_i}$ for some $i \in \{1, \ldots, n\}$, $|\beta| = k-1$. We establish that

$$|D^\alpha u(x_0)| \leq \frac{nk}{r} \left\|D^\beta u\right\|_{L^\infty\left(\partial B\left(x_0, \frac{r}{k}\right)\right)}$$

If $x \in \partial B\left(x_0, \frac{r}{k}\right)$, then $B\left(x, \frac{k-1}{k} r\right) \subset B(x_0, r) \subset U$. Thus (1), (2) for $k-1$ imply

$$\left|D^\beta u(x)\right| \leq \frac{\left(2^{n+1} n(k-1)\right)^{k-1}}{\alpha(n)\left(\frac{k-1}{k} r\right)^{n+k-1}} \|u\|_{L^1(B(x_0,r))}$$

Combining the two previous estimates yields the bound

$$|D^\alpha u(x_0)| \leq \frac{\left(2^{n+1} nk\right)^k}{\alpha(n) r^{n+k}} \|u\|_{L^1(B(x_0,r))}$$

This confirms (1), (2) for $|\alpha| = k$. □

Corollary 3.9 (Liouville's Theorem)**.** *Suppose $u : \mathbb{R}^n \to \mathbb{R}$ is harmonic and bounded. Then u is constant.*

Theorem 3.10 (Analyticity)**.** *Assume u is harmonic in U. Then u is analytic in U.*

Proof. 1. Fix any point $x_0 \in U$. We must show u can be represented by a convergent power series in some neighborhood of x_0.

Let $r := \frac{1}{4} \operatorname{dist}(x_0, \partial U)$. Then $M := \frac{1}{\alpha(n) r^n} \|u\|_{L^1(B(x_0,2r))} < \infty$.

2. Since $B(x, r) \subset B(x_0, 2r) \subset U$ for each $x \in B(x_0, r)$, Theorem 7 provides the bound

$$\begin{aligned} \|D^\alpha u\|_{L^\infty(B(x_0,r))} &\leq M \left(\frac{2^{n+1} n}{r}\right)^{|\alpha|} |\alpha|^{|\alpha|} \\ &\leq M \left(\frac{2^{n+1} n^2 e}{r}\right)^{|\alpha|} \alpha! \end{aligned}$$

since

$$|\alpha|^{|\alpha|} \leq e^{|\alpha|} n^{|\alpha|} \alpha!$$

3. The Taylor series for u at x_0 is

$$\sum_\alpha \frac{D^\alpha u(x_0)}{\alpha!} (x - x_0)^\alpha$$

the sum taken over all multiindices. We assert this power series converges, provided

$$|x - x_0| < \frac{r}{2^{n+2} n^3 e}$$

To verify this, let us compute for each N the remainder term:

$$\begin{aligned} R_N(x) &:= u(x) - \sum_{k=0}^{N-1} \sum_{|\alpha|=k} \frac{D^\alpha u\left(x_0\right)\left(x-x_0\right)^\alpha}{\alpha!} \\ &= \sum_{|\alpha|=N} \frac{D^\alpha u\left(x_0+\theta_x\left(x-x_0\right)\right)\left(x-x_0\right)^\alpha}{\alpha!} \end{aligned}$$

for some $0 \leq \theta_x \leq 1$. We establish this formula by writing out the first N terms and the error in the Taylor expansion about 0 for the function of one variable $g(t) := u\left(x_0+t\left(x-x_0\right)\right)$, at $t=1$. Then, we can estimate

$$\begin{aligned} |R_N(x)| &\leq M \sum_{|\alpha|=N} \left(\frac{2^{n+1} n^2 e}{r}\right)^N \left(\frac{r}{2^{n+2} n^3 e}\right)^N \\ &\leq M n^N \frac{1}{(2n)^N} = \frac{M}{2^N} \to 0 \quad \text{as } N \to \infty \end{aligned}$$

□

§4 Subharmonic function

Definition 4.1. *We say $v \in C^2(\bar{U})$ is **subharmonic** if*

$$-\Delta v \leq 0 \quad \text{in } U$$

Theorem 4.2. *Suppoes subharmonic v in U, then*

$$v(x) \leq \fint_{\partial B(x,r)} v \, \mathrm{d}y$$

for all $B(x,r) \subset U$. Therefore,

$$\max_{\bar{U}} v = \max_{\partial U} v$$

Corollary 4.3. *Let U be a bounded, open subset of $\mathbb{R}^n$. Prove that there exists a constant C, depending only on U, such that*

$$\max_{\bar{U}} |u| \leq C \left(\max_{\partial U} |g| + \max_{\bar{U}} |f|\right)$$

whenever u is a smooth solution of

$$\begin{cases} -\Delta u = f \text{ in } U \\ \quad u = g \text{ on } \partial U \end{cases}$$

Proof: Apply previous theorem to the subharmonic funcion $u + \frac{|x|^2}{2n}\lambda$, where $\lambda = \max_{\bar{U}} |f|$.

Theorem 4.4. *Let $\phi : \mathbb{R} \to \mathbb{R}$ be smooth and convex. Assume u is harmonic and $v := \phi(u)$, then v is subharmonic.*

Consequently, $v := |Du|^2$ is subharmonic, whenever u is harmonic.

§5 Green's funcion

Lemma 5.1. *Assume U is open, bounded, and ∂U is C^1. Suppose $u \in C^2(\bar{U})$ is an arbitrary function, we have the indentity*

$$u(x) = \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial \nu}(y) - u(y)\frac{\partial \Phi}{\partial \nu}(y-x)\,\mathrm{d}S(y) - \int_U \Phi(y-x)\Delta u(y)\,\mathrm{d}y$$

for any $x \in U$.

Proof. Fix $x \in U$, choose $\varepsilon > 0$ so small that $B(x,\varepsilon) \subset U$, and apply Green's formula on the region $V_\varepsilon := U - B(x,\varepsilon)$ to $u(y)$ and $\Phi(y-x)$. We thereby compute

$$\begin{aligned}&\int_{V_\varepsilon} u(y)\Delta\Phi(y-x) - \Phi(y-x)\Delta u(y)dy\\ =&\int_{\partial V_\varepsilon} u(y)\frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)dS(y)\end{aligned}$$

that is

$$\begin{aligned}&\int_U - \int_{B(x,\varepsilon)} -\Phi(y-x)\Delta u(y)dy\\ =&\int_{\partial U_\varepsilon} - \int_{\partial B_{(x,\varepsilon)}} u(y)\frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)dS(y)\end{aligned}$$

We observe also

$$\left|\int_{\partial B(x,\varepsilon)} \Phi(y-x)\frac{\partial u}{\partial \nu}(y)dS(y)\right| \le C\varepsilon^{n-1}\left|\Phi(\varepsilon)\right| \to 0$$

as $\varepsilon \to 0$. Furthermore

$$\int_{\partial B(x,\varepsilon)} u(y)\frac{\partial \Phi}{\partial \nu}(y-x)dS(y) = \fint_{\partial B(x,\varepsilon)} u(y)dS(y) \to u(x)$$

Hence our sending $\varepsilon \to 0$ yields the formula

$$u(x) = \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial \nu}(y) - \frac{\partial \Phi}{\partial \nu}(y-x)u(y)dS(y) - \int_U \Phi(y-x)\Delta u(y)dy$$

□

Definition 5.2. *The **Green's function** for the region U is*

$$G(x,y) := \Phi(y-x) - \phi^x(y) \quad (x,y \in U, x \neq y)$$

where corrector function $\phi^x = \phi^x(y)$, solving the boundary-value problem

$$\begin{cases} \Delta\phi^x(y) = 0 & \text{in } U \\ \phi^x(y) = \Phi(y-x) & \text{on } \partial U \end{cases}$$

Let us apply Green's formula once more, to compute

$$\begin{aligned} -\int_U \phi^x(y)\Delta u(y)\,\mathrm{d}y &= \int_U u(y)\Delta\phi^x(y)\,\mathrm{d}y - \phi^x(y)\Delta u(y)\,\mathrm{d}y \\ &= \int_{\partial U} u(y)\frac{\partial\phi^x}{\partial\nu}(y) - \phi^x(y)\frac{\partial u}{\partial\nu}(y)dS(y) \\ &= \int_{\partial U} u(y)\frac{\partial\phi^x}{\partial\nu}(y) - \Phi(y-x)\frac{\partial u}{\partial\nu}(y)dS(y) \end{aligned}$$

and obtain

$$\Phi(y-x)\frac{\partial u}{\partial\nu}(y)dS(y) = \int_{\partial U} u(y)\frac{\partial\phi^x}{\partial\nu}(y) + \int_U \phi^x(y)\Delta u(y)dy$$

Then we find that

$$\begin{aligned} &u(x) \\ &= \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial\nu}(y) - u(y)\frac{\partial\Phi}{\partial\nu}(y-x)dS(y) - \int_U \Phi(y-x)\Delta u(y)dy \\ &= \int_{\partial U} u(y)\left[\frac{\partial\phi^x}{\partial\nu}(y) - \frac{\partial\Phi}{\partial\nu}(y-x)\right]dS(y) + \int_U \left[\phi^x(y) - \Phi(y-x)\right]\Delta u(y)\,\mathrm{d}y \\ &= -\int_{\partial U} u(y)\frac{\partial G}{\partial\nu}(x,y)dS(y) - \int_U G(x,y)\Delta u(y)\,\mathrm{d}y \end{aligned}$$

Theorem 5.3 (Representation formula using Green's function)**.** *Suppose now $u \in C^2(\bar{U})$ solves the boundary-value problem*

$$\begin{cases} -\Delta u = f \text{ in } U \\ u = g \text{ on } \partial U \end{cases}$$

for given continuous functions f, g, then

$$u(x) = -\int_{\partial U} g(y)\frac{\partial G}{\partial\nu}(x,y)\,\mathrm{d}S(y) + \int_U f(y)G(x,y)\,\mathrm{d}y$$

Theorem 5.4 (Symmetry of Green's function)**.** *For all $x, y \in U, x \neq y$, we have*

$$G(y,x) = G(x,y)$$

Proof. Fix $x, y \in U, x \neq y$. Write

$$v(z) := G(x,z), w(z) := G(y,z)$$

for $z \in U$. Then $\Delta v(z) = 0 (z \neq x), \Delta w(z) = 0 (z \neq y)$ and $w = v = 0$ on ∂U. Thus our applying

Green's identity on $V := U - [B(x,\varepsilon) \cup B(y,\varepsilon)]$ for sufficiently small $\varepsilon > 0$ yields

$$\int_{\partial B(x,\varepsilon)} \frac{\partial v}{\partial \nu} w - \frac{\partial w}{\partial \nu} v dS(z) = \int_{\partial B(y,\varepsilon)} \frac{\partial w}{\partial \nu} v - \frac{\partial v}{\partial \nu} w dS(z)$$

Now w is smooth near x, whence

$$\left| \int_{\partial B(x,\varepsilon)} \frac{\partial w}{\partial \nu} v dS \right| = o(1) \quad \text{as } \varepsilon \to 0$$

On the other hand, $v(z) = \Phi(z-x) - \phi^x(z)$, where ϕ^x is smooth in U. Thus

$$\lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} \frac{\partial v}{\partial \nu} w dS = \lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(x-z) w(z) dS = w(x)$$

Thus the left-hand side converges to $w(x)$ as $\varepsilon \to 0$. Likewise the right-hand side converges to $v(y)$. Consequently

$$G(y,x) = w(x) = v(y) = G(x,y)$$

□

§5.1 (Upper) Half-Space

Definition 5.5 (Green's function for the half-space)**.** *Let $\phi^x(y) = \Phi(y - \tilde{x})$ in $\mathbb{R}^n_+$,*

$$G(x,y) := \Phi(y-x) - \Phi(y-\tilde{x})$$

Then if $y \in \partial\mathbb{R}^n_+$,

$$\frac{\partial G}{\partial \nu}(x,y) = -G_{y_n}(x,y) = -\frac{2x_n}{n\alpha(n)} \frac{1}{|x-y|^n}$$

The function

$$K(x,y) := \frac{2x_n}{n\alpha(n)} \frac{1}{|x-y|^n} \quad \left(x \in \mathbb{R}^n_+, y \in \partial\mathbb{R}^n_+\right)$$

*is **Poisson's kernel for** $\mathbb{R}^n_+$.*

Theorem 5.6 (Poisson's formula for half-space)**.** *Suppose the boundary-value problem*

$$\begin{cases} \Delta u = 0 \text{ in } \mathbb{R}^n_+ \\ u = g \text{ on } \partial\mathbb{R}^n_+ \end{cases}$$

$g \in C\left(\mathbb{R}^{n-1}\right) \cap L^\infty\left(\mathbb{R}^{n-1}\right)$, and define u by

$$u(x) = \frac{2x_n}{n\alpha(n)} \int_{\partial\mathbb{R}^n_+} \frac{g(y)}{|x-y|^n} \, \mathrm{d}y \quad \left(x \in \mathbb{R}^n_+\right)$$

Then

1. $u \in C^\infty\left(\mathbb{R}_+^n\right) \cap L^\infty\left(\mathbb{R}_+^n\right)$,

2. $\Delta u = 0$ *in* $\mathbb{R}_+^n$,

3. $\lim\limits_{\substack{x \to x^0 \\ x \in \mathbb{R}_+^n}} u(x) = g\left(x^0\right)$ *for each point* $x^0 \in \partial\mathbb{R}_+^n$.

§5.2 Ball

Definition 5.7 (Green's function for ball)**.** *Let*

$$\phi^x(y) = \Phi\left(\frac{|x|}{r}(y - \tilde{x})\right)$$

for $x, y \in B(0,r), x \neq y$, *where* $\tilde{x} = r^2 \cdot \frac{x}{|x|}$

$$G(x,y) := \Phi(y - x) - \Phi\left(\frac{|x|}{r}(y - \tilde{x})\right)$$

Accordingly

$$\begin{aligned}
\frac{\partial G}{\partial \nu}(x,y) &= \sum_{i=1}^n y_i G_{y_i}(x,y) \\
&= \frac{-1}{n\alpha(n)} \frac{1}{|x-y|^n} \sum_{i=1}^n y_i \left((y_i - x_i) - y_i|x|^2 + x_i\right) \\
&= \frac{-1}{n\alpha(n)} \frac{1 - |x|^2}{|x-y|^n}
\end{aligned}$$

for each $y \in \partial B(0,1)$ *The function*

$$K_r(x,y) := \frac{r^2 - |x|^2}{n\alpha(n)r} \frac{1}{|x-y|^n} \qquad (x \in B(0,r), y \in \partial B(0,r))$$

is ***Poisson's kernel for the ball*** $B(0,r)$.

Theorem 5.8 (Poisson's formula for ball)**.** *Assume* $g \in C(\partial B(0,r))$ *and Suppose now* u *solves the boundary-value problem*

$$\begin{cases} \Delta u = 0 \text{ in } B(0,r) \\ \quad u = g \text{ on } \partial B(0,r) \end{cases}$$

Then

$$u(x) = \frac{r^2 - |x|^2}{n\alpha(n)r} \int_{\partial B(0,r)} \frac{g(y)}{|x-y|^n} \, \mathrm{d}S(y) \qquad \left(x \in B^0(0,r)\right)$$

1. $u \in C^\infty\left(B(0,r)\right)$

2. $\Delta u = 0$ *in* $B(0,r)$

3. $\lim\limits_{\substack{x \to x^0 \\ x \in B(0,r)}} u(x) = g\left(x^0\right)$ *for each point* $x^0 \in \partial B(0,r)$.

Theorem 5.9 (Reflection principle)**.** *Let U^+ denote the open half-ball $\mathbb{R}^n_+ \cap B$. Assume $u \in C^2(U^+) \cap C\left(\overline{U^+}\right)$ is harmonic in U^+, with $u = 0$ on $\bar{U}^+ \cap \{x_n = 0\}$. Set*

$$v(x) := \begin{cases} u(x) & \text{if } x_n \geq 0 \\ -u(x_1, \ldots, x_{n-1}, -x_n) & \text{if } x_n < 0 \end{cases}$$

for $x \in U = B(0,1)$. Then $v \in C^2(U) \cap C(\bar{U})$ and thus v is harmonic in U.

Proof. Noted that $v \in C(\bar{U})$. Suppoes poission integral relative to v

$$\tilde{v}(x) = \int_{\partial B(0,1)} K(x,y)v(y)dS(y)$$

then $\tilde{v} \in C^2(U) \cap C(\bar{U})$ is harmonic within $B(0,1)$. Also, we observe that

$$\tilde{v} = v \quad \text{on } \partial U^+$$

Thus we have $\tilde{v} = v$ in U^+ since $\tilde{v} - v$ is harmonic within U^+. Similarly, we conclude that $\tilde{v} = v$ in U^-. □

§6 Energy Methods

Theorem 6.1 (Uniqueness)**.** *Assume U is open, bounded, and ∂U is C^1. Consider first the boundary-value problem*

$$\begin{cases} -\Delta u = f \text{ in } U \\ \quad u = g \text{ on } \partial U \end{cases} \tag{1}$$

There exists at most one solution $u \in C^2(\bar{U})$.

Proof. Assume $\tilde{u}$ is another solution and set $w := u - \tilde{u}$. Then $\Delta w = 0$ in U, and so an integration by parts shows

$$0 = \int_U w\Delta w \,\mathrm{d}x = \int_{\partial U} w\frac{\partial w}{\partial \nu}dS - \int_U |Dw|^2 \,\mathrm{d}x = -\int_U |Dw|^2 \,\mathrm{d}x$$

Thus $Dw \equiv 0$ in U, and, since $w = 0$ on ∂U, we deduce $w = u - \tilde{u} \equiv 0$ in U. □

Theorem 6.2 (Dirichlet's principle)**.** *For this, we define the energy functional*

$$I[w] := \int_U \frac{1}{2}|Dw|^2 - wf \,\mathrm{d}x \tag{2}$$

w belonging to the admissible set

$$\mathcal{A} := \left\{w \in C^2(\bar{U}) \mid w = g \text{ on } \partial U\right\}.$$

Assume $u \in C^2(\bar{U})$ solves (1). Then

$$I[u] = \min_{w \in \mathcal{A}} I[w].$$

Conversely, if $u \in \mathcal{A}$ satisfies (2), then u solves the boundary-value problem (1).

Proof. 1. Assume u solves (1) and choose $w \in \mathcal{A}$. Then we have

$$0 = \int_U (-\Delta u - f)(u - w)\,\mathrm{d}x$$

An integration by parts yields

$$0 = \int_U Du \cdot D(u - w) - f(u - w)\,\mathrm{d}x$$

and there is no boundary term since $u - w = g - g \equiv 0$ on ∂U. Hence

$$\begin{aligned}\int_U |Du|^2 - uf\,\mathrm{d}x &= \int_U Du \cdot Dw - wf\,\mathrm{d}x \\ &\leq \int_U \frac{1}{2}|Du|^2\,\mathrm{d}x + \int_U \frac{1}{2}|Dw|^2 - wf\,\mathrm{d}x\end{aligned}$$

Rearranging, we conclude

$$I[u] \leq I[w] \quad (w \in \mathcal{A}).$$

2. Now, conversely, suppose (2) holds. Fix any $v \in C_c^\infty(U)$ and write

$$\begin{aligned}i(\tau) &:= I[u + \tau v] \\ &= \int_U \frac{1}{2}|Du + \tau Dv|^2 - (u + \tau v)f\,\mathrm{d}x \\ &= \int_U \frac{1}{2}|Du|^2 + \tau Du \cdot Dv + \frac{\tau^2}{2}|Dv|^2 - (u + \tau v)f\,\mathrm{d}x\end{aligned}$$

Since $u + \tau v \in \mathcal{A}$ for each τ, the scalar function $i(\cdot)$ has a minimum at zero, and thus

$$0 = i'(0) = \int_U Du \cdot Dv - vf\,\mathrm{d}x = \int_U (-\Delta u - f)v\,\mathrm{d}x$$

This identity is valid for each function $v \in C_c^\infty(U)$ and so $-\Delta u = f$ in U. $\square$

Chapter III

Heat Equation

§1 Fundmental Solution

Definition 1.1. *The function*

$$\Phi(x,t) := \begin{cases} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}} & (x \in \mathbb{R}^n, t > 0) \\ 0 & (x \in \mathbb{R}^n, t < 0) \end{cases}$$

is called the ***fundamental solution of the heat equation****.*

Theorem 1.2 (Solution of initial-value problem)**.** *We now employ Φ to fashion a solution to the Cauchy problem*

$$\begin{cases} u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,\infty) \\ u = g \text{ on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Assume $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, and define u by

$$u(x,t) = \int_{\mathbb{R}^n} \Phi(x-y,t) g(y) \, \mathrm{d}y \quad (x \in \mathbb{R}^n, t > 0)$$

Then

1. $u \in C^\infty(\mathbb{R}^n \times (0,\infty))$,
2. $u_t(x,t) - \Delta u(x,t) = 0 \quad (x \in \mathbb{R}^n, t > 0)$
3. $\lim\limits_{\substack{(x,t)\to(x^0,0) \\ x\in\mathbb{R}^n, t>0}} u(x,t) = g(x^0)$ *for each point* $x^0 \in \mathbb{R}^n$.

Remark *We sometimes write*

$$\begin{cases} \Phi_t - \Delta\Phi = 0 & \text{in } \mathbb{R}^n \times (0,\infty) \\ \Phi = \delta_0 & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

δ_0 denoting the Dirac measure on $\mathbb{R}^n$ giving unit mass to the point 0.

Theorem 1.3 (Solution of nonhomogeneous problem)**.** *Now let us turn our attention to the nonhomogeneous initial-value problem*

$$\begin{cases} u_t - \Delta u = f(x,t) \text{ in } \mathbb{R}^n \times (0,\infty) \\ \qquad\qquad u = 0 \text{ on } \mathbb{R}^n \times \{t=0\}. \end{cases}$$

assuming $f \in C_1^2(\mathbb{R}^n \times [0,\infty))$ *and* f *has compact support. We have*

$$\begin{aligned} u(x,t) &= \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s) f(y,s)\,\mathrm{d}y\,\mathrm{d}s \\ &= \int_0^t \frac{1}{(4\pi(t-s))^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{4(t-s)}} f(y,s)\,\mathrm{d}y\,\mathrm{d}s \end{aligned}$$

for $x \in \mathbb{R}^n, t > 0$*. Then*

1. $u \in C_1^2(\mathbb{R}^n \times (0,\infty))$*,*
2. $u_t(x,t) - \Delta u(x,t) = f(x,t) \quad (x \in \mathbb{R}^n, t > 0)$*,*
3. $\displaystyle\lim_{\substack{(x,t)\to(x^0,0)\\ x\in\mathbb{R}^n, t>0}} u(x,t) = 0$ *for each point* $x^0 \in \mathbb{R}^n$*.*

Proof. 1. First we change variables, to write

$$u(x,t) = \int_0^t \int_{\mathbb{R}^n} \Phi(y,s) f(x-y,t-s)\,\mathrm{d}y\,\mathrm{d}s$$

As $f \in C_1^2(\mathbb{R}^n \times [0,\infty))$ has compact support and $\Phi = \Phi(y,s)$ is smooth near $s = t > 0$, we compute

$$u_t(x,t) = \int_0^t \int_{\mathbb{R}^n} \Phi(y,s) f_t(x-y,t-s)\,\mathrm{d}y ds + \int_{\mathbb{R}^n} \Phi(y,t) f(x-y,0) dy$$

and

$$u_{x_i x_j}(x,t) = \int_0^t \int_{\mathbb{R}^n} \Phi(y,s) f_{x_i x_j}(x-y,t-s)\,\mathrm{d}y\,\mathrm{d}s \quad (i,j = 1,\dots,n)$$

Thus u_t, $D_x^2 u$, and likewise u, $D_x u$, belong to $C(\mathbb{R}^n \times (0,\infty))$.

2. We then calculate

$$\begin{aligned}
u_t(x,t) - \Delta u(x,t) =& \int_0^t \int_{\mathbb{R}^n} \Phi(y,s) \left[\left(\frac{\partial}{\partial t} - \Delta_x \right) f(x-y,t-s) \right] \mathrm{d}y \, \mathrm{d}s \\
&+ \int_{\mathbb{R}^n} \Phi(y,t) f(x-y,0) dy \\
=& \int_\varepsilon^t \int_{\mathbb{R}^n} \Phi(y,s) \left[\left(-\frac{\partial}{\partial s} - \Delta_y \right) f(x-y,t-s) \right] \mathrm{d}y \, \mathrm{d}s \\
&+ \int_0^\varepsilon \int_{\mathbb{R}^n} \Phi(y,s) \left[\left(-\frac{\partial}{\partial s} - \Delta_y \right) f(x-y,t-s) \right] \mathrm{d}y \, \mathrm{d}s \\
&+ \int_{\mathbb{R}^n} \Phi(y,t) f(x-y,0) dy \\
=& I_\varepsilon + J_\varepsilon + K
\end{aligned}$$

Now

$$|J_\varepsilon| \leq \left(\|f_t\|_{L^\infty} + \left\| D^2 f \right\|_{L^\infty} \right) \int_0^\varepsilon \int_{\mathbb{R}^n} \Phi(y,s) \, \mathrm{d}y \, \mathrm{d}s \leq \varepsilon C$$

Integrating by parts, we also find

$$\begin{aligned}
I_\varepsilon =& \int_\varepsilon^t \int_{\mathbb{R}^n} \left[\left(\frac{\partial}{\partial s} - \Delta_y \right) \Phi(y,s) \right] f(x-y,t-s) \, \mathrm{d}y \, \mathrm{d}s \\
&+ \int_{\mathbb{R}^n} \Phi(y,\varepsilon) f(x-y,t-\varepsilon) dy \\
&- \int_{\mathbb{R}^n} \Phi(y,t) f(x-y,0) dy \\
=& \int_{\mathbb{R}^n} \Phi(y,\varepsilon) f(x-y,t-\varepsilon) dy - K
\end{aligned}$$

since Φ solves the heat equation we ascertain

$$\begin{aligned}
u_t(x,t) - \Delta u(x,t) &= \lim_{\varepsilon \to 0} \int_{\mathbb{R}^n} \Phi(y,\varepsilon) f(x-y,t-\varepsilon) dy \\
&= \lim_{\varepsilon \to 0} f(x,t-\varepsilon) \\
&= f(x,t) \quad (x \in \mathbb{R}^n, t > 0)
\end{aligned}$$

the limit as $\varepsilon \to 0$ being computed. Finally note $\|u(\cdot,t)\|_{L^\infty} \leq t\|f\|_{L^\infty} \to 0$. □

§2 Mean-value formula

Definition 2.1. *Let $U \subset \mathbb{R}^n$ be open amd bounded, and fix a time $T > 0$.*

(1) We define the parabolic cylinder

$$U_T := U \times (0,T]$$

(2) The parabolic boundary of U_T is

$$\Gamma_T := \bar{U}_T - U_T$$

Definition 2.2. *For fixed $x \in \mathbb{R}^n, t \in \mathbb{R}, r > 0$, we define*

$$E(x,t;r) := \left\{ (y,s) \in \mathbb{R}^{n+1} \mid s \le t, \Phi(x-y,t-s) \ge \frac{1}{r^n} \right\}.$$

$E(x,t;r)$ is sometimes called a "heat ball".

Noted that $t - \frac{r^2}{4\pi} \le s \le t$ and $E(x,t;r)$ is bounded.

Theorem 2.3 (A mean-value property for the heat equation)**.** *Let $u \in C_1^2(U_T)$ solve the heat equation $\left(\frac{\partial}{\partial t} - \Delta\right) u = 0$. Then*

$$u(x,t) = \frac{1}{4r^n} \iint_{E(x,t;r)} u(y,s) \frac{|x-y|^2}{(t-s)^2} \, \mathrm{d}y \, \mathrm{d}s$$

for each $E(x,t;r) \subset U_T$.

Proof: Step 1. Shift the space and time coordinates so that $x = 0$ and $t = 0$. Upon mollifying if necessary, we may assume u is smooth. Write $E(r) = E(0,0;r)$ and set

$$\begin{aligned} \phi(r) &:= \frac{1}{r^n} \iint_{E(r)} u(y,s) \frac{|y|^2}{s^2} \, \mathrm{d}y \, \mathrm{d}s \\ &= \iint_{E(1)} u\left(ry, r^2 s\right) \frac{|y|^2}{s^2} \, \mathrm{d}y \, \mathrm{d}s \end{aligned}$$

We compute

$$\begin{aligned} \phi'(r) &= \iint_{E(1)} \sum_{i=1}^n u_{y_i} y_i \frac{|y|^2}{s^2} + 2r u_s \frac{|y|^2}{s} \, \mathrm{d}y \, \mathrm{d}s \\ &= \frac{1}{r^{n+1}} \iint_{E(r)} \sum_{i=1}^n u_{y_i} y_i \frac{|y|^2}{s^2} + 2 u_s \frac{|y|^2}{s} \, \mathrm{d}y \, \mathrm{d}s \\ &=: A + B \end{aligned}$$

Step 2. Also, let us introduce the useful function

$$\psi := -\frac{n}{2} \log(-4\pi s) + \frac{|y|^2}{4s} + n \log r$$

and observe $\psi = 0$ on $\partial E(r) - \{(0,0)\}$, $E(r)^\circ = \{\psi > 0\}$. since $\Phi(y,-s) = r^{-n}$ on $\partial E(r)$. We utilize (21) to write

$$\begin{aligned} B &= \frac{1}{r^{n+1}} \iint_{E(r)} 4 u_s \sum_{i=1}^n y_i \psi_{y_i} \, \mathrm{d}y \, \mathrm{d}s \\ &= -\frac{1}{r^{n+1}} \iint_{E(r)} 4 n u_s \psi + 4 \sum_{i=1}^n u_{s y_i} y_i \psi \, \mathrm{d}y \, \mathrm{d}s \end{aligned}$$

there is no boundary term since $\psi = 0$ on $\partial E(r) - \{(0,0)\}$. Integrating by parts with respect to s, we discover

$$\begin{aligned} B &= \frac{1}{r^{n+1}} \iint_{E(r)} -4nu_s\psi + 4\sum_{i=1}^{n} u_{y_i} y_i \psi_s \,\mathrm{d}y\,\mathrm{d}s \\ &= \frac{1}{r^{n+1}} \iint_{E(r)} -4nu_s\psi + 4\sum_{i=1}^{n} u_{y_i} y_i \left(-\frac{n}{2s} - \frac{|y|^2}{4s^2}\right) \mathrm{d}y\,\mathrm{d}s \\ &= \frac{1}{r^{n+1}} \iint_{E(r)} -4nu_s\psi - \frac{2n}{s}\sum_{i=1}^{n} u_{y_i} y_i \,\mathrm{d}y\,\mathrm{d}s - A \end{aligned}$$

Consequently, since u solves the heat equation,

$$\begin{aligned} \phi'(r) &= A + B \\ &= \frac{1}{r^{n+1}} \iint_{E(r)} -4n\Delta u\psi - \frac{2n}{s}\sum_{i=1}^{n} u_{y_i} y_i \,\mathrm{d}y\,\mathrm{d}s \\ &= \sum_{i=1}^{n} \frac{1}{r^{n+1}} \iint_{E(r)} 4nu_{y_i}\psi_{y_i} - \frac{2n}{s} u_{y_i} y_i \,\mathrm{d}y\,\mathrm{d}s \\ &= 0 \end{aligned}$$

Thus ϕ is constant, and therefore

$$\phi(r) = \lim_{t\to 0}\phi(t) = u(0,0)\left(\lim_{t\to 0}\frac{1}{t^n}\iint_{E(t)} \frac{|y|^2}{s^2}\,\mathrm{d}y\,\mathrm{d}s\right) = 4u(0,0)$$

§3 Properties of solution

Definition 3.1. *Let $U \subset \mathbb{R}^n$ be open amd bounded, and fix a time $T > 0$.*

(1) We define the parabolic cylinder

$$U_T := U \times (0,T] = U \times (0,T) \bigcup U \times \{T\}$$

(2) The parabolic boundary of U_T is

$$\Gamma_T := \bar{U}_T - U_T = \partial U \times [0.T] \bigcup \bar{U} \times \{0\}$$

Noted that $\bar{U}_T = \bar{U} \times [0,T]$ and

Theorem 3.2 (Strong maximum principle for the heat equation)**.** *Suppoes $U \subset\subset \mathbb{R}^n$ and assume $u \in C_1^2(U_T) \cap C\left(\bar{U}_T\right)$ solves the heat equation in U_T. Then*

(1)

$$\max_{\bar{U}_T} u = \max_{\Gamma_T} u$$

(2) Furthermore, if U is connected and there exists a point $(x_0, t_0) \in U_T$ such that

$$u(x_0, t_0) = \max_{\bar{U}_T} u$$

then u is constant in $\bar{U}_{t_0} = \bar{U} \times [0, t_0]$.

Proof: Step 1. Suppose there exists a point $(x_0, t_0) \in U_T$ with

$$u(x_0, t_0) = M := \max_{\bar{U}_T} u$$

Then for all sufficiently small $r > 0$, $E(x_0, t_0; r) \subset U_T$; and we employ the mean-value property to deduce

$$M = u(x_0, t_0) = \frac{1}{4r^n} \iint_{E(x_0,t_0;r)} u(y, s) \frac{|x_0 - y|^2}{(t_0 - s)^2} dyds \leq M$$

since

$$1 = \frac{1}{4r^n} \iint_{E(x_0,t_0;r)} \frac{|x_0 - y|^2}{(t_0 - s)^2} dyds$$

Equality holds only if u is identically equal to M within $E(x_0, t_0; r)$. Consequently

$$u(y, s) = M \quad \text{for all } (y, s) \in E(x_0, t_0; r).$$

Step 2. Draw any line segment L in U_T connecting (x_0, t_0) with some other point $(y_0, s_0) \in U_T$, with $s_0 < t_0$. Consider

$$r_0 := \min \{s \geq s_0 \mid u(x, t) = M \text{ for all points } (x, t) \in L, s \leq t \leq t_0\}$$

Since u is continuous, the minimum is attained. Assume $r_0 > s_0$. Then $u(z_0, r_0) = M$ for some point (z_0, r_0) on $L \cap U_T$ and so $u \equiv M$ on $E(z_0, r_0; r)$ for all sufficiently small $r > 0$. Since $E(z_0, r_0; r)$ contains $L \cap \{r_0 - \sigma \leq t \leq r_0\}$ for some small $\sigma > 0$, we have a contradiction. Thus $r_0 = s_0$, and hence $u \equiv M$ on L.

2. Now fix any point $x \in U$ and any time $0 \leq t < t_0$. There exist points $\{x_0, x_1, \ldots, x_m = x\}$ such that the line segments in $\mathbb{R}^n$ connecting x_{i-1} to x_i lie in U for $i = 1, \ldots, m$. (This follows since the set of points in U which can be so connected to x_0 by a polygonal path is nonempty, open and relatively closed in U.) Select times $t_0 > t_1 > \cdots > t_m = t$. Then the line segments in $\mathbb{R}^{n+1}$ connecting (x_{i-1}, t_{i-1}) to (x_i, t_i) $(i = 1, \ldots, m)$ lie in U_T. According to step $1, u \equiv M$ on each such segment and so $u(x, t) = M$.

Theorem 3.3 (Uniqueness on bounded domains)**.** *Let $g \in C(\Gamma_T)$, $f \in C(U_T)$. Then there exists at most one solution $u \in C_1^2(U_T) \cap C(\bar{U}_T)$ of the initial/boundary-value nonhomogeneous problem*

$$\begin{cases} u_t - \Delta u = f \text{ in } U_T \\ \qquad\quad u = g \text{ on } \Gamma_T. \end{cases}$$

Theorem 3.4 (Smoothness). *Suppose $U \subset \mathbb{R}^n$ is open and $u \in C_1^2(U_T)$ solves the heat equation in U_T. Then*

$$u \in C^\infty(U_T)$$

Theorem 3.5 (Local Estimates on derivatives). *There exists for each pair of integers $k, l = 0, 1, \ldots$ a constant $C_{k,l}$ such that*

$$\max_{C(x,t;r/2)} \left| D_x^k D_t^l u \right| \leq \frac{C_{k,l}}{r^{k+2l+n+2}} \|u\|_{L^1(C(x,t;r))}$$

for all cylinders $C(x,t;r/2) \subset C(x,t;r) \subset U_T$ and all solutions u of the heat equation in U_T.

§4 Cauchy Problem

Theorem 4.1 (Maximum principle for the Cauchy problem). *Suppose $u \in C_1^2(\mathbb{R}^n \times (0,T]) \cap C(\mathbb{R}^n \times [0,T])$ solves*

$$\begin{cases} u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,T) \\ u = g \text{ on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

and satisfies the growth estimate

$$u(x,t) \leq Ae^{a|x|^2} \quad (x \in \mathbb{R}^n, 0 \leq t \leq T)$$

for constants $A, a > 0$. Then

$$\sup_{\mathbb{R}^n \times [0,T]} u = \sup_{\mathbb{R}^n} g.$$

Corollary 4.2 (Uniqueness for Cauchy problem). *Let $g \in C(\mathbb{R}^n)$ and $f \in C(\mathbb{R}^n \times [0,T])$. Then there exists at most one solution $u \in C_1^2(\mathbb{R}^n \times (0,T]) \cap C(\mathbb{R}^n \times [0,T])$ of the initial-value problem*

$$\begin{cases} u_t - \Delta u = f \text{ in } \mathbb{R}^n \times (0,T) \\ u = g \text{ on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

satisfying the growth estimate

$$|u(x,t)| \leq Ae^{a|x|^2} \quad (x \in \mathbb{R}^n, \quad 0 \leq t \leq T)$$

for constants $A, a > 0$.

§5 Energy Methods

Theorem 5.1. *We investigate again the initial/boundary-value problem*

$$\begin{cases} u_t - \Delta u = f \text{ in } U_T \\ u = g \text{ on } \Gamma_T \end{cases}$$

We assume as usual that $U \subset \mathbb{R}^n$ is open and bounded and that ∂U is C^1. The terminal time $T > 0$ is given. Set energy

$$e(t) := \int_U u^2(x,t)\,\mathrm{d}x$$

for $0 \le t \le T$.

Theorem 5.2 (Uniqueness)**.** *Assume that $U \subset \mathbb{R}^n$ is open and bounded and that ∂U is C^1. The terminal time $T > 0$ is given. There exists only one solution $u \in C_1^2\left(\bar{U}_T\right)$ of the initial/boundary-value nonhomogeneous problem*

$$\begin{cases} u_t - \Delta u = f \text{ in } U_T \\ \qquad\quad u = g \text{ on } \Gamma_T \end{cases}$$

Proof. 1. If $\tilde{u}$ is another solution, $w := u - \tilde{u}$ solves

$$\begin{cases} w_t - \Delta w = 0 \text{ in } U_T \\ \qquad\quad w = 0 \text{ on } \Gamma_T. \end{cases}$$

Set

$$e(t) := \int_U w^2(x,t)\,\mathrm{d}x \quad (0 \le t \le T)$$

Then

$$\begin{aligned} \dot{e}(t) &= 2\int_U w w_t\,\mathrm{d}x \\ &= 2\int_U w\Delta w\,\mathrm{d}x \\ &= -2\int_U |Dw|^2\,\mathrm{d}x \le 0 \end{aligned}$$

and so

$$e(t) \le e(0) = 0 \quad (0 \le t \le T)$$

Consequently $w = u - \tilde{u} \equiv 0$ in U_T.

Theorem 5.3 (Backwards uniqueness)**.** *Suppose $u, \tilde{u} \in C^2\left(\bar{U}_T\right)$ solve*

$$\begin{cases} u_t - \Delta u = 0 \text{ in } U_T \\ \qquad\quad u = g \text{ on } \partial U \times [0,T] \end{cases}$$

If

$$u(x,T) = \tilde{u}(x,T) \quad \text{for all } x \in U$$

then

$$u \equiv \tilde{u} \quad \text{within } U_T.$$

Proof. 1. Write $w := u - \tilde{u}$ and set

$$e(t) := \int_U w^2(x,t)\,\mathrm{d}x \quad (0 \le t \le T)$$

As before

$$\dot{e}(t) = -2 \int_U |Dw|^2 \, \mathrm{d}x$$

Furthermore

$$\begin{aligned} \ddot{e}(t) &= -4 \int_U Dw \cdot Dw_t \, \mathrm{d}x \\ &= 4 \int_U \Delta w w_t \, \mathrm{d}x \\ &= 4 \int_U (\Delta w)^2 \, \mathrm{d}x \end{aligned}$$

Now since $w = 0$ *on* ∂U*,*

$$\begin{aligned} \int_U |Dw|^2 \, \mathrm{d}x &= -\int_U w \Delta w \, \mathrm{d}x \\ &\leq \left(\int_U w^2 \, \mathrm{d}x \right)^{1/2} \left(\int_U (\Delta w)^2 \, \mathrm{d}x \right)^{1/2} \end{aligned}$$

Thus

$$(\dot{e}(t))^2 \leq e(t) \ddot{e}(t)$$

2. Now if $e(t) = 0$ *for all* $0 \leq t \leq T$*, we are done. Otherwise there exists an interval* $[t_1, t_2] \subset [0, T]$*, with*

$$e(t) > 0 \quad \textit{for } t_1 \leq t < t_2$$

and

$$e\left(t_2\right) = 0$$

3. Consequently

$$f(t) := \log e(t) \quad (t_1 \leq t < t_2)$$

is convex on the interval (t_1, t_2) *and* $f(t_2^-) = -\infty$*, a contradiction.*

Chapter IV

Wave Equation

§1 Solution for n=1, d'Alembert's formula

Theorem 1.1. *Assume* $g \in C^2(\mathbb{R})$, $h \in C^1(\mathbb{R})$,

$$\begin{cases} u_{tt} - u_{xx} = 0 \text{ in } \mathbb{R} \times (0, \infty) \\ u = g, u_t = h \text{ on } \mathbb{R} \times \{t = 0\} \end{cases}$$

We define u *by* ***d'Alembert's formula***

$$u(x,t) = \frac{1}{2}[g(x+t) + g(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} h(y)dy \quad (x \in \mathbb{R}, t \geq 0)$$

Then

(i) $u \in C^2(\mathbb{R} \times [0, \infty))$

(ii) $u_{tt} - u_{xx} = 0$ *in* $\mathbb{R} \times (0, \infty)$

(iii) $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u(x,t) = g(x^0)$, $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u_t(x,t) = h(x^0)$ *for each point* $x^0 \in \mathbb{R}$.

Remark *(i) Our solution* u *has the form*

$$u(x,t) = F(x+t) + G(x-t)$$

for appropriate functions F *and* G. *Conversely any function of this form solves* $u_{tt} - u_{xx} = 0$.

(ii) If $g \in C^k$ *and* $h \in C^{k-1}$, *then* $u \in C^k$ *but is not in general smoother.*

Theorem 1.2 (Reflection Method)**.** *Let us consider this initial/boundary-value problem on the half-line* $\mathbb{R}_+ = \{x > 0\}$ *:*

$$\begin{cases} u_{tt} - u_{xx} = 0 \text{ in } \mathbb{R}_+ \times (0, \infty) \\ u = g, u_t = h \text{ on } \mathbb{R}_+ \times \{t = 0\} \\ \qquad u = 0 \text{ on } \{x = 0\} \times (0, \infty) \end{cases}$$

where g, h are given, with $g(0) = h(0) = 0$ and $g''(0) = 0$.

Proof: We set

$$\tilde{u}(x,t) := \begin{cases} u(x,t) & (x \geq 0, t \geq 0) \\ -u(-x,t) & (x \leq 0, t \geq 0) \end{cases}$$

$$\tilde{g}(x) := \begin{cases} g(x) & (x \geq 0) \\ -g(-x) & (x \leq 0) \end{cases}$$

$$\tilde{h}(x) := \begin{cases} h(x) & (x \geq 0) \\ -h(-x) & (x \leq 0) \end{cases}$$

Then (9) becomes

$$\begin{cases} \tilde{u}_{tt} = \tilde{u}_{xx} & \textit{in } \mathbb{R} \times (0,\infty) \\ \tilde{u} = \tilde{g}, \tilde{u}_t = \tilde{h} & \textit{on } \mathbb{R} \times \{t = 0\} \end{cases}$$

and $g \in C^2(\mathbb{R})$, $f \in C^1(\mathbb{R})$. Hence d'Alembert's formula implies

$$\tilde{u}(x,t) = \frac{1}{2}[\tilde{g}(x+t) + \tilde{g}(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} \tilde{h}(y)dy$$

§2

Theorem 2.1 (Spherical means; Euler-Poisson-Darboux equation)**.** *Now suppose $n \geq 2, m \geq 2$, and $u \in C^m(\mathbb{R}^n \times [0,\infty))$ solves the initial-value problem*

$$\begin{cases} u_{tt} - \Delta u = 0 & \textit{in } \mathbb{R}^n \times (0,\infty) \\ u = g, u_t = h & \textit{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Let $x \in \mathbb{R}^n, t > 0, r > 0$. Define

$$\begin{cases} U(x;r,t) := \fint_{\partial B(x,r)} u(y,t)dS(y) \\ G(x;r) := \fint_{\partial B(x,r)} g(y)dS(y) \\ H(x;r) := \fint_{\partial B(x,r)} h(y)dS(y) \end{cases}$$

Fix $x \in \mathbb{R}^n$, then $U \in C^m\left(\overline{\mathbb{R}}_+ \times [0,\infty)\right)$ and

$$\begin{cases} U_{tt} - U_{rr} - \dfrac{n-1}{r}U_r = 0 \textit{ in } \mathbb{R}_+ \times (0,\infty) \\ U = G, U_t = H \textit{ on } \mathbb{R}_+ \times \{t = 0\} \end{cases}$$

Proof: 1. First, we compute for $r > 0$

$$U_r(x;r,t) = \frac{r}{n} \fint_{B(x,r)} \Delta_y u(y,t)dy$$

From this equality we deduce $\lim_{r\to 0^+} U_r(x;r,t) = 0$. *We next differentiate it*

$$U_{rr}(x;r,t) = \fint_{\partial B(x,r)} \Delta u dS + \left(\frac{1}{n} - 1\right) \fint_{B(x,r)} \Delta u dy$$

Thus $\lim_{r\to 0^+} U_{rr}(x;r,t) = \frac{1}{n}\Delta u(x,t)$. *Using this formula, we can similarly compute* U_{rrr}, *etc., and so verify that* $U \in C^m\left(\overline{\mathbb{R}}_+ \times [0,\infty)\right)$.

2. Continuing the calculation above, we see from that

$$\begin{aligned} U_r &= \frac{r}{n} \fint_{B(x,r)} u_{tt}\, \mathrm{d}y \\ &= \frac{1}{n\alpha(n)} \frac{1}{r^{n-1}} \int_{B(x,r)} u_{tt} dy \end{aligned}$$

Thus

$$r^{n-1} U_r = \frac{1}{n\alpha(n)} \int_{B(x,r)} u_{tt} dy$$

and so

$$\left(r^{n-1}U_r\right)_r = \frac{1}{n\alpha(n)} \int_{\partial B(x,r)} u_{tt} dS = r^{n-1} \fint_{\partial B(x,r)} u_{tt} dS = r^{n-1} U_{tt}$$

§3 Solution for even n=2k+1

Lemma 3.1. *Let* $\phi : \mathbb{R} \to \mathbb{R}$ *be* C^{k+1}. *Then for* $k = 1, 2, \ldots$

(i)

$$\left(\frac{d^2}{dr^2}\right)\left(\frac{1}{r}\frac{d}{dr}\right)^{k-1}\left(r^{2k-1}\phi(r)\right) = \left(\frac{1}{r}\frac{d}{dr}\right)^{k}\left(r^{2k}\frac{d\phi}{dr}(r)\right)$$

(ii)

$$\left(\frac{1}{r}\frac{d}{dr}\right)^{k-1}\left(r^{2k-1}\phi(r)\right) = \sum_{j=0}^{k-1} \beta_j^k r^{j+1} \frac{d^j\phi}{dr^j}(r)$$

where the constants $\beta_j^k (j = 0, \ldots, k-1)$ *are independent of* ϕ.

(iii) $\beta_0^k = 1 \cdot 3 \cdot 5 \cdots (2k-1)$.

Theorem 3.2. *Now assume* $n \geq 3$ *is an odd integer and set*

$$n = 2k + 1 \quad (k \geq 1)$$

Henceforth suppose $u \in C^{k+1}(\mathbb{R}^n \times [0,\infty))$ *solves the initial-value problem*

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0,\infty) \\ u = g, u_t = h & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Then the function U *defined is* C^{k+1}.

We write

$$\begin{cases} \tilde{U}(r,t) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}U(x;r,t)\right) \\ \tilde{G}(r) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}G(x;r)\right) \quad (r > 0, t \geq 0). \\ \tilde{H}(r) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}H(x;r)\right) \end{cases}$$

Then $\tilde{U}(r,t)$ *solves the wave equation on the half-line. We have*

$$\begin{cases} \tilde{U}_{tt} - \tilde{U}_{rr} = 0 & \text{in } \mathbb{R}_+ \times (0,\infty) \\ \tilde{U} = \tilde{G}, \tilde{U}_t = \tilde{H} & \text{on } \mathbb{R}_+ \times \{t = 0\} \\ \tilde{U} = 0 & \text{on } \{r = 0\} \times (0,\infty) \end{cases}$$

We conclude for $0 \leq r \leq t$ *that*

$$\tilde{U}(r,t) = \frac{1}{2}[\tilde{G}(r+t) - \tilde{G}(t-r)] + \frac{1}{2}\int_{t-r}^{t+r} \tilde{H}(y)dy$$

But recall $u(x,t) = \lim_{r\to 0} U(x;r,t)$. *Furthermore Lemma 2(ii) asserts*

$$\begin{aligned} \tilde{U}(r,t) &= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}U(x;r,t)\right) \\ &= \sum_{j=0}^{k-1} \beta_j^k r^{j+1} \frac{\partial^j}{\partial r^j} U(x;r,t) \end{aligned}$$

and so

$$\lim_{r\to 0} \frac{\tilde{U}(r,t)}{\beta_0^k r} = \lim_{r\to 0} U(x;r,t) = u(x,t)$$

Thus (30) implies

$$\begin{aligned} u(x,t) &= \frac{1}{\beta_0^k} \lim_{r\to 0}\left[\frac{\tilde{G}(t+r) - \tilde{G}(t-r)}{2r} + \frac{1}{2r}\int_{t-r}^{t+r} \tilde{H}(y)dy\right] \\ &= \frac{1}{\beta_0^k}\left[\tilde{G}'(t) + \tilde{H}(t)\right] \end{aligned}$$

Proof: If $r > 0$,

$$\begin{aligned}
\tilde{U}_{rr} &= \left(\frac{\partial^2}{\partial r^2}\right)\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}U\right) \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k}\left(r^{2k}U_r\right) \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left[r^{2k-1}U_{rr} + 2kr^{2k-2}U_r\right] \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left[r^{2k-1}\left(U_{rr} + \frac{n-1}{r}U_r\right)\right] \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1}\left(r^{2k-1}U_{tt}\right) = \tilde{U}_{tt}
\end{aligned}$$

Using previous lemma (ii) conclude as well that $\tilde{U} = 0$ *on* $\{r = 0\}$.

§4 Nonhomogenous Problem

Theorem 4.1 (Solution of nonhomogeneous wave equation)**.** *Assume that* $n \geq 2$ *and* $f \in C^{[\frac{n}{2}]+1}\left(\mathbb{R}^n \times [0,\infty)\right)$

$$\begin{cases} u_{tt} - \Delta u = f(x,t) & \text{in } \mathbb{R}^n \times (0,\infty) \\ u = 0, u_t = 0 & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Motivated by Duhamel's principle, we define $u = u(x,t;s)$ *to be the solution of*

$$\begin{cases} u_{tt}(\cdot;s) - \Delta u(\cdot;s) = 0 & \text{in } \mathbb{R}^n \times (s,\infty) \\ u(\cdot;s) = 0, u_t(\cdot;s) = f(\cdot,s) & \text{on } \mathbb{R}^n \times \{t = s\} \end{cases}$$

Now set

$$u(x,t) := \int_0^t u(x,t;s)ds \quad (x \in \mathbb{R}^n, t \geq 0)$$

Duhamel's principle asserts this is a solution of

$$\begin{cases} u_{tt} - \Delta u = f & \text{in } \mathbb{R}^n \times (0,\infty) \\ u = 0, u_t = 0 & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

and we have

(i) $u \in C^2\left(\mathbb{R}^n \times [0,\infty)\right)$,

(ii) $u_{tt} - \Delta u = f$ *in* $\mathbb{R}^n \times (0,\infty)$

(iii) $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ x\in\mathbb{R}^n, t>0}} u(x,t) = 0$, $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ x\in\mathbb{R}^n, t>0}} u_t(x,t) = 0$ *for each point* $x^0 \in \mathbb{R}^n$.

Proof. 1. If n *is odd,* $\left[\frac{n}{2}\right]+1 = \frac{n+1}{2}$. *According to Theorem 2, we have* $u(\cdot,\cdot;s) \in C^2\left(\mathbb{R}^n \times [s,\infty)\right)$ *for each* $s \geq 0$, *and so* $u \in C^2\left(\mathbb{R}^n \times [0,\infty)\right)$. *If* n *is even,* $\left[\frac{n}{2}\right] + 1 = \frac{n+2}{2}$. *Hence* $u \in C^2\left(\mathbb{R}^n \times [0,\infty)\right)$, *according to Theorem 3 .*

2. We then compute respectively

$$u_t(x,t) = u(x,t;t) + \int_0^t u_t(x,t;s)ds = \int_0^t u_t(x,t;s)ds$$

and

$$u_{tt}(x,t) = u_t(x,t;t) + \int_0^t u_{tt}(x,t;s)ds = f(x,t) + \int_0^t u_{tt}(x,t;s)ds$$

Furthermore

$$\Delta u(x,t) = \int_0^t \Delta u(x,t;s)ds = \int_0^t u_{tt}(x,t;s)ds$$

Thus

$$u_{tt}(x,t) - \Delta u(x,t) = f(x,t) \quad (x \in \mathbb{R}^n, t > 0)$$

and clearly $u(x,0) = u_t(x,0) = 0$ *for* $x \in \mathbb{R}^n$.

§5 Energy Methods

Theorem 5.1 (Uniqueness for wave equation)**.** *Let* $U \subset \mathbb{R}^n$ *be a bounded, open set with a smooth boundary* ∂U*. There exists at most one function* $u \in C^2\left(\bar{U}_T\right)$ *solving the initial/boundary-value problem*

$$\begin{cases} u_{tt} - \Delta u = f \text{ in } U_T \\ u = g \text{ on } \Gamma_T \\ u_t = h \text{ on } U \times \{t = 0\} \end{cases}$$

Proof: If $\tilde{u}$ *is another such solution, then* $w := u - \tilde{u}$ *solves*

$$\begin{cases} w_{tt} - \Delta w = 0 \text{ in } U_T \\ w = 0 \text{ on } \Gamma_T \\ w_t = 0 \text{ on } U \times \{t = 0\} \end{cases}$$

Define the energy

$$E(t) := \frac{1}{2}\int_U w_t^2(x,t) + |Dw(x,t)|^2 \,\mathrm{d}x \quad (0 \le t \le T)$$

We compute

$$\begin{aligned} \dot{E}(t) &= \int_U w_t w_{tt} + Dw \cdot Dw_t \,\mathrm{d}x \\ &= \int_U w_t \left(w_{tt} - \Delta w\right) \,\mathrm{d}x = 0 \end{aligned}$$

There is no boundary term since $w = 0$*, and hence* $w_t = 0$*, on* $\partial U \times [0,T]$.

Theorem 5.2 (Finite propagation speed)**.** *Fix* $x_0 \in \mathbb{R}^n, t_0 > 0$ *and consider the backwards wave*

cone with apex (x_0, t_0)

$$K(x_0, t_0) := \{(x,t) : 0 \le t \le t_0, |x - x_0| \le t_0 - t\}$$

If $u \equiv u_t \equiv 0$ *on* $B(x_0, t_0) \times \{t = 0\}$*, then* $u \equiv 0$ *within the cone* $K(x_0, t_0)$*.*

Proof: Define the local energy

$$e(t) := \frac{1}{2} \int_{B(x_0, t_0 - t)} u_t^2(x,t) + |Du(x,t)|^2 \, \mathrm{d}x \quad (0 \le t \le t_0).$$

Then

$$\begin{aligned}
\dot{e}(t) &= \int_{B(x_0,t_0-t)} u_t u_{tt} + Du \cdot Du_t \, \mathrm{d}x - \frac{1}{2} \int_{\partial B(x_0,t_0-t)} u_t^2 + |Du|^2 dS \\
&= \int_{B(x_0,t_0-t)} u_t \left(u_{tt} - \Delta u\right) \, \mathrm{d}x \\
&\quad + \int_{\partial B(x_0,t_0-t)} \frac{\partial u}{\partial \nu} u_t dS - \frac{1}{2} \int_{\partial B(x_0,t_0-t)} u_t^2 + |Du|^2 dS \\
&= \int_{\partial B(x_0,t_0-t)} \frac{\partial u}{\partial \nu} u_t - \frac{1}{2} u_t^2 - \frac{1}{2} |Du|^2 dS \le 0
\end{aligned}$$

by the Cauchy-Schwarz and Cauchy inequalities. We find $\dot{e}(t) \le 0$*; and so* $e(t) \le e(0) = 0$ *for all* $0 \le t \le t_0$*. Thus* u_t*,* $Du \equiv 0$*, and consequently* $u \equiv 0$ *within the cone* $K(x_0, t_0)$*.*

Part II

Chapter V

Nolinear First-Order PDE

§1 Complete integrals, Envelopes

§1.1 Complete integrals

Definition 1.1. *In this chapter we investigate general nonlinear first-order partial differential equations of the form*

$$F(Du, u, x) = 0$$

where $x \in U$ and U is an open subset of $\mathbb{R}^n$. Here

$$F : \mathbb{R}^n \times \mathbb{R} \times \bar{U} \to \mathbb{R}$$

is given, and $u : \bar{U} \to \mathbb{R}$ is the unknown, $u = u(x)$.

Let us write

$$F = F(p, z, x) = F(p_1, \ldots, p_n, z, x_1, \ldots, x_n)$$

for $p \in \mathbb{R}^n, z \in \mathbb{R}, x \in U$. We also assume hereafter that F is smooth and set

$$\begin{cases} D_pF = (F_{p_1}, \ldots, F_{p_n}) \\ D_zF = F_z \\ D_xF = (F_{x_1}, \ldots, F_{x_n}) \end{cases}$$

We are concerned with discovering solutions u of the boundary-value PDE

$$\begin{cases} F(Du, u, x) = 0 & \text{in } U \\ u = g & \text{on } \Gamma \end{cases}$$

where Γ is some given subset of ∂U and $g : \Gamma \to \mathbb{R}$ is prescribed.

Definition 1.2 (Complete integrals)**.** *We begin our analysis of the nonlinear first-order PDE*

$$F(Du, u, x) = 0$$

Suppose $A \subset \mathbb{R}^n$ is an open set. Assume for each parameter $a = (a_1, \dots, a_n) \in A$ we have a C^2 solution $u = u(x; a)$.

A C^2 function $u = u(x; a)$ is called a ***complete integral*** *in $U \times A$ provided*

$$u(x; a) \text{ solves the PDE for each } a \in A$$

and

$$\text{rank}\left(D_a u, D^2_{xa} u\right) = n \quad (x \in U, a \in A)$$

where

$$\left(D_a u, D^2_{xa} u\right) := \begin{pmatrix} u_{a_1} & u_{x_1 a_1} & \cdots & u_{x_n a_1} \\ \vdots & \vdots & \ddots & \vdots \\ u_{a_n} & u_{x_1 a_n} & \cdots & u_{x_n a_n} \end{pmatrix}_{n \times (n+1)}.$$

Definition 1.3. *Let $u = u(x; a)$ be a C^1 function of $x \in U$, parameter $a \in A$, where $U \subset \mathbb{R}^n$ and $A \subset \mathbb{R}^m$ are open sets. Consider the vector equation*

$$D_a u(x; a) = 0 \quad (x \in U, a \in A).$$

Suppose that we can solve it for the parameter a,

$$a = \phi(x)$$

thus

$$D_a u(x; \phi(x)) = 0 \quad (x \in U).$$

We then call

$$v(x) := u(x; \phi(x)) \quad (x \in U)$$

the ***envelope*** *of the functions $\{u(\cdot; a)\}_{a \in A}$.*

Theorem 1.4 (Construction of new solutions)**.** *Suppose for each $a \in A$ as above that $u = u(\cdot; a)$ solves the PDE. Assume further that the envelope v, defined above, exists and is a C^1 function. Then v solves PDE as well called a* ***singular integral*** *of PDE.*

Proof. We have $v(x) = u(x; \phi(x))$; and so for $i = 1, \dots, n$

$$\begin{aligned} v_{x_i}(x) &= u_{x_i}(x; \phi(x)) + \sum_{j=1}^{m} u_{a_j}(x, \phi(x)) \phi^j_{x_i}(x) \\ &= u_{x_i}(x; \phi(x)) \end{aligned}$$

Hence for each $x \in U$,

$$F(Dv(x), v(x), x) = F(Du(x; \phi(x)), u(x; \phi(x)), x) = 0.$$

§2 Characteristics

§2.1 Derivation of Characteristic ODE

Theorem 2.1 (Structure of Characteristic ODE)**.** *We return to our basic nonlinear first-order PDE*

$$F(Du, u, x) = 0 \quad \text{in } U \tag{1}$$

subject now to the boundary condition

$$u = g \quad \text{on } \Gamma \tag{2}$$

where $\Gamma \subseteq \partial U$ and $g : \Gamma \to \mathbb{R}$ are given. We hereafter suppose that F, g are smooth functions and assume u is a C^2 solution.

Let us suppose the curve is described parametrically by the function

$$\mathbf{x}(s) = \big(x^1(s), \dots, x^n(s)\big)^T \tag{3}$$

the parameter s lying in some subinterval $I \subseteq \mathbb{R}$. We define also

$$z(s) := u(\mathbf{x}(s)) \tag{4}$$

In addition, set

$$\mathbf{p}(s) := Du(\mathbf{x}(s)) \tag{5}$$

that is, $\mathbf{p}(s) = (p^1(s), \dots, p^n(s))$, where

$$p^i(s) = u_{x_i}(\mathbf{x}(s)) \quad (i = 1, \dots, n) \tag{5'}$$

records the values of the gradient Du.

We must choose the function $\mathbf{x}(\cdot)$ in such a way that we can compute $z(\cdot)$ and $\mathbf{p}(\cdot)$. Indeed, we set $\mathbf{x}$ solves the ODE

$$\dot{\mathbf{x}}(s) = F_p(\mathbf{p}(s), z(s), \mathbf{x}(s)) \tag{6}$$

that is,

$$\dot{x}^j(s) = F_{p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s)) \tag{6'}$$

Then we differentiate (5)

$$\dot{\mathbf{p}}(s) = D^2u(\mathbf{x}(s))\dot{\mathbf{x}}(s) \tag{7}$$

that is,

$$\dot{p}^i(s) = \sum_{j=1}^{n} u_{x_i x_j}(\mathbf{x}(s))\dot{x}^j(s) \tag{7'}$$

On the other hand, we can also differentiate the PDE (1) with respect to x_i:

$$F_p D^2 u + F_z Du + F_x = \mathbf{0} \tag{8}$$

that is,

$$\sum_{j=1}^{n} F_{p_j}(Du, u, x) u_{x_j x_i} + F_z(Du, u, x) u_{x_i} + F_{x_i}(Du, u, x) = 0 \tag{8'}$$

We evaluate (8) at $x = \mathbf{x}(s)$, obtaining the identity:

$$F_p(\mathbf{p}(s), z(s), \mathbf{x}(s)) D^2 u(\mathbf{x}(s)) + F_z(\mathbf{p}(s), z(s), \mathbf{x}(s))\mathbf{p}(s) + F_x(\mathbf{p}(s), z(s), \mathbf{x}(s))$$

that is,

$$\begin{aligned} &\sum_{j=1}^{n} F_{p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s)) u_{x_i x_j}(\mathbf{x}(s)) \\ &\quad + F_z(\mathbf{p}(s), z(s), \mathbf{x}(s)) p^i(s) + F_{x_i}(\mathbf{p}(s), z(s), \mathbf{x}(s)) = 0 \end{aligned}$$

Substitute this expression and (8) into (6): (9)

$$\dot{p}^i(s) = -F_{x_i}(\mathbf{p}(s), z(s), \mathbf{x}(s)) - F_z(\mathbf{p}(s), z(s), \mathbf{x}(s)) p^i(s) \tag{9}$$

Finally we differentiate (3):

$$\dot{z}(s) = \sum_{j=1}^{n} u_{x_j}(\mathbf{x}(s)) \dot{x}^j(s) = \sum_{j=1}^{n} p^j(s) F_{p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s))$$

the second equality holding by (5) and (8).

We summarize by rewriting equations (8)-(10) in vector notation:

$$\begin{cases} \dot{\mathbf{p}}(s) = -D_x F(\mathbf{p}(s), z(s), \mathbf{x}(s)) - D_z F(\mathbf{p}(s), z(s), \mathbf{x}(s))\mathbf{p}(s) \\ \dot{z}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \cdot \mathbf{p}(s) \\ \dot{\mathbf{x}}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \end{cases}$$

Furthermore,

$$F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \equiv 0$$

These identities hold for $s \in I$.

The important system (11) of $2n+1$ first-order ODE comprises the characteristic equations of the first-order PDE (1). The functions $\mathbf{p}(\cdot) = (p^1(\cdot), \ldots, p^n(\cdot))$, $z(\cdot)$, $\mathbf{x}(\cdot) = (x^1(\cdot), \ldots, x^n(\cdot))$ are called the characteristics. We will sometimes refer to $\mathbf{x}(\cdot)$ as the projected characteristic: it is the projection of the full characteristics $(\mathbf{p}(\cdot), z(\cdot), \mathbf{x}(\cdot)) \subset \mathbb{R}^{2n+1}$ onto the physical region $U \subset \mathbb{R}^n$.

§2.2 Boundary Condition

Lemma 2.2 (Straightening the boundary)**.** *To simplify subsequent calculations, it is convenient first to change variables, so as to "flatten out" part of the boundary ∂U. To accomplish this, we first fix any point $x^0 \in \partial U$. Then we find smooth mappings $\mathbf{\Phi}, \mathbf{\Psi} : \mathbb{R}^n \to \mathbb{R}^n$ such that $\mathbf{\Psi} = \mathbf{\Phi}^{-1}$ and $\mathbf{\Phi}$ straightens out ∂U near x^0.*

Given any function $u : U \to \mathbb{R}$, let us write $V := \mathbf{\Phi}(U)$ and set

$$v(y) := u(\mathbf{\Psi}(y)) \quad (y \in V)$$

Then

$$u(x) = v(\mathbf{\Phi}(x)) \quad (x \in U)$$

Now suppose that u is a C^1 solution of our boundary-value problem (1), (2) in U. What PDE does v then satisfy in V ?

According to (25), we see

$$u_{x_i}(x) = \sum_{k=1}^{n} v_{y_k}(\mathbf{\Phi}(x))\Phi^k_{x_i}(x) \quad (i = 1, \dots, n)$$

that is,

$$Du(x) = Dv(y)D\Phi(x)$$

Thus (1) implies

$$F(Dv(y)D\Phi(\Psi(y)), v(y), \Psi(y)) = F(Du(x), u(x), x) = 0$$

This is an expression having the form

$$G(Dv(y), v(y), y) = 0 \quad \textit{in } V$$

In addition $v = h$ on Δ, where $\Delta := \mathbf{\Phi}(\Gamma)$ and $h(y) := g(\mathbf{\Psi}(y))$. In summary, our problem (1), (2) transforms to read

$$\begin{cases} G(Dv, v, y) = 0 \textit{ in } V \\ \qquad\qquad v = h \textit{ on } \Delta \end{cases}$$

for G, h as above. The point is that if we change variables to straighten out the boundary near x^0, the boundary-value problem (1), (2) converts into a problem having the same form.

Lemma 2.3 (Compatibility conditions on boundary data)**.** *We intend now to utilize the characteristic ODE to construct a solution at least near $x^0 \in \Gamma$, lying in the plane $\{x_n = 0\}$, and for this we must discover appropriate initial conditions*

$$\mathbf{p}(0) = p^0, z(0) = z^0, \mathbf{x}(0) = x^0$$

Now clearly if the curve $\mathbf{x}(\cdot)$ *passes through* x^0*, we should insist that*

$$z^0 = g\left(x^0\right) \tag{1}$$

Since boundary-values implies $u\left(x_1, \ldots x_{n-1}, 0\right) = g\left(x_1, \ldots, x_{n-1}\right)$ *near* x^0*, we may differentiate to find*

$$Du\left(x^0\right) = \left(Dg\left(x^0\right), 0\right)$$

As we also want the PDE to hold, we should therefore insist $p^0 = (p_1^0, \ldots, p_n^0)$ *satisfies these relations:*

$$\begin{cases} p_i^0 = g_{x_i}\left(x^0\right) \quad (i = 1, \ldots, n-1) \\ F\left(p^0, z^0, x^0\right) = 0 \end{cases} \tag{2}$$

These identities provide n *equations for the* n *quantities* $p^0 = (p_1^0, \ldots, p_n^0)$*.*

We call (1) and (2) the compatibility conditions. A triple $(p^0, z^0, x^0) \in \mathbb{R}^{2n+1}$ *verifying (1), (2) is admissible.*

Remark *Note* z^0 *is uniquely determined by the boundary condition and our choice of the point* x^0*, but a vector* p^0 *satisfying (2) may not exist or may not be unique.*

Lemma 2.4 (Noncharacteristic boundary conditions)**.** *Given a point* $y = (y_1, \ldots, y_{n-1}, 0) \in \Gamma$*, with* y *close to* x^0*, we intend to solve the characteristic ODE*

$$\begin{cases} \dot{\mathbf{p}}(s) = -D_x F(\mathbf{p}(s), z(s), \mathbf{x}(s)) - D_z F(\mathbf{p}(s), z(s), \mathbf{x}(s))\mathbf{p}(s) \\ \dot{z}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \cdot \mathbf{p}(s) \\ \dot{\mathbf{x}}(s) = D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \end{cases}$$

with the initial conditions

$$\mathbf{p}(0) = \mathbf{q}(y), z(0) = g(y), \mathbf{x}(0) = y$$

Our task then is to find a function $\mathbf{q}(\cdot) = (q^1(\cdot), \ldots, q^n(\cdot))$*, so that*

$$\mathbf{q}\left(x^0\right) = p^0$$

and $(\mathbf{q}(y), g(y), y)$ *is admissible for every given* y *near* x*; that is, the compatibility conditions*

$$\begin{cases} q^i(y) = g_{x_i}(y) \quad (i = 1, \ldots, n-1) \\ F(\mathbf{q}(y), g(y), y) = 0 \end{cases}$$

hold for all $y \in \Gamma$ *close to* x^0*.*

There exists a unique solution $\mathbf{q}(\cdot)$ *of for all* $y \in \Gamma$ *sufficiently close to* x^0*, provided*

$$F_{p_n}\left(p^0, z^0, x^0\right) \neq 0$$

We say admissible (p^0, z^0, x^0) *is* ***noncharacteristic*** *if* $F_{p_n}\left(p^0, z^0, x^0\right) \neq 0$

Proof: Our problem is to find $q^n(y)$ so that

$$F(\mathbf{q}(y), g(y), y) = 0$$

where $q^i(y) = g_{x_i}(y)$ for $i = 1, \ldots, n-1$. Since $F(p^0, z^0, x^0) = 0$, the Implicit Function Theorem implies we can indeed locally and uniquely solve for $q^n(y)$, provided that the noncharacteristic condition is valid.

Remark *If Γ is not flat near x^0, the condition that Γ be noncharacteristic reads*

$$D_pF\left(p^0, z^0, x^0\right) \cdot \boldsymbol{\nu}\left(x^0\right) \neq 0$$

$\boldsymbol{\nu}(x^0)$ denoting the outward unit normal to ∂U at x^0.

§2.3 Local Solution

Definition 2.5. *Suppose further that (p^0, z^0, x^0) is an admissible triple of boundary data, which is noncharacteristic. According to Lemma 1 there is a function $\mathbf{q}(\cdot)$ so that $p^0 = \mathbf{q}(x^0)$ and the triple $(\mathbf{q}(y), g(y), y)$ is admissible, for all y sufficiently close to x^0.*

Given any such point $y = (y_1, \ldots, y_{n-1}, 0)$ near x^0, we solve the characteristic ODE

$$\begin{cases} \dot{\mathbf{p}}(s) = -D_xF(\mathbf{p}(s), z(s), \mathbf{x}(s)) - D_zF(\mathbf{p}(s), z(s), \mathbf{x}(s))\mathbf{p}(s) \\ \dot{z}(s) = D_pF(\mathbf{p}(s), z(s), \mathbf{x}(s)) \cdot \mathbf{p}(s) \\ \dot{\mathbf{x}}(s) = D_pF(\mathbf{p}(s), z(s), \mathbf{x}(s)) \end{cases}$$

subject to initial conditions

$$\mathbf{p}(0) = \mathbf{q}(y), z(0) = g(y), \mathbf{x}(0) = y$$

Let us write

$$\begin{cases} \mathbf{p}(s) = \mathbf{p}(y, s) = \mathbf{p}\left(y_1, \ldots, y_{n-1}, s\right) \\ z(s) = z(y, s) = z\left(y_1, \ldots, y_{n-1}, s\right) \\ \mathbf{x}(s) = \mathbf{x}(y, s) = \mathbf{x}\left(y_1, \ldots, y_{n-1}, s\right) \end{cases}$$

to display the dependence of the solution on s and y. Also, we will henceforth when convenient regard x^0 as lying in $\mathbb{R}^{n-1}$.

Lemma 2.6 (Local invertibility)**.** *Assume we have the noncharacteristic condition $F_{p_n}(p^0, z^0, x^0) \neq 0$. Then there exist an open interval $I \subseteq \mathbb{R}$ containing 0, a neighborhood W of x^0 in $\Gamma \subset \mathbb{R}^{n-1}$, and a neighborhood V of x^0 in $\mathbb{R}^n$, such that for each $x \in V$ there exist unique $s \in I, y \in W$ such that*

$$x = \mathbf{x}(y, s)$$

The mappings $x \mapsto (s, y)$ are C^2.

Proof: We have $\mathbf{x}(x^0, 0) = x^0$. Consequently the Inverse Function Theorem gives the result,

provided

$$\det \frac{\partial \mathbf{x}}{\partial(y,s)}\left(x^0,0\right) \neq 0$$

Now

$$\mathbf{x}(y,0) = (y,0) \quad (y \in \Gamma)$$

and so if $i = 1, \ldots, n-1$,

$$x^j_{y_i}\left(x^0,0\right) = \begin{cases} \delta_{ij}(j = 1, \ldots, n-1) \\ 0(j = n) \end{cases}$$

Furthermore equation (31)(c) implies

$$x^j_s\left(x^0,0\right) = F_{p_j}\left(p^0, z^0, x^0\right).$$

Thus

$$D\mathbf{x}\left(x^0,0\right) = \begin{pmatrix} 1 & & 0 & \\ \ddots & \vdots & \vdots & F_{p_n}\left(p^0,z^0,x^0\right) \\ 0 & 1 & \vdots & \\ 0 & \cdots & 0 & \end{pmatrix}_{n\times n}$$

whence $\det D\mathbf{x}\left(x^0,0\right) \neq 0$ *follows from the noncharacteristic condition (35).*

In view of Lemma 2 for each $x \in V$, *we can locally uniquely solve the equation*

$$\begin{cases} x = \mathbf{x}(y,s) \\ \textit{for } y = \mathbf{y}(x), s = s(x) \end{cases}$$

Finally, let us define

$$\begin{cases} u(x) := z(\mathbf{y}(x), s(x)) \\ \mathbf{p}(x) := \mathbf{p}(\mathbf{y}(x), s(x)) \end{cases}$$

for $x \in V$ *and* s, y *as in (37). We come finally to our principal assertion, namely, that we can locally weave together the solutions of the characteristic ODE into a solution of the PDE.*

§3 Hamilton-Jacobi Equation

§3.1

Definition 3.1. *We study in some detail the initial-value problem for the Hamilton-Jacobi equation:*

$$\begin{cases} u_t + H(Du) = 0 \textit{ in } \mathbb{R}^n \times (0,\infty) \\ \qquad\qquad u = g \textit{ on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Here $u : \mathbb{R}^n \times [0, \infty) \to \mathbb{R}$ *is the unknown,* $u = u(x, t)$*, and* $Du = D_x u = (u_{x_1}, \ldots, u_{x_n})$*. We are given the Hamiltonian* $H : \mathbb{R}^n \to \mathbb{R}$ *and the initial function* $g : \mathbb{R}^n \to \mathbb{R}$*.*

Definition 3.2. *Assume that* $L : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ *is a given smooth function, hereafter called the* ***Lagrangian****. We write*

$$L = L(v, x) = L(v_1, \ldots, v_n, x_1, \ldots, x_n) \quad (v, x \in \mathbb{R}^n)$$

and

$$\begin{cases} D_v L = (L_{v_1} \cdots L_{v_n}) \\ D_x L = (L_{x_1} \cdots L_{x_n}) . \end{cases}$$

Now fix two points $x, y \in \mathbb{R}^n$ *and a time* $t > 0$*. We introduce then the* ***action functional***

$$I[\mathbf{w}(\cdot)] := \int_0^t L(\dot{\mathbf{w}}(s), \mathbf{w}(s)) ds$$

defined for functions $\mathbf{w}(\cdot) = (w^1(\cdot), w^2(\cdot), \ldots, w^n(\cdot))$ *belonging to the admissible class*

$$\mathcal{A} := \left\{ \mathbf{w}(\cdot) \in C^2([0, t]; \mathbb{R}^n) \mid \mathbf{w}(0) = y, \mathbf{w}(t) = x \right\}$$

Theorem 3.3 (Euler-Lagrange equations)**.** *We assume that there in fact exists a function* $\mathbf{x}(\cdot) \in \mathcal{A}$ *satisfying our calculus of variations Problem*

$$I[\mathbf{x}] = \min_{\mathbf{w} \in \mathcal{A}} I[\mathbf{w}]$$

Then $\mathbf{x}(\cdot)$ *solves the system of Euler-Lagrange equations*

$$-\frac{d}{ds} \left(D_v L(\dot{\mathbf{x}}(s), \mathbf{x}(s)) \right) + D_x L(\dot{\mathbf{x}}(s), \mathbf{x}(s)) = \mathbf{0}$$

for $0 \leq s \leq t$*.*

Proof: 1. Choose a smooth function $\mathbf{y} : [0, t] \to \mathbb{R}^n$ *satisfying*

$$\mathbf{y}(0) = \mathbf{y}(t) = \mathbf{0}$$

and define for $\tau \in \mathbb{R}$

$$\mathbf{w}(\cdot) := \mathbf{x}(\cdot) + \tau \mathbf{y}(\cdot)$$

Then $\mathbf{w}(\cdot) \in \mathcal{A}$ *and so*

$$I[\mathbf{x}(\cdot)] \leq I[\mathbf{w}(\cdot)]$$

Thus the real-valued function

$$i(\tau) := I[\mathbf{x}(\cdot) + \tau \mathbf{y}(\cdot)]$$

has a minimum at $\tau = 0$, and consequently

$$i'(0) = 0$$

provided $i'(0)$ exists.

2. We explicitly compute this derivative. Observe

$$i(\tau) = \int_0^t L(\dot{\mathbf{x}}(s) + \tau\dot{\mathbf{y}}(s), \mathbf{x}(s) + \tau\mathbf{y}(s))ds$$

and so

$$i'(\tau) = \int_0^t \sum_{i=1}^n L_{v_i}(\dot{\mathbf{x}} + \tau\dot{\mathbf{y}}, x + \tau\mathbf{y})\dot{y}^i + L_{x_i}(\dot{\mathbf{x}} + \tau\dot{\mathbf{y}}, x + \tau\mathbf{y})y^i ds$$

Set $\tau = 0$:

$$0 = i'(0) = \int_0^t \sum_{i=1}^n L_{v_i}(\dot{\mathbf{x}}, \mathbf{x})\dot{y}^i + L_{x_i}(\dot{\mathbf{x}}, \mathbf{x})y^i ds$$

We then integrate by parts in the first term inside the integral, to discover

$$0 = \sum_{i=1}^n \int_0^t \left[-\frac{d}{ds}\left(L_{v_i}(\dot{\mathbf{x}}, \mathbf{x})\right) + L_{x_i}(\dot{\mathbf{x}}, \mathbf{x}) \right] y^i ds$$

This identity is valid for all smooth functions $\mathbf{y}(\cdot) = (y^1(\cdot), \dots, y^n(\cdot))$ satisfying the boundary conditions $\mathbf{y}(0) = \mathbf{y}(t) = \mathbf{0}$, and so for $0 \le s \le t$

$$-\frac{d}{ds}\left(L_{v_i}(\dot{\mathbf{x}}, \mathbf{x})\right) + L_{x_i}(\dot{\mathbf{x}}, \mathbf{x}) = 0 \quad (i = 1, \dots, n)$$

Remark *If curve $\mathbf{x}(\cdot) \in \mathcal{A}$ solves the Euler-Lagrange equations, we say $\mathbf{x}(\cdot)$ is a* ***critical point*** *of $I[\cdot]$. So every minimizer is a critical point, but a critical point need not be a minimizer.*

Definition 3.4. *We hereafter assume the C^2 function $\mathbf{x}(\cdot)$ is a critical point of the action functional and thus solves the Euler-Lagrange equations.*

First we set

$$\mathbf{p}(s) := D_v L(\dot{\mathbf{x}}(s), \mathbf{x}(s)) \quad (0 \le s \le t)$$

$\mathbf{p}(\cdot)$ is called the ***generalized momentum*** *corresponding to the position $\mathbf{x}(\cdot)$ and velocity $\dot{\mathbf{x}}(\cdot)$. We next make this important hypothesis:*

$$\begin{cases} \text{Suppose for all } x, p \in \mathbb{R}^n \text{ that the equation} \\ \qquad p = D_v L(v, x) \\ \text{can be uniquely solved for } v \text{ as a smooth} \\ \text{function of } p \text{ and } x, v = \mathbf{v}(p, x) \end{cases}$$

*The **Hamiltonian** H associated with the Lagrangian L is*

$$H(p,x) := p \cdot \mathbf{v}(p,x) - L(\mathbf{v}(p,x),x) \quad (p,x \in \mathbb{R}^n)$$

where the function $\mathbf{v}(\cdot)$ is defined implicitly

Theorem 3.5 (Derivation of Hamilton's ODE)**.** *The functions $\mathbf{x}(\cdot)$ and $\mathbf{p}(\cdot)$ satisfy Hamilton's equations:*

$$\begin{cases} \dot{\mathbf{x}}(s) = D_p H(\mathbf{p}(s), \mathbf{x}(s)) \\ \dot{\mathbf{p}}(s) = -D_x H(\mathbf{p}(s), \mathbf{x}(s)) \end{cases}$$

for $0 \le s \le t$.

Furthermore, the mapping $s \mapsto H(\mathbf{p}(s), \mathbf{x}(s))$ is constant.

§3.2 Legendre Transform, Hopf-Lax Formula

Definition 3.6 (Legendre transform)**.** *We suppose the Lagrangian $L : \mathbb{R}^n \to \mathbb{R}$ satisfies these conditions: the mapping $L(v)$ is convex (hence continuous) and*

$$\lim_{|v| \to \infty} \frac{L(v)}{|v|} = +\infty$$

*The **Legendre transform** of L is*

$$L^*(p) := \sup_{v \in \mathbb{R}^n} \{p \cdot v - L(v)\} \quad (p \in \mathbb{R}^n)$$

*This is also referred to as the **Fenchel transform**.*

Theorem 3.7 (Convex duality of Hamiltonian and Lagrangian)**.** *Assume L satisfies those conditions and define $H = L^*$.*

(i) Then the mapping $p \mapsto H(p)$ is convex and

$$\lim_{|p| \to \infty} \frac{H(p)}{|p|} = +\infty$$

(ii) Furthermore

$$L = H^* = (L^*)^*$$

*We say H and L are **dual convex functions**.*

The identity (15) implies that the three statements

$$\begin{cases} p \cdot v = L(v) + H(p) \\ \quad p = DL(v) \\ \quad v = DH(p) \end{cases}$$

are equivalent provided H is differentiable at p and L is differentiable at v.

Part III

Sobolev Space

Chapter VI

Sobolev Space

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§1 Hölder spaces $C^{k,\gamma}$

§1.1 Preliminaries and notation

Throughout the text X denotes a subset of $\mathbb{R}^n$ and $\Omega \subset \mathbb{R}^n$ an open set. We use $C(X)$ for the space of real-valued continuous functions on X (when X is equipped with the subspace topology of $\mathbb{R}^n$). For a multi-index $\alpha \in \mathbb{N}^n$ we write $|\alpha| = \alpha_1 + \cdots + \alpha_n$ and $\partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}$.

§1.2 The space C^m

Lemma 1.1. *Let X be subset of $\mathbb{R}^n$.*

1. *Then*
$$C_b(X) = \left\{ f \in C(X) : \sup_{x \in X} |f(x)| < \infty \right\}$$
is a Banach space with the norm $\|f\|_{C_b(X)} := \sup_{x \in X} |f(x)|$.

   Remark. *Noted that if X is compact, then $C_b(X) = C(X)$.*

2. *One can view $C\left(\overline{X}\right)$ as the set*
$$\left\{ f \in C(X) : \exists \text{ continuous } \tilde{f} : \overline{X} \to \mathbb{R} \text{ s.t. } \tilde{f}\Big|_X = f \right\}$$

   Remark. *The extension $\tilde{f}$ is unique if it exists thus we identify f with $\tilde{f}$ if f can be extended to $\overline{X}$.*

Definition 1.2. *Let Ω be a open set in $\mathbb{R}^n$, $m \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ and $0 < \gamma \leq 1$*

1. *The space*
$$C^m(\Omega) := \{u : \Omega \to \mathbb{R} \mid \partial^\alpha u \text{ exists and is continuous for all } |\alpha| \leq m\}$$

2. *The space*
$$C_b^m(\Omega) := \{u \in C^m(\Omega) \mid \partial^\alpha u \text{ bounded for all } |\alpha| \leq m\}$$
is a Banach space with the norm $\|u\|_{C_b^m(\Omega)} := \sum_{|\alpha| \leq m} \sup_{x \in \Omega} |\partial^\alpha u(x)|$

3. *The space*
$$C^m(\bar{\Omega}) := \left\{u \in C^m(\Omega) \mid \partial^\alpha u \text{ admits a continuous extension to } \bar{\Omega} \text{ for all } |\alpha| \leq m\right\}$$

4. *The space*
$$C_b^m\left(\bar{\Omega}\right) := C^m(\bar{\Omega}) \cap C_b^m(\Omega)$$

Remark. *Sometimes, we write $\left(C^m(\Omega), \|\cdot\|_{C^m(\Omega)}\right)$ to denote $\left(C_b^m(\Omega), \|\cdot\|_{C_b^m(\Omega)}\right)$, and $\left(C^m(\bar{\Omega}), \|\cdot\|_{C^m(\bar{\Omega})}\right)$ to denote $\left(C_b^m(\bar{\Omega}), \|\cdot\|_{C_b^m(\bar{\Omega})}\right)$ for simplicity.*

§1.3 Hölder spaces $C^{k,\gamma}\left(\bar{\Omega}\right)$

Definition 1.3. *Let Ω be a open set in $\mathbb{R}^n$, $m \in \mathbb{Z}_{\geq 0}$ and $0 < \gamma \leq 1$. If $u : \Omega \to \mathbb{R}$, we define*

1. *The γth -**Hölder seminorm** of u is*

$$[u]_{C^{0,\gamma}(\Omega)} := \sup_{\substack{x,y\in\Omega \\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^{\gamma}}$$

2. *The **Hölder space***

$$C^{k,\gamma}(\Omega)$$

consists of all functions $u \in C^k(\bar{\Omega})$ for which the norm

$$\|u\|_{C^{k,\gamma}(\Omega)} := \sum_{|\alpha|\leq k} \|D^{\alpha}u\|_{C(\Omega)} + \sum_{|\alpha|=k} [D^{\alpha}u]_{C^{0,\gamma}(\Omega)}$$

is finite.

Remark. *Noted that $C^{k,\gamma}(\Omega) \subset C^k(\bar{\Omega})$ since u^{β} is uniformly continuous for all $|\beta| = k$. And if $u \in C^{k,\gamma}(\Omega)$ is extended to $\bar{\Omega}$, then the extension is also in $C^{k,\gamma}(\bar{\Omega})$. Thus*

$$C^{k,\gamma}(\Omega) = C^{k,\gamma}(\bar{\Omega}) := \left\{u \in C^k(\bar{\Omega}) : \|D^{\alpha}u\|_{C(\bar{\Omega})}, [u]_{C^{0,\gamma}(\bar{\Omega})} < \infty\right\}$$

and the norms are equal. We use $C^{k,\gamma}(\Omega)$ and $C^{k,\gamma}(\bar{\Omega})$ interchangeably, but always mean the latter.

Theorem 1.4. *The space $C^{k,\gamma}(\bar{\Omega})$ is a Banach space*

§2 Positive Sobolev Spaces $W^{m,p}$

Definition 2.1. *Let Ω be an open set in $\mathbb{R}^n$, $1 \leq p \leq \infty$ and $k \in \mathbb{Z}_{\geq 0}$.*

1. *The **Sobolev space***

$$W^{m,p}(\Omega) = \{f \in \mathscr{D}'(\Omega) : D^{\alpha}f \in L^p(\Omega), 0 \leq |\alpha| \leq k\}$$

with the norm

$$\|u\|_{W^{m,p}(\Omega)} := \begin{cases} \left(\sum_{|\alpha|\leq k} \int_{\Omega} |D^{\alpha}u|^p \, dx\right)^{1/p} & (1 \leq p < \infty) \\ \sum_{|\alpha|\leq k} \operatorname{ess\,sup}_{\Omega} |D^{\alpha}u| & (p = \infty) \end{cases}$$

2. *$H^{m,p}(\Omega) \equiv$ the completion of $\left\{u \in C^m(\Omega) : \|u\|_{m,p} < \infty\right\}$ with respect to the norm $\|\cdot\|_{m,p}$*
3. *$W_0^{m,p}(\Omega) \equiv$ the closure of $C_0^{\infty}(\Omega)$ in the space $W^{m,p}(\Omega)$.*

Theorem 2.2. *Let $k \in \mathbb{Z}_{\geq 0}$ and $p \in [1, \infty]$, then*

1. *$W^{m,p}(\Omega)$ is a Banach space.*

2. *$W^{m,p}(\Omega)$ is separable if $1 \le p < \infty$.*

3. *$W^{m,p}(\Omega)$ is uniformly convex and reflexive if $1 < p < \infty$.*

4. *In particular, $W^{m,2}(\Omega)$ is a separable Hilbert space with inner product*

$$(u, v)_{W^{m,2}(\Omega)} = \sum_{0 \le |\alpha| \le m} (D^\alpha u, D^\alpha v)_{L^2} \tag{1}$$

Proof. Step 1. Let $\{f_n\}$ be a Cauchy sequence in $W^{m,p}(\Omega)$, then $\{D^\alpha f_n\}$ are Cauchy sequence in $L^p(\Omega)$ for all index $|\alpha| \le k$. By the completeness of L^p, there exist f_α such that

$$D^\alpha f_n \to f_\alpha \quad \text{in } L^p(\Omega)$$

Considering in $\mathscr{D}(\Omega)'$, $f_n \to f$ in $\mathscr{D}(\Omega)'$, thus $D^\alpha f_n \to D^\alpha f$ in $\mathscr{D}(\Omega)'$. Then we have $D^\alpha f = f_\alpha$ and

$$f_n \to f \quad \text{in } W^{m,p}(\Omega)$$

Step 2. Suppose that $p < \infty$. Let $N = \sum_{|\alpha| < k} 1$ be the number of all distinct index $|\alpha| \le k$, define the map

$$u \mapsto \{D^\alpha u\}_{|\alpha| \le m}$$

from $W^{m,p}(\Omega)$ to $(L^p(\Omega))^N$. Thus $W^{m,p}(\Omega)$ can be viewed as a closed subspace of $(L^p(\Omega))^N$ which is reflexive, then so $W^{m,p}(\Omega)$. □

§3 Negative Sobolev Spaces $W^{-m,p'}(\Omega)$

Definition 3.1. *Let Ω be an open region in $\mathbb{R}^n$, $1 \le p < \infty$ and $m \in \mathbb{Z}_{\ge 0}$. We define the dual space of $W_0^{m,p}(\Omega)$,*

$$W^{-m,p'}(\Omega) := (W_0^{m,p}(\Omega))^*$$

where $\frac{1}{p'} + \frac{1}{p} = 1$. Noted that $p' \in (1, \infty]$

Theorem 3.2. *Let Ω an open set, $k \in \mathbb{Z}_{\ge 0}$, $p' \in (1, \infty]$, we have*

$$W^{-m,p'}(\Omega) = \left\{ f = \sum_{|\alpha| \le k} (-1)^{|\alpha|} D^\alpha f_\alpha : f_\alpha \in L^{p'}(\Omega) \right\}$$

where the functional f acts on $\varphi \in \mathscr{D}(\Omega)$ by

$$\langle f, \varphi \rangle = \sum_{|\alpha| \le m} \int_\Omega f_\alpha D^\alpha \varphi \, \mathrm{d}x$$

and extend to $W_0^{m,p'}(\Omega)$ by density. Moreover, the norm of f in $W^{-m,p'}(\Omega)$ is equivalent to

$$\inf\left\{\left(\sum_{|\alpha|\le m}\|g_\alpha\|_{L^{p'}(\Omega)}^{p'}\right)^{1/p'} : f = g \text{ in } W^{-m,p'}(\Omega)\right\}$$

§4 Extensions

Theorem 4.1 (Extension Theorem)**.** *Let Ω be a bounded Lipschitz open set; $1 \le p \le \infty$. Select a bounded open set V such that $\Omega \subset\subset V$. Then there exists a bounded linear operator*

$$E : W^{k,p}(\Omega) \longrightarrow W^{k,p}(\mathbb{R}^n)$$

such that for each $u \in W^{k,p}(\Omega)$:

(i) $Eu|_\Omega = u$ in $W^{k,p}(\Omega)$,

(ii) Eu has support within V,

(iii) E is bounded: $\|Eu\|_{W^{k,p}(\mathbb{R}^n)} \le C\|u\|_{W^{k,p}(\Omega)}$ where the constant $C = C(n,p,\Omega,V)$ depends only on p, Ω, and V.

*We call Eu an **extension** of u to $\mathbb{R}^n$.*

Proof. **Step 1: Local flattening of the boundary.** Since Ω is Lipschitz, for each point $x_i^0 \in \partial\Omega$, there exists a radius $r_i > 0$ and a Lipschitz function $\gamma_i : \mathbb{R}^{n-1} \to \mathbb{R}$ such that, upon a suitable rotation of coordinates,

$$\Omega \cap B(x_i^0, r_i) = \{x = (x', x_n) \in B(x_i^0, r_i) : x_n > \gamma_i(x')\}.$$

By compactness of $\partial\Omega$, finitely many such neighborhoods $B(x_i^0, r_i)$ cover $\partial\Omega$.

Step 2: Local extensions. On each neighborhood $B(x_i^0, r_i)$, define a local extension $\tilde{u}_i$ of u to the whole $B(x_i^0, r_i)$ (or a slightly larger set) by reflection across the Lipschitz boundary:

$$\tilde{u}_i(x', x_n) := \begin{cases} u(x', x_n), & x_n > \gamma_i(x'), \\ u(x', 2\gamma_i(x') - x_n), & x_n \le \gamma_i(x'). \end{cases}$$

Standard estimates show that

$$\|\tilde{u}_i\|_{W^{k,p}(B(x_i^0,r_i))} \le C\|u\|_{W^{k,p}(\Omega\cap B(x_i^0,r_i))}.$$

Step 3: Partition of unity. Take smooth functions $\{\phi_i\}_{i=0}^N \subset C_c^\infty(\mathbb{R}^n)$ forming a partition of

unity subordinate to $\{B(x_i^0, r_i)\}_{i=1}^N$ and an interior set $B_0 \subset \Omega$, such that

$$\phi_0 + \sum_{i=1}^N \phi_i = 1 \quad \text{on } \overline{\Omega}.$$

Step 4: Global extension. Define

$$Eu := \phi_0 u + \sum_{i=1}^N \phi_i \tilde{u}_i.$$

By construction:

(a) $(Eu)|_\Omega = u$, since on $\Omega \cap B(x_i^0, r_i)$ we have $\tilde{u}_i = u$,

(b) $\operatorname{supp}(Eu) \subset V$, by choosing $B(x_i^0, r_i) \subset V$ and $\operatorname{supp} \phi_i \subset B(x_i^0, r_i)$,

(c) $\|Eu\|_{W^{k,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{k,p}(\Omega)}$, by Leibniz formula and finite overlap of supports.

Therefore, E is a bounded linear extension operator satisfying all required properties. □

§5 Density Theorem

§5.1 Approximation by Functions in $C^\infty(\Omega)$

Lemma 5.1 (Local approximation by smooth functions)**.** *Let Ω be an open set, $1 \leq p < \infty$. Then for every $u \in W^{k,p}(\Omega)$ there exist local smooth function u_n such that*

$$u_m \to u \quad \textit{in } W^{k,p}_{\text{loc}}(\Omega)$$

Proof. Define the

$$u^\varepsilon = \eta_\varepsilon * u \quad \text{in } \Omega_\varepsilon$$

Then we have $u^\varepsilon \in C^\infty(\Omega_\varepsilon)$ and $D^\alpha(\eta_\varepsilon * u) = \eta_\varepsilon * D^\alpha u$ in Ω_ε for all $|\alpha| \leq k$ and $\varepsilon > 0$. By the the consequence of mollifier,

$$D^\alpha u^\varepsilon = (D^\alpha u)^\varepsilon \to D^\alpha u \quad \text{in } L^p_{\text{loc}}(\Omega)$$

if we choose any open set $V \subset\subset \Omega$, then

$$\|u^\varepsilon - u\|^p_{W^{k,p}(V)} = \sum_{|\alpha| \leq k} \|D^\alpha(u^\varepsilon - u)\|^p_{L^p(V)} = \sum_{|\alpha| \leq k} \|(D^\alpha u)^\varepsilon - D^\alpha u\|^p_{L^p(V)} \to 0$$

as $\varepsilon \to 0$. This proves that $u_m \to u \quad$ in $W^{k,p}_{\text{loc}}(\Omega)$. □

Theorem 5.2 ($H = W$)**.** *Let Ω be an open set and $1 \leq p < \infty$. Then for every $u \in W^{m,p}(\Omega)$ there exist $u_m \in H^{m,p} = C^\infty(\Omega) \cap W^{m,p}(\Omega)$ such that*

$$u_m \to u \quad \text{in } W^{m,p}(\Omega)$$

Proof. Step 1. Let $\{V_i\}_{i=1}^\infty$ be an exhaustion by open sets of Ω and $\{\zeta_i\}_{i=1}^\infty$ be a smooth partition of unity subordinate to $\{V_i\}_{i=1}^\infty$. Next, choose any function $u \in W^{m,p}(\Omega)$. We have $\operatorname{supp}(\zeta_i u) \subset \operatorname{supp}\zeta_i \subset V_i$ is compact, $\zeta_i u \in W^{m,p}(\Omega)$, and

$$u = \sum_{i=1}^\infty u^i$$

where $u^i := \zeta_i u$

Step 2. Fix $\delta > 0$. For every i, choose $\varepsilon_i > 0$ small that $\operatorname{supp} u^i \subset V_i^{\varepsilon_i}$ ($\operatorname{dist}(\operatorname{supp} u^i, \partial V_i) > 0$) and v^i such that

$$\left\|v^i - u^i\right\|_{W^{m,p}(\Omega)} = \left\|v^i - u^i\right\|_{W^{m,p}(V_i)} \leq \frac{\delta}{2^{i+1}}$$

Write $v := \sum_{i=1}^\infty v^i$. This function is well defined and belongs to $C^\infty(\Omega)$ since $\operatorname{supp} u^i \subset \operatorname{supp}\zeta_i$ and $\{\operatorname{supp} u^i\}$ is locally finite. Then for any compact set $\Omega' \subset \Omega$, we have $\Omega' \subset \bigcup_{i=1}^N V_i$ for some N depending on Ω' and

$$\begin{aligned}
\|v - u\|_{W^{k,p}(\Omega')} &= \left\|\sum_{i=1}^N v^i - \sum_{i=1}^N u^i\right\|_{W^{k,p}(\Omega')} \\
&\leq \sum_{i=1}^\infty \left\|v^i - u^i\right\|_{W^{k,p}(\Omega)} \\
&\leq \delta \sum_{i=1}^\infty \frac{1}{2^i} \\
&= \delta
\end{aligned}$$

Take the supremum over compact sets $\Omega' \subset \Omega$, to conclude $\|v - u\|_{W^{k,p}(\Omega)} \leq \delta$ by Fatou's lemma. □

§5.2 Approximation by functions in $C^\infty(\bar{\Omega})$

Theorem 5.3. *Let Ω be a bounded Lipschitz open set; $1 \leq p < \infty$. Then for erery $u \in W^{k,p}(\Omega)$, then there exist $u_m \in C^\infty(\bar{\Omega})$ such that*

$$u_m \to u \quad \text{in } W^{k,p}(\Omega)$$

Proof. **Step 1: Extension to the whole space.** Since Ω is Lipschitz, there exists a bounded linear extension operator

$$E : W^{k,p}(\Omega) \to W^{k,p}(\mathbb{R}^n)$$

such that $(Eu)|_\Omega = u$ and $\|Eu\|_{W^{k,p}(\mathbb{R}^n)} \leq C\,\|u\|_{W^{k,p}(\Omega)}$. Set $v := Eu$.

Step 2: Mollification in $\mathbb{R}^n$. By theorem 5.2 , there exist $v_\varepsilon \in C^\infty(\mathbb{R}^n)$ such that

$$\|v_\varepsilon - v\|_{W^{k,p}(\mathbb{R}^n)} \longrightarrow 0$$

Step 3: Restriction back to Ω. Set $u_\varepsilon := v_\varepsilon|_\Omega$. Since v_ε is C^∞ on $\mathbb{R}^n$, we have $u_\varepsilon \in C^\infty(\overline{\Omega})$. Moreover,

$$\|u_\varepsilon - u\|_{W^{k,p}(\Omega)} = \|v_\varepsilon - v\|_{W^{k,p}(\Omega)} \leq \|v_\varepsilon - v\|_{W^{k,p}(\mathbb{R}^n)} \longrightarrow 0.$$

Choosing a sequence $\varepsilon_m \downarrow 0$ and writing $u_m := u_{\varepsilon_m}$ yields the desired approximating sequence.

□

Proof. 1. Fix any point $x^0 \in \partial\Omega$. As $\partial\Omega$ is C^1, there exist, a radius $r > 0$ and a C^1 function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that-upon relabeling the coordinate axes if necessary, we have

$$\Omega \cap B\left(x^0, r\right) = \left\{x \in B\left(x^0, r\right) \mid x_n > \gamma\left(x_1, \ldots, x_{n-1}\right)\right\}.$$

Set $V := \Omega \cap B\left(x^0, r/2\right)$.

2. Define the shifted point

$$x^\varepsilon := x + \varepsilon\lambda e_n \quad (x \in V, \varepsilon > 0),$$

and observe that for some fixed, sufficiently large number $\lambda > 0$ the ball $B\left(x^\varepsilon, \varepsilon\right) \subset \Omega \cap B\left(x^0, r\right)$ for all $x \in V$ and all small $\varepsilon > 0$.

Now define $u_\varepsilon(x) := u\left(x^\varepsilon\right) = u(x + \varepsilon\lambda e_n) \quad x \in V$. Next write $v^\varepsilon = \eta_\varepsilon * u_\varepsilon$. Clearly $v^\varepsilon \in C^\infty(\bar{V})$.

3. We now claim

$$v^\varepsilon \to u \quad \text{in } W^{k,p}(V)$$

To confirm this, take α to be any multiindex with $|\alpha| \leq k$. Then

$$\begin{aligned}
\|D^\alpha v^\varepsilon - D^\alpha u\|_{L^p(V)} &= \|\eta_\varepsilon * (D^\alpha u_\varepsilon - D^\alpha u) + \eta_\varepsilon * D^\alpha u - D^\alpha u\|_{L^p(V)} \\
&\leq \|\eta_\varepsilon * (D^\alpha u_\varepsilon - D^\alpha u)\|_{L^p(V)} + \|\eta_\varepsilon * D^\alpha u - D^\alpha u\|_{L^p(V)} \\
&\leq \|\eta_\varepsilon\| \cdot \|(D^\alpha u)_\varepsilon - D^\alpha u\| + \|\eta_\varepsilon * D^\alpha u - D^\alpha u\|
\end{aligned}$$

4. Select $\delta > 0$. Since $\partial\Omega$ is compact, we can find finitely many points $x_i^0 \in \partial\Omega$, radius $r_i > 0$, corresponding sets $V_i = \Omega \cap B^0\left(x_i^0, \frac{r_i}{2}\right)$, and functions $v_i \in C^\infty\left(\bar{V}_i\right)$ $(i = 1, \ldots, N)$ such that $\partial\Omega \subset \bigcup_{i=1}^N B^0\left(x_i^0, \frac{r_i}{2}\right)$ and

$$\|v_i - u\|_{W^{k,p}(V_i)} \leq \delta$$

Take an open set

$$V_0 = \Omega_\delta \subset\subset V_0' = \Omega_{\delta'} \subset\subset \Omega$$

such that $\Omega \subset \bigcup_{i=0}^{N} V_i$ ($0 < \delta' < \delta < \min\{\frac{r_i}{2}\}$) and select, using Theorem 1, a function $v_0 \in C^\infty(V_0') \subset C^\infty(\bar{V}_0)$ satisfying

$$\|v_0 - u\|_{W^{k,p}(V_0)} \leq \delta$$

5. Now let $\{\zeta_i\}_{i=0}^{N}$ be a smooth partition of unity on $\bar{\Omega}$, subordinate to the open sets $\left\{V_0, B^0\left(x_1^0, \frac{r_1}{2}\right), \cdots, B^0\right.$ Define $v := \sum_{i=0}^{N} \zeta_i v_i$. Then clearly $v \in C^\infty(\bar{\Omega})$. In addition, since $u = \sum_{i=0}^{N} \zeta_i u$, we see using Leibniz formula that for each $|\alpha| \leq k$

$$\begin{aligned}\|D^\alpha v - D^\alpha u\|_{L^p(\Omega)} &\leq \sum_{i=0}^{N} \|D^\alpha(\zeta_i v_i) - D^\alpha(\zeta_i u)\|_{L^p(V_i)} \\ &\leq C_{\zeta,\alpha} \sum_{i=0}^{N} \|v_i - u\|_{W^{k,p}(V_i)} = C(N+1)\delta\end{aligned}$$

□

§6 Trace

Theorem 6.1 (Trace Theorem). *Let Ω be a bounded Lipschitz open set; $1 \leq p \leq \infty$.; $1 \leq p < \infty$. Then there exists a bounded linear operator*

$$T : W^{1,p}(\Omega) \to L^p(\partial\Omega)$$

such that

(i) $Tu = u|_{\partial\Omega}$ if $u \in C(\bar{\Omega}) \cap W^{1,p}(\Omega)$

(ii) T is bounded : $\|Tu\|_{L^p(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)}$ with the constant C depending only on p and Ω.

*We call Tu the **trace** of u on $\partial\Omega$.*

Remark. *By the theorem 5.3 , the trace operator is uniquely determined by property (i).*

Proof. **Step 1: Local flattening of the boundary.** Since Ω is Lipschitz, for each point $x_i^0 \in \partial\Omega$, there exists a neighborhood U_i and a Lipschitz function $\gamma_i : \mathbb{R}^{n-1} \to \mathbb{R}$ such that, after a suitable rotation of coordinates,

$$\Omega \cap U_i = \{x = (x', x_n) \in U_i : x_n > \gamma_i(x')\}.$$

By compactness of $\partial\Omega$, finitely many such neighborhoods U_i cover $\partial\Omega$.

Step 2: Reduction to the upper half-space. Consider the local change of coordinates

$$\Phi_i : U_i \to \tilde{U}_i \subset \mathbb{R}^n, \quad \Phi_i(x', x_n) = (x', x_n - \gamma_i(x')).$$

Then locally $\Phi_i(\Omega \cap U_i) \subset \mathbb{R}^n_+ := \{(x', x_n) : x_n > 0\}$.

For $u \in W^{1,p}(\Omega)$, define $v_i = u \circ \Phi_i^{-1}$ in $\mathbb{R}^n_+$. Classical one-dimensional argument (Hardy inequality) shows

$$\|v_i(\cdot, 0)\|_{L^p(\mathbb{R}^{n-1})} \leq C\|v_i\|_{W^{1,p}(\mathbb{R}^n_+)}.$$

Step 3: Partition of unity. Take a smooth partition of unity $\{\phi_i\}_{i=1}^N$ subordinate to $\{U_i\}$, such that $\sum_i \phi_i = 1$ near $\partial\Omega$.

Define the local traces:

$$T_i u := v_i(\cdot, 0) = (u \circ \Phi_i^{-1})(\cdot, 0) \in L^p(\mathbb{R}^{n-1}).$$

Then the global trace is

$$Tu := \sum_{i=1}^N (\phi_i T_i u) \circ \Phi_i \in L^p(\partial\Omega).$$

Step 4: Verification.

(a) <u>Linearity and boundedness:</u>

$$\|Tu\|_{L^p(\partial\Omega)} \leq C \sum_i \|T_i u\|_{L^p(\mathbb{R}^{n-1})} \leq C \sum_i \|v_i\|_{W^{1,p}(\mathbb{R}^n_+)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

(b) <u>Consistency with smooth functions:</u> If $u \in C(\overline{\Omega}) \cap W^{1,p}(\Omega)$, then locally $T_i u = u|_{\partial\Omega \cap U_i}$, so globally

$$Tu = u|_{\partial\Omega}.$$

Hence T defines a bounded linear trace operator on $W^{1,p}(\Omega)$. □

Theorem 6.2 (Trace-zero functions in $W^{1,p}$)**.** *Let Ω be a bounded Lipschitz open set; $1 \leq p < \infty$. Suppose furthermore that $u \in W^{1,p}(\Omega)$. Then*

$$u \in W_0^{1,p}(\Omega) \quad \textit{iff} \quad Tu = 0 \textit{ on } \partial\Omega.$$

§7 The space $W^{k,2}$

§7.1 $W^{k,2}(\mathbb{R}^n)$

Lemma 7.1. *Suppoes $u \in L_{\text{loc}}(\mathbb{R}^n)$ has a weak derivative $D^\alpha u \in L^1(\mathbb{R}^n)$ or $L^2(\mathbb{R}^n)$, then*

$$\mathcal{F}D^\alpha u = (iy)^\alpha \hat{u}$$

Theorem 7.2 (Characterization of $W^{k,2}(\mathbb{R}^n)$ by Fourier transform)**.** *Let k be a nonnegative integer.*

A function $u \in L^2(\mathbb{R}^n)$ belongs to $W^{k,2}(\mathbb{R}^n)$ if and only if

$$\left(1+|y|^k\right)\hat{u} \in L^2(\mathbb{R}^n)$$

Remark. *In addition, we have the inequalities*

$$\left\|\left(1+|y|^k\right)\hat{u}\right\|_{L^2(\mathbb{R}^n)} \leq \sqrt{2}\,\|u\|_{W^{k,2}(\mathbb{R}^n)}$$

for each $u \in W^{k,2}(\mathbb{R}^n)$. Thus the norm $\|\cdot\| := \left\|\left(1+|y|^k\right)\mathcal{F}\cdot\right\|_{L^2(\mathbb{R}^n)}$ is equivalent to $\|\cdot\|_{W^{k,2}(\mathbb{R}^n)}$ by the norm equivalent theorem.

Proof. Assume first $u \in W^{k,2}(\mathbb{R}^n)$. Then for each multiindex $|\alpha| \leq k$, we have $D^\alpha u \in L^2(\mathbb{R}^n)$ and

$$\mathcal{F}(D^\alpha u) = (iy)^\alpha \hat{u}$$

belongs to $L^2(\mathbb{R}^n)$. Also,

$$\|D^\alpha u\|_{L^2(\mathbb{R}^n)} = \|(iy)^\alpha \hat{u}\|_{L^2(\mathbb{R}^n)}$$

Thus

$$\begin{aligned}
\|\left(1+|y|^k\right)\hat{u}\|_{L^2(\mathbb{R}^n)} &= \left(\int_{\mathbb{R}^n}\left(1+2|y|^k+|y|^{2k}\right)\hat{u}^2\,\mathrm{d}y\right)^{\frac{1}{2}} \\
&\leq \left(\int_{\mathbb{R}^n} 2\left(1+|y|^{2k}\right)\hat{u}^2\,\mathrm{d}y\right)^{\frac{1}{2}} \\
&\leq \sqrt{2}\left(\|\hat{u}\|_{L^2(\mathbb{R})} + \sum_{|\alpha|=k}\|y^\alpha\hat{u}\|_{L^2(\mathbb{R})}\right) \\
&= \sqrt{2}\left(\|u\|_{L^2(\mathbb{R})} + \sum_{|\alpha|=k}\|D^\alpha\hat{u}\|_{L^2(\mathbb{R})}\right) \\
&\leq \sqrt{2}\,\|u\|_{H^k(\mathbb{R}^n)}
\end{aligned}$$

belongs to $L^2(\mathbb{R}^n)$ and

2. Suppose conversely $\left(1+|y|^k\right)\hat{u} \in L^2(\mathbb{R}^n)$ and $|\alpha| \leq k$. Then

$$\|(iy)^\alpha\hat{u}\|^2_{L^2(\mathbb{R}^n)} \leq \int_{\mathbb{R}^n}|y|^{2|\alpha|}|\hat{u}|^2 dy \leq C_\alpha\left\|\left(1+|y|^k\right)\hat{u}\right\|^2_{L^2(\mathbb{R}^n)} < \infty$$

thus $(iy)^\alpha\hat{u} \in L^2(\mathbb{R}^n)$. Set

$$u_\alpha := \mathcal{F}^{-1}\left((iy)^\alpha\hat{u}\right)$$

Then for each $\phi \in C_c^\infty(\mathbb{R}^n)$

$$(D^\alpha\phi, u) = (\mathcal{F}D^\alpha\phi, \hat{u}) = \left((iy)^\alpha\hat{\phi}, \hat{u}\right) = (-1)^{|\alpha|}\left(\hat{\phi}, (iy)^\alpha\hat{u}\right) = (-1)^{|\alpha|}(\phi, u_\alpha)$$

Thus $u_\alpha = D^\alpha u$ in the weak sense and $D^\alpha u \in L^2(\mathbb{R}^n)$. Hence $u \in H^k(\Omega)$, as required. □

Definition 7.3. *Let $s > 0$ be a noninteger real number. We define the fractional Sobolev space $H^s(\mathbb{R}^n)$ as follows:*

$$H^s(\mathbb{R}^n) := \left\{ u \in L^2(\mathbb{R}^n) : (1 + |y|^s)\, \hat{u} \in L^2(\mathbb{R}^n) \right\}$$

with the norm

$$\|u\|_{H^s(\mathbb{R}^n)} := \|(1 + |y|^s)\, \hat{u}\|_{L^2(\mathbb{R}^n)}$$

§7.2 Gelfand triple

Theorem 7.4. *Let $\Omega \subset \mathbb{R}^n$ be a open set. Then the following **Gelfand triple** holds:*

$$H_0^k(\Omega) \hookrightarrow L^2(\Omega) \hookrightarrow H^{-k}(\Omega)$$

where $H^{-k}(\Omega)$ denotes the dual space of $H_0^k(\Omega)$. Moreover, both embeddings are continuous and dense.

Chapter VII

Sobolev Inequality; Imbedding Theorem

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§1 Sobolev inequalities

Theorem 1.1. *Let Ω be a a bounded Lipschitz domain in $\mathbb{R}^n$. If $u \in W_0^{k,p}(\Omega)$ and $kp < n$, then there exists a constant $C = C(k, p, n, \Omega)$ such that*

$$\|u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}$$

where $p^* = \frac{np}{n-kp}$ ($\frac{1}{p^*} = \frac{1}{p} - \frac{k}{n}$) is the Sobolev conjugate of p.

§1.1 Proof of Sobolev inequalities

Definition 1.2. *Let n-multiindex $\mathbf{p}_i, \boldsymbol{\theta}_i$ and $\mathbf{p}$ be given such that*

$$\frac{\boldsymbol{\theta}}{\mathbf{p}} = \sum_{i=1}^{k} \frac{\boldsymbol{\theta}_i}{\mathbf{p}_i}$$

meaning that $\frac{\boldsymbol{\theta}^j}{\mathbf{p}^j} = \sum_{i=1}^k \frac{\boldsymbol{\theta}_i^j}{\mathbf{p}_i^j}$ *for each component* $j = 1, \ldots, n.$ *, and let* f_i

$$\left\| \prod f_i \right\|_{L^{\mathbf{p}}} \leq \prod \|f_i\|_{L^{\mathbf{p}_i}}^{\boldsymbol{\theta}_i}$$

where

$$\|u\|_{L^{\boldsymbol{\theta}}_{\mathbf{p}}} := \left\| \cdots \left\| \|f\|_{L^{\mathbf{p}^1}(dx_1)}^{\boldsymbol{\theta}^1} \right\|_{L^{\mathbf{p}^2}(dx_2)}^{\boldsymbol{\theta}^2} \cdots \right\|_{L^{\mathbf{p}^n}(dx_n)}^{\boldsymbol{\theta}^n}$$

Definition 1.3 (Multi-index Hölder inequality)**.** *Let* n*-multiindices* $\mathbf{p}_i, \boldsymbol{\theta}_i$ *and* $\mathbf{p}$ *be given such that*

$$\frac{\boldsymbol{\theta}}{\mathbf{p}} = \sum_{i=1}^k \frac{\boldsymbol{\theta}_i}{\mathbf{p}_i},$$

meaning that

$$\frac{\boldsymbol{\theta}^j}{\mathbf{p}^j} = \sum_{i=1}^k \frac{\boldsymbol{\theta}_i^j}{\mathbf{p}_i^j}, \quad j = 1, \ldots, n.$$

For functions f_i*, define the nested* $L^{\mathbf{p}}$*-norm with weights* $\boldsymbol{\theta}$ *as*

$$\|f\|_{L^{\boldsymbol{\theta}}_{\mathbf{p}}} := \left\| \cdots \left\| \|f\|_{L^{\mathbf{p}^1}(dx_1)}^{\boldsymbol{\theta}^1} \right\|_{L^{\mathbf{p}^2}(dx_2)}^{\boldsymbol{\theta}^2} \cdots \right\|_{L^{\mathbf{p}^n}(dx_n)}^{\boldsymbol{\theta}^n}.$$

Then the multi-index Hölder inequality states that

$$\left\| \prod_{i=1}^k f_i \right\|_{L^{\mathbf{p}}} \leq \prod_{i=1}^k \|f_i\|_{L^{\mathbf{p}_i}}^{\boldsymbol{\theta}_i}.$$

Theorem 1.4 (Estimates for $C_c^1(\mathbb{R}^n)$)**.** *Assume* $1 \leq p < n$*. There exists a constant* $C = \frac{p(n-1)}{n-p}$*, such that*

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)}$$

for all $u \in C_c^1(\mathbb{R}^n)$*.*

Proof. 1. First Assume $p = 1$. Since u has compact support, for each $i = 1, \ldots, n$ and $x \in \mathbb{R}^n$ we have

$$u(x) = \int_{-\infty}^{x_i} u_{x_i}(x_1, \ldots, x_{i-1}, y_i, x_{i+1}, \ldots, x_n) \, \mathrm{d}y_i$$

and so

$$|u(x)| \leq \int_{-\infty}^{\infty} |Du(x_1, \ldots, y_i, \ldots, x_n)| \, \mathrm{d}y_i$$

Consequently

$$|u(x)|^{\frac{n}{n-1}} \leq \prod_{i=1}^n \left(\int_{-\infty}^{\infty} |Du(x_1, \ldots, y_i, \ldots, x_n)| \, \mathrm{d}y_i \right)^{\frac{1}{n-1}} \tag{1}$$

Integrate with respect to x_1 :

$$\begin{aligned}\int_{-\infty}^{\infty}|u|^{\frac{n}{n-1}}\,\mathrm{d}x_1 &\leq \int_{-\infty}^{\infty}\prod_{i=1}^{n}\left(\int_{-\infty}^{\infty}|Du|\,\mathrm{d}y_i\right)^{\frac{1}{n-1}}\,\mathrm{d}x_1\\ &= \left(\int_{-\infty}^{\infty}|Du|\,\mathrm{d}y_1\right)^{\frac{1}{n-1}}\int_{-\infty}^{\infty}\prod_{i=2}^{n}\left(\int_{-\infty}^{\infty}|Du|\,\mathrm{d}y_i\right)^{\frac{1}{n-1}}\,\mathrm{d}x_1\\ &\leq \left(\int_{-\infty}^{\infty}|Du|\,\mathrm{d}y_1\right)^{\frac{1}{n-1}}\prod_{i=2}^{n}\left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|Du|\,\mathrm{d}x_1\,\mathrm{d}y_i\right)^{\frac{1}{n-1}}\end{aligned}$$

Then integrate this inequality with respect to x_2:

We continue by integrating with respect to $x_3,\ldots,x_n$, eventually to find

$$\begin{aligned}\int_{\mathbb{R}^n}|u|^{\frac{n}{n-1}}\,\mathrm{d}x &\leq \prod_{i=1}^{n}\left(\int_{\mathbb{R}^n}|Du|dx_1\ldots dy_i\ldots dx_n\right)^{\frac{1}{n-1}}\\ &= \left(\int_{\mathbb{R}^n}|Du|dx\right)^{\frac{n}{n-1}}\end{aligned}$$

This is estimate for $p=1$.

2. Consider now the case that $1<p<n$. We apply previous estimate to $v:=|u|^{\gamma}$, where $\gamma=\frac{p(n-1)}{n-p}>1$. Then

$$\begin{aligned}\left(\int_{\mathbb{R}^n}|u|^{p^*}\,\mathrm{d}x\right)^{\frac{n-1}{n}} &= \left(\int_{\mathbb{R}^n}|u|^{\frac{\gamma n}{n-1}}dx\right)^{\frac{n-1}{n}}\\ &\leq \int_{\mathbb{R}^n}|D\,|u|^{\gamma}|\,\mathrm{d}x\\ &= \gamma\int_{\mathbb{R}^n}|u|^{\gamma-1}|Du|dx\\ &\leq \gamma\left(\int_{\mathbb{R}^n}|u|^{(\gamma-1)\frac{p}{p-1}}dx\right)^{\frac{p-1}{p}}\|Du\|_p\\ &= \gamma\left(\int_{\mathbb{R}^n}|u|^{p^*}\,\mathrm{d}x\right)^{\frac{p-1}{p}}\|Du\|_p\end{aligned}$$

in which case $\frac{\gamma n}{n-1}=(\gamma-1)\frac{p}{p-1}=\frac{np}{n-p}=p^*$. So we have

$$\left(\int_{\mathbb{R}^n}|u|^{p^*}dx\right)^{\frac{1}{p^*}}\leq\gamma\left(\int_{\mathbb{R}^n}|Du|^p dx\right)^{\frac{1}{p}}$$

□

Theorem 1.5 (Estimates for $W_0^{1,p}$)**.** *Assume Ω is a bounded, open subset of $\mathbb{R}^n$. Suppose $u\in W_0^{1,p}(\Omega)$ for some $1\leq p<n$. Then we have the estimate*

$$\|u\|_{L^{p^*}(\Omega)}\leq C\|Du\|_{L^p(\Omega)}$$

Furthermore, $\|u\|_{L^q(\Omega)} \leq C'\|Du\|_{L^p(\Omega)}$ *for each* $q \in [1, p^*]$*, the constant* C' *depending only on* p, q, n *and* Ω*.*

Proof: Since $u \in W_0^{1,p}(\Omega)$*, there exist functions* $u_m \in C_c^\infty(\Omega)$ *converging to* u *in* $W^{1,p}(\Omega)$*. We extend each function* u_m *to be 0 on* $\mathbb{R}^n - \bar{\Omega}$ *(that* $C_c^\infty(\Omega) \subset C_c^\infty(\mathbb{R}^n)$*) and apply the estimate for* $C_c^\infty(\mathbb{R}^n)$ *to discover*

$$\|u_m\|_{L^{p^*}(\Omega)} = \|u_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|Du_m\|_{L^p(\mathbb{R}^n)} = C\|Du_m\|_{L^p(\Omega)}$$

Thus

$$\|u_m - u_n\|_{L^{p^*}(\Omega)} \leq C\|Du_m - Du_n\|_{L^p(\Omega)} \leq C\|u_m - u_n\|_{W^{m,p}(\Omega)} \to 0$$

Since $L^{p^*}(\Omega)$ *is complete, we conclude that* $u_m \xrightarrow{L^{p^*}(\Omega)} u$ *and*

$$\|u\|_{L^{p^*}(\Omega)} \leq C\|Du\|_{L^p(\Omega)}$$

Remark *In view of this estimate, on* $W_0^{1,p}(\Omega)$ *the norm* $\|Du\|_{L^p(\Omega)}$ *is equivalent to* $\|u\|_{W^{1,p}(\Omega)}$*, if* Ω *is a bounded, open subset of* $\mathbb{R}^n$*.*

$$\|Du\|_{L^p(\Omega)} \leq \|u\|_{W^{1,p}(\Omega)}$$

$$\|u\|^p_{W^{1,p}(\Omega)} = \|u\|^p_{L^p(\Omega)} + \|Du\|^p_{L^p(\Omega)} \leq C\|Du\|^p_{L^p(\Omega)}$$

since $p \in [1, p^*]$*.*

Theorem 1.6 (Estimates for $W^{1,p}, 1 \leq p < n$)**.** *Let* Ω *be a bounded, open subset of* $\mathbb{R}^n$*, and suppose* $\partial\Omega$ *is* C^1*. If* $1 \leq p < n$*. and* $u \in W^{1,p}(\Omega)$*. Then* $u \in L^{p^*}(\Omega)$*, with the estimate*

$$\|u\|_{L^{p^*}(\Omega)} \leq C\,\|u\|_{W^{1,p}(\Omega)}$$

the constant C *depending only on* p, n*, and* Ω*.*

Proof: Since $\partial\Omega$ *is* C^1*, there exists an extension* $Eu = \bar{u} \in W^{1,p}(\mathbb{R}^n)$*, such that*

$$\begin{cases} \bar{u} = u \text{ in } \Omega, \bar{u} \text{ has compact support within } V \\ \|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\,\|u\|_{W^{1,p}(\Omega)} \end{cases}$$

Because $\bar{u}$ *has compact support within* V *(*$V \subset \Omega_\varepsilon$*), we know that there exist functions* $u_m \in C^\infty(V) \subset C_c^\infty(\mathbb{R}^n)$ *such that*

$$u_m \to \bar{u} \quad \text{in } W^{1,p}(\mathbb{R}^n)$$

Now according to inequality for $C_c^\infty(\mathbb{R}^n)$*,*

$$\|u_m - u_l\|_{L^{p^*}(\mathbb{R}^n)} \leq C\,\|Du_m - Du_l\|_{L^p(\mathbb{R}^n)} \leq C\,\|u_m - u_l\|_{W^{1,p}(\mathbb{R}^n)}$$

and

$$\|u_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du_m\|_{L^p(\mathbb{R}^n)}$$

for all $l, m \geq 1$. *Thus*

$$u_m \xrightarrow{L^{p^*}(\mathbb{R}^n)} \bar{u}$$

and

$$\|\bar{u}\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|D\bar{u}\|_{L^p(\mathbb{R}^n)}$$

Then we conclude that

$$\|u\|_{L^{p^*}(\Omega)} \leq \|\bar{u}\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|D\bar{u}\|_{L^p(\mathbb{R}^n)} \leq C\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\Omega)}$$

§2 Morrey's inequality

Theorem 2.1. *Let* Ω *be a a bounded Lipschitz domain in* $\mathbb{R}^n$. *If* $u \in W^{k,p}(\Omega)$ *and* $kp > n$ *then there exists a constant* $C = C(k, p, n, \Omega)$ *such that*

$$\|u\|_{C^{m,\gamma}(\bar{\Omega})} \leq C \|u\|_{W^{k,p}(\Omega)}$$

where $m = k - 1 - \left[\frac{n}{p}\right]$ *and* $\gamma = \frac{n}{p} - \left[\frac{n}{p}\right]$.

§2.1 Proof of Morrey's inequality

Lemma 2.2. *For all* $u \in C^1$, *we claim there exists a constant* $C = \frac{1}{n\alpha(n)}$, *depending only on* n, *such that*

$$\fint_{B(x,r)} |u(y) - u(x)| \, \mathrm{d}y \leq C \int_{B(x,r)} \frac{|Du(y)|}{|y-x|^{n-1}} \, \mathrm{d}y$$

$$\begin{aligned}
\fint_{B(x,r)} |u(y) - u(x)| \, \mathrm{d}y &= \frac{1}{\alpha(n)r^n} \int_0^r \int_{\partial B(0,1)} |u(x+s\omega) - u(x)| s^{n-1} \, \mathrm{d}S_\omega \, \mathrm{d}s \\
&= \frac{1}{\alpha(n)r^n} \int_0^r s^{n-1} \int_{\partial B(0,1)} \left| \int_0^s \frac{d}{dt} u(x+tw) \, \mathrm{d}t \right| \mathrm{d}S_\omega \, \mathrm{d}s \\
&\leq \frac{1}{\alpha(n)r^n} \int_0^r s^{n-1} \int_{\partial B(0,1)} \int_0^s |Du(x+tw)| \, \mathrm{d}t \, \mathrm{d}S_\omega \, \mathrm{d}s \\
&= \frac{1}{\alpha(n)r^n} \int_0^r s^{n-1} \int_{B(x,s)} \frac{|Du(y)|}{|y-x|^{n-1}} \, \mathrm{d}y ds \\
&\leq \frac{1}{\alpha(n)r^n} \int_0^r s^{n-1} \int_{B(x,r)} \frac{|Du(y)|}{|y-x|^{n-1}} \, \mathrm{d}y ds \\
&= \frac{1}{n\alpha(n)} \int_{B(x,r)} \frac{|Du(y)|}{|y-x|^{n-1}} \, \mathrm{d}y
\end{aligned}$$

Theorem 2.3 ($C^1(\mathbb{R}^n) \cap W^{1,p}(\mathbb{R}^n)$)**.** *Assume* $n < p \leq \infty$. *Then there exists a constant* C,

depending only on p and n, such that

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$$

for all $u \in C^1(\mathbb{R}^n)$, where $\gamma = 1 - n/p$

Proof: 1. Now fix $x \in \mathbb{R}^n$. We apply lemma as follows:

$$\begin{aligned}
|u(x)| &\leq \fint_{B(x,r)} |u(x) - u(y)| \, \mathrm{d}y + \fint_{B(x,r)} |u(y)| \, \mathrm{d}y \\
&\leq C \int_{B(x,r)} \frac{|Du(y)|}{|x-y|^{n-1}} \, \mathrm{d}y + C \|u\|_{L^p(B(x,r))} \\
&\leq C\|Du\|_{L^p(\mathbb{R}^n)} \left(\int_{B(x,r)} \frac{1}{|x-y|^{(n-1)\frac{p}{p-1}}} \, \mathrm{d}y \right)^{\frac{p-1}{p}} + C \|u\|_{L^p(\mathbb{R}^n)} \\
&\leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}
\end{aligned}$$

The last estimate holds since $p > n$ implies $(n-1)\frac{p}{p-1} < n$. As $x \in \mathbb{R}^n$ is arbitrary, it follows that

$$\|u\|_{C(\bar{\Omega})} = \sup_{x \in \mathbb{R}^n} |u| \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$$

2. Next, choose any two points $x, y \in \mathbb{R}^n$ and write $r := |x - y|$. Let $W := B(x,r) \cap B(y,r)$. Then

$$|u(x) - u(y)| \leq \fint_W |u(x) - u(z)| \, \mathrm{d}z + \fint_W |u(y) - u(z)| \, \mathrm{d}z$$

But lemma allows us to estimate

$$\begin{aligned}
\fint_W |u(x) - u(z)| \, \mathrm{d}z &\leq \fint_{B(x,r)} |u(x) - u(z)| \, \mathrm{d}z \\
&\leq C \int_{B(x,r)} \frac{|Du(z)|}{|z-x|^{n-1}} \, \mathrm{d}z \\
&\leq C\|Du\|_{L^p(\mathbb{R}^n)} \left(\int_{B(x,r)} \frac{dz}{|x-z|^{(n-1)\frac{p}{p-1}}} \right)^{\frac{p-1}{p}} \\
&= C \frac{n\alpha(n)(p-1)}{pn} r^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)}
\end{aligned}$$

Likewise,

$$\fint_W |u(y) - u(z)| \, \mathrm{d}z \leq C r^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)}$$

These estimates yield

$$|u(x) - u(y)| \leq C r^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)} = C|x-y|^\gamma \|Du\|_{L^p(\mathbb{R}^n)}$$

Thus

$$[u]_{C^{0,1-n/p}(\mathbb{R}^n)} = \sup_{x\neq y}\left\{\frac{|u(x)-u(y)|}{|x-y|^{1-n/p}}\right\} \leq C\|Du\|_{L^p(\mathbb{R}^n)}$$

This complete the proof .

Theorem 2.4 (Estimates for $W^{1,p}$)**.** *Let Ω be a bounded, open subset of $\mathbb{R}^n$, and suppose $\partial\Omega$ is C^1. Assume $n < p \leq \infty$ and $u \in W^{1,p}(\Omega)$. Then u has a version $u^* \in C^{0,\gamma}(\bar{\Omega})$, for $\gamma = 1 - \frac{n}{p}$, with the estimate*

$$\|u^*\|_{C^{0,\gamma}(\bar{\Omega})} \leq C\,\|u\|_{W^{1,p}(\Omega)}$$

The constant C depends only on p, n and Ω. In view of this Theorem, we will henceforth always identify a function $u \in W^{1,p}(\Omega)$ $(p > n)$ with its continuous version.

Proof: Since $\partial\Omega$ is C^1, there exists an extension $Eu = \bar{u} \in W^{1,p}(\mathbb{R}^n)$ such that

$$\begin{cases} \bar{u} = u \text{ in } \Omega \\ \bar{u} \text{ has compact support, and} \\ \|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\,\|u\|_{W^{1,p}(\Omega)} \end{cases}$$

Assume first $n < p < \infty$. Since $\bar{u}$ has compact support, we obtain functions $u_m \in C_c^\infty(\mathbb{R}^n)$ such that

$$u_m \xrightarrow{W^{1,p}(\mathbb{R}^n)} \bar{u}$$

Now according to previous theorem ,

$$\|u_m - u_l\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\,\|u_m - u_l\|_{W^{1,p}(\mathbb{R}^n)}$$

and

$$\|u_m\|_{C^{0,\gamma}} \leq C\,\|u_m\|_{W^{1,p}(\mathbb{R}^n)}$$

for all $l, m \geq 1$, whence there exists a function $u^* \in C^{0,\gamma}(\mathbb{R}^n)$ such that

$$u_m \xrightarrow{C^{0,\gamma}(\mathbb{R}^n)} u^*$$

We see that $u^* = u$ a.e. on Ω, so that u^* is a version of u and

$$\|u^*\|_{C^{0,1-n/p}(\mathbb{R}^n)} \leq C\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\,\|u\|_{W^{1,p}(\Omega)}$$

2. If $p = \infty$, $\gamma = 1$ we can assume that $\bar{u} \leq \|\bar{u}\|_{L^\infty(\mathbb{R}^n)}$ for all $x \in \mathbb{R}^n$. Then

$$\sup_{x\in\mathbb{R}^n} \bar{u} \leq \|\bar{u}\|_{L^\infty(\mathbb{R}^n)}$$

and

$$
\begin{aligned}
|\bar{u}(y)-\bar{u}(x)| &= \left|\int_0^{|y-x|} \frac{d}{dt}\bar{u}(x+t\frac{y-x}{|y-x|})\,\mathrm{d}t\right| \\
&\le \left|\frac{y-x}{|y-x|}\cdot D\bar{u}(x+t\frac{y-x}{|y-x|})\right| |y-x| \\
&\le \|D\bar{u}\|_{L^\infty(\mathbb{R}^n)}\,|y-x|
\end{aligned}
$$

for erery $x, y \in \mathbb{R}^n$. *Thus* $\bar{u} \in C^{0,1}(\mathbb{R}^n)$ *and* $\bar{u}|_\Omega$ *is a version of* u. *Also,*

$$
\|u\|_{C^{0,1}(\bar{\Omega})} \le \|\bar{u}\|_{C^{0,1}(\mathbb{R}^n)} \le \|\bar{u}\|_{W^{1,\infty}(\mathbb{R}^n)} \le C\,\|u\|_{W^{1,\infty}(\Omega)}
$$

§3 Poincaré's inequality

Theorem 3.1 (Poincaré's inequality)**.** *Let* Ω *be a bounded, connected, open subset of* $\mathbb{R}^n$*, with a* C^1 *boundary* $\partial\Omega$. *Assume* $1 \le p \le \infty$. *Then there exists a constant* C*, depending only on* n, p *and* Ω*, such that*

$$
\|u-(u)_\Omega\|_{L^p(\Omega)} \le C\|Du\|_{L^p(\Omega)}
$$

for each function $u \in W^{1,p}(\Omega)$.

Proof: We argue by contradiction. Were the stated estimate false, there would exist for each integer $k = 1, \ldots$ *a function* $u_k \in W^{1,p}(\Omega)$ *satisfying*

$$
\|u_k-(u_k)_\Omega\|_{L^p(\Omega)} > k\,\|Du_k\|_{L^p(\Omega)}
$$

We renormalize by defining

$$
v_k := \frac{u_k-(u_k)_\Omega}{\|u_k-(u_k)_\Omega\|_{L^p(\Omega)}} \quad (k=1,\ldots)
$$

Then

$$
(v_k)_\Omega = 0, \|v_k\|_{L^p(\Omega)} = 1
$$

and (2) implies

$$
\|Dv_k\|_{L^p(\Omega)} < \frac{1}{k} \quad (k=1,2,\ldots)
$$

In particular the functions $\{v_k\}_{k=1}^\infty$ *are bounded in* $W^{1,p}(\Omega)$. *Then there exist a subsequence* $\{v_{k_j}\}_{j=1}^\infty \subset \{v_k\}_{k=1}^\infty$ *and a function* $v \in L^p(\Omega)$ *such that*

$$
v_{k_j} \to v \quad \text{in } L^p(\Omega)
$$

It follows that

$$
(v)_\Omega = 0, \|v\|_{L^p(\Omega)} = 1
$$

On the other hand, since $\|Dv_k\|_{L^p(\Omega)} \to 0$

$$\int_\Omega v\phi_{x_i}dx = \lim_{k_j\to\infty}\int_\Omega v_{k_j}\phi_{x_i}dx = -\lim_{k_j\to\infty}\int_\Omega v_{k_j,x_i}\phi dx = 0$$

for each $\phi \in C_c^\infty(\Omega)$. *Consequently* $v \in W^{1,p}(\Omega)$, *with* $Dv = 0$ *a.e.*

Thus v *is constant, since* Ω *is connected. Since* v *is constant and* $(v)_\Omega = 0$, *we must have* $v \equiv 0$, *in which case* $\|v\|_{L^p(\Omega)} = 0$. *This contradiction establishes estimate.*

§4 Compactness

Definition 4.1. *Let* X *and* Y *be Banach spaces,* $X \subset Y$. *We say that* X *is* ***compactly embedded*** *in* Y, *written*

$$X \subset\subset Y$$

provided the imbedding operator is a linear compact operator i.e.

(i) $\|u\|_Y \le C\|u\|_X$ *for all* $u \in X$, *where* C *is a constant,*

(ii) each bounded sequence in X *is precompact in* Y.

Theorem 4.2 (Rellich-Kondrachov Compactness Theorem)**.** *Assume* Ω *is a bounded open subset of* $\mathbb{R}^n$ *and* $\partial\Omega$ *is* C^1. *Suppose* $1 \le p < n$. *Then*

$$W^{1,p}(\Omega) \subset\subset L^q(\Omega)$$

for each $1 \le q < p^*$.

Proof: Step 1. Fix $1 \le q < p^*$ *and note that since* Ω *is bounded and* $\partial\Omega$ *is* C^1, *Gagliardo-Nirenberg-Sobolev inequality for* $W^{1,p}(\Omega)$ *implies*

$$W^{1,p}(\Omega) \subset L^q(\Omega), \quad \|u\|_{L^q(\Omega)} \le C\|u\|_{W^{1,p}(\Omega)}.$$

It remains therefore to show that if $\{u_m\}_{m=1}^\infty$ *is a bounded sequence in* $W^{1,p}(\Omega)$, *there exists a subsequence* $\{u_{m_j}\}_{j=1}^\infty$ *which converges in* $L^q(\Omega)$.

Step 2. In view of the Extension Theorem , we may with no loss of generality assume that $\Omega = \mathbb{R}^n$ *and the functions* $\{u_m\}_{m=1}^\infty$ *all have compact support in some bounded open set* $V \subset \mathbb{R}^n$. *We also may assume*

$$\sup_m \|u_m\|_{W^{1,p}(V)} < \infty.$$

Step 3. Let us first study the smoothed functions

$$u_m^\varepsilon := \eta_\varepsilon * u_m \quad (\varepsilon > 0,\ m = 1, 2, \ldots),$$

where η_ε *denotes the usual mollifier. We may suppose the functions* $\{u_m^\varepsilon\}_{m=1}^\infty$ *all have support in* V *as well.*

Step 4. We first claim

$$u_m^\varepsilon \xrightarrow{L^q(V)} u_m \text{ as } \varepsilon \to 0, \text{ uniformly in } m.$$

To prove this, we first note that if u_m is smooth, then

$$\begin{aligned} u_m^\varepsilon(x) - u_m(x) &= \frac{1}{\varepsilon^n} \int_{B(x,\varepsilon)} \eta\left(\frac{x-z}{\varepsilon}\right)(u_m(z) - u_m(x))\,dz \\ &= \int_{B(0,1)} \eta(y)(u_m(x-\varepsilon y) - u_m(x))\,dy \\ &= \int_{B(0,1)} \eta(y) \int_0^1 \frac{d}{dt}(u_m(x-\varepsilon ty))\,dt\,dy \\ &= -\varepsilon \int_{B(0,1)} \eta(y) \int_0^1 Du_m(x-\varepsilon ty)\cdot y\,dt\,dy \end{aligned}$$

Thus

$$\begin{aligned} \int_V |u_m^\varepsilon(x) - u_m(x)|\,dx &\le \varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \int_V |Du_m(x-\varepsilon ty)|\,dx\,dt\,dy \\ &\le \varepsilon \int_V |Du_m(z)|\,dz. \end{aligned}$$

By approximation, this estimate holds if $u_m \in W^{1,p}(V)$. Hence

$$\|u_m^\varepsilon - u_m\|_{L^1(V)} \le \varepsilon \|Du_m\|_{L^1(V)} \le \varepsilon C \|Du_m\|_{L^p(V)},$$

We thereby discover

$$u_m^\varepsilon \to u_m \quad \text{in } L^1(V), \text{ uniformly in } m.$$

But then since $1 \le q < p^*$, we see using the interpolation inequality for L^p-norms that

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \le \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta \|u_m^\varepsilon - u_m\|_{L^{p^*}(V)}^{1-\theta},$$

where $\frac{1}{q} = \frac{\theta}{1} + \frac{1-\theta}{p^*}, \quad 0 < \theta < 1$. Consequently,the Gagliardo-Nirenberg-Sobolev inequality imply

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \le C \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta,$$

whence assertion (2) follows from (3).

Step 5. Next we claim

$$\begin{cases} \text{for each fixed } \varepsilon > 0, \text{ the sequence } \{u_m^\varepsilon\}_{m=1}^\infty \\ \text{is uniformly bounded and equicontinuous.} \end{cases}$$

Indeed, if $x \in \mathbb{R}^n$, then

$$|u_m^\varepsilon(x)| \le \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)|u_m(y)|\,dy$$

$$\leq \|\eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)}\|u_m\|_{L^1(V)} \leq \frac{C}{\varepsilon^n} < \infty$$

for $m = 1, 2, \ldots$. *Similarly,*

$$|Du_m^\varepsilon(x)| \leq \int_{B(x,\varepsilon)} |D\eta_\varepsilon(x-y)||u_m(y)|\,dy$$

$$\leq \|D\eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)}\|u_m\|_{L^1(V)} \leq \frac{C}{\varepsilon^{n+1}} < \infty,$$

for $m = 1, \ldots$. *Assertion (4) follows from these two estimates.*

Step 6. We now observe that since the functions $\{u_m\}_{m=1}^\infty$, *and thus the functions* $\{u_m^\varepsilon\}_{m=1}^\infty$, *have support in some fixed bounded set* $V \subset \mathbb{R}^n$, *we may utilize (4) and the Arzela-Ascoli compactness criterion, to obtain a subsequence* $\left\{u_{m_j}^\varepsilon\right\}_{j=1}^\infty \subset \{u_m^\varepsilon\}_{m=1}^\infty$ *which converges uniformly on* V. *In particular therefore*

$$\limsup_{j,k\to\infty} \left\|u_{m_j}^\varepsilon - u_{m_k}^\varepsilon\right\|_{L^q(V)} = 0$$

Now fix $\delta > 0$. *We will show there exists a subsequence* $\{u_{m_j}\}_{j=1}^\infty \subset \{u_m\}_{m=1}^\infty$ *such that*

$$\limsup_{j,k\to\infty} \|u_{m_j} - u_{m_k}\|_{L^q(V)} \leq \delta.$$

To see this, let us first employ assertion (2) to select $\varepsilon > 0$ *so small that*

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \leq \delta/2$$

for $m = 1, 2, \ldots$.

But then (6) and (7) imply

$$\limsup_{j,k\to\infty} \left\|u_{m_j} - u_{m_k}\right\|_{L^q(V)} \leq \delta$$

and so (5) is proved.

Step 7. We next employ assertion (5) with $\delta = 1, \frac{1}{2}, \frac{1}{3}, \ldots$ *and use a standard diagonal argument to extract a subsequence* $\{u_{m_l}\}_{l=1}^\infty \subset \{u_m\}_{m=1}^\infty$ *satisfying*

$$\limsup_{l,k\to\infty} \|u_{m_l} - u_{m_k}\|_{L^q(V)} = 0.$$

Corollary 4.3. *Assume* Ω *is a bounded open subset of* $\mathbb{R}^n$ *and* $\partial\Omega$ *is* C^1. *We have in particular*

$$W^{1,p}(\Omega) \subset\subset L^p(\Omega)$$

for all $1 \leq p \leq \infty$.

Proof: 1. If $1 \leq p < n$, *it is obvious from Rellich-Kondrachov Compactness Theorem.*

2. If $n < p \leq \infty$, *there is a version* $u_n^* \in C^{0,\gamma}(\bar{\Omega})$ *of* u_n *for bounded sequence* $\{u_n\}$ *in* $W^{1,p}(\Omega)$

and also

$$\|u_n^*\|_{C^{0,\gamma}(\bar{\Omega})} \leq C\|u_n\|_{W^{1,p}(\Omega)} \leq M$$

which implies u_n^* *is uniformly bounded and equicontinuous. By Arzela-Ascoli theorem,* $\{u_{n_k}\}$ *convergent uniformly to* $u \in C(\Omega)$ *and also*

$$u_{n_k} \to u \quad \text{in } L^p(\Omega)$$

since Ω *is bounded.*

Theorem 4.4. *Assume* Ω *is a bounded open subset of* $\mathbb{R}^n$*. Suppose* $1 \leq p < n$*. Then*

$$W^{1,p}(\Omega) \subset\subset L^q(\Omega)$$

for each $1 \leq q < p^*$*.*

§5 Difference quotients and $\mathrm{W}^{1,p}$

Definition 5.1. *Assume* $u : \Omega \to \mathbb{R}$ *is a locally summable function and* $V \subset\subset \Omega$*.*

(1) The ith *-difference quotient of size* h *is*

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h} \quad (i = 1, \ldots, n)$$

for $x \in V$ *and* $h \in \mathbb{R}, 0 < |h| < \operatorname{dist}(V, \partial\Omega)$*.*

(2) $D^h u := \left(D_1^h u, \ldots, D_n^h u\right)$*.*

Proposition 5.2. *Choose* $i = 1, \ldots, n, \phi \in C_c^\infty(V)$*, and note for small enough* h *that*

$$\int_V u(x)\left[\frac{\phi(x + he_i) - \phi(x)}{h}\right] dx = -\int_V \left[\frac{u(x) - u(x - he_i)}{h}\right] \phi(x) dx$$

that is,

$$\int_V u\left(D_i^h \phi\right) dx = -\int_V \left(D_i^{-h} u\right) \phi dx$$

Theorem 5.3 (Difference quotients and weak derivatives)**.** *Let* Ω *be a open sets of* $\mathbb{R}^n$*.*

(1) Suppose $1 \leq p < \infty$ *and* $u \in W^{1,p}(\Omega)$*. Then for each* $V \subset\subset \Omega$

$$\left\|D^h u\right\|_{L^p(V)} \leq C\|Du\|_{L^p(\Omega)}$$

for some constant C *depending on* Ω, V *and all* $0 < |h| < \frac{1}{2}\operatorname{dist}(V, \partial\Omega)$*.*

(2) Assume $1 < p < \infty, u \in L^p(V)$*, and there exists a constant* C *such that*

$$\left\|D^h u\right\|_{L^p(V)} \leq C$$

for all $0 < |h| < \frac{1}{2}\operatorname{dist}(V, \partial\Omega)$. *Then*

$$u \in W^{1,p}(V), \quad \textit{with} \quad \|Du\|_{L^p(V)} \leq C$$

Proof: Step1. Assume $1 \leq p < \infty$, *and temporarily suppose* u *is smooth. Then for each* $x \in V, i = 1, \ldots, n$, *and* $0 < |h| < \frac{1}{2}\operatorname{dist}(V, \partial\Omega)$, *we have*

$$u(x + he_i) - u(x) = h \int_0^1 u_{x_i}(x + the_i)\, dt$$

so that

$$|u(x + he_i) - u(x)| \leq |h| \int_0^1 |Du(x + the_i)|\, dt$$

Consequently

$$\begin{aligned}\int_V \left|D^h u\right|^p dx &\leq C \sum_{i=1}^n \int_V \int_0^1 |Du(x + the_i)|^p\, dt dx \\ &= C \sum_{i=1}^n \int_0^1 \int_V |Du(x + the_i)|^p\, dx dt\end{aligned}$$

Thus

$$\int_V \left|D^h u\right|^p dx \leq C \int_\Omega |Du|^p dx$$

This estimate holds should u *be smooth, and thus is valid by approximation for arbitrary* $u \in W^{1,p}(\Omega)$.

2. Now suppose estimate holds for all $0 < |h| < \frac{1}{2}\operatorname{dist}(V, \partial\Omega)$ *and some constant* C. *Estimate (9) implies*

$$\sup_h \left\|D_i^{-h} u\right\|_{L^p(V)} < \infty$$

and therefore, since $1 < p < \infty$, *there exists a function* $v_i \in L^p(V)$ *and a subsequence* $h_k \to 0$ *such that*

$$D_i^{-h_k} u \rightharpoonup v_i \quad \textit{weakly in } L^p(V)$$

But then

$$\begin{aligned}\int_V u\phi_{x_i} dx &= \int_\Omega u\phi_{x_i} dx \\ &= \lim_{h_k \to 0} \int_\Omega u D_i^{h_k} \phi dx \\ &= -\lim_{h_k \to 0} \int_V D_i^{-h_k} u\phi dx \\ &= -\int_V v_i \phi dx \\ &= -\int_\Omega v_i \phi dx\end{aligned}$$

Thus $v_i = u_{x_i}$ *in the weak sense (* $i = 1, \ldots, n$ *), and so* $Du \in L^p(V)$. *As* $u \in L^p(V)$, *we deduce therefore that* $u \in W^{1,p}(V)$.

§5.1 Lipschitz functions and $\mathrm{W}^{1,\infty}$

Theorem 5.4 (Characterization of $W^{1,\infty}$)**.** *Let Ω be open and bounded, with $\partial\Omega$ of class C^1. Then $u : \Omega \to \mathbb{R}$ is Lipschitz continuous if and only if $u \in W^{1,\infty}(\Omega)$.*

Proof. Step 1. First assume $\Omega = \mathbb{R}^n$ and u has compact support. Suppose $u \in W^{1,\infty}(\mathbb{R}^n) \subset C^{0,\gamma}$. Then $u^\varepsilon := \eta_\varepsilon * u$ is smooth and satisfies

$$\begin{cases} u^\varepsilon \to u \text{ uniformly as } \varepsilon \to 0 \\ \|Du^\varepsilon\|_{L^\infty(\mathbb{R}^n)} \le \|Du\|_{L^\infty(\mathbb{R}^n)} \end{cases}$$

Choose any two points $x, y \in \mathbb{R}^n, x \neq y$. We have

$$\begin{aligned} u^\varepsilon(x) - u^\varepsilon(y) &= \int_0^1 \frac{d}{dt} u^\varepsilon(tx + (1-t)y)dt \\ &= \int_0^1 Du^\varepsilon(tx + (1-t)y)dt \cdot (x-y) \end{aligned}$$

and so

$$|u^\varepsilon(x) - u^\varepsilon(y)| \le \|Du^\varepsilon\|_{L^\infty(\mathbb{R}^n)} |x-y| \le \|Du\|_{L^\infty(\mathbb{R}^n)} |x-y|.$$

We let $\varepsilon \to 0$ to discover

$$|u(x) - u(y)| \le \|Du\|_{L^\infty(\mathbb{R}^n)} |x-y|.$$

Hence u is Lipschitz continuous.

Step 2. On the other hand assume now u is Lipschitz continuous; we must prove that u has essentially bounded weak first derivatives. Since u is Lipschitz, we see

$$\left\|D_i^{-h} u\right\|_{L^\infty(\mathbb{R}^n)} \le \mathrm{Lip}(u)$$

and thus there exists a function $v_i \in L^\infty(\mathbb{R}^n)$ and a subsequence $h_k \to 0$ such that

$$D_i^{-h_k} u \rightharpoonup v_i \quad \text{weakly in } L^2_{loc}(\mathbb{R}^n)$$

Consequently

$$\begin{aligned} \int_{\mathbb{R}^n} u\phi_{x_i} dx &= \lim_{h_k \to 0} \int_{\mathbb{R}^n} u D_i^{h_k} \phi dx \\ &= -\lim_{h_k \to 0} \int_{\mathbb{R}^n} D_i^{-h_k} u \phi dx \\ &= -\int_{\mathbb{R}^n} v_i \phi dx \end{aligned}$$

by (12). The above equality holds for all $\phi \in C_c^\infty(\mathbb{R}^n)$, and so $v_i = u_{x_i}$ in the weak sense ($i = 1, \ldots, n$). Consequently $u \in W^{1,\infty}(\mathbb{R}^n)$.

3. In the general case that Ω is bounded, with $\partial\Omega$ of class C^1, we as usual extend u to $Eu = \bar{u}$ and apply the above argument.

Corollary 5.5. *Let Ω be a open set $u \in W^{1,\infty}_{loc}(\Omega)$ if and only if u is locally Lipschitz continuous in Ω.*

Theorem 5.6 (Differentiability almost everywhere)**.** *Assume $u \in W^{1,p}_{\text{loc}}(\Omega)$ for some $n < p \leq \infty$. Then u is differentiable a.e. in Ω, and its gradient equals its weak gradient a.e.*

Proof: Recall that we always identify u with its continuous version.

1. Assume first $n < p < \infty$. From the remark after the proof of Theorem 4 in §5.6.2, we recall Morrey's estimate

$$|v(y) - v(x)| \leq Cr^{1-\frac{n}{p}} \left(\int_{B(x,2r)} |Dv(z)|^p dz \right)^{1/p} \qquad (y \in B(x,r))$$

valid for any C^1 function v and thus, by approximation, for any $v \in W^{1,p}$.

2. Choose $u \in W^{1,p}_{loc}(\Omega)$. Now for a.e. $x \in \Omega$, a version of Lebesgue's Differentiation Theorem (§E.4) implies

$$f_{B(x,r)} |Du(x) - Du(z)|^p dz \to 0$$

as $r \to 0$, Du denoting as usual the weak derivative of u. Fix any such point x and set

$$v(y) := u(y) - u(x) - Du(x) \cdot (y - x)$$

in estimate (14), where

$$r = |x - y|$$

We find

$$\begin{aligned} &| \, u(y) - u(x) - Du(x) \cdot (y - x) \, | \\ &\quad \leq Cr^{1-n/p} \left(\int_{B(x,2r)} |Du(x) - Du(z)|^p dz \right)^{1/p} \\ &\quad \leq Cr \left(f_{B(x,2r)} |Du(x) - Du(z)|^p dz \right)^{1/p} \\ &\quad = o(r) \quad \textit{by (15)} \\ &\quad = o(|x - y|) \quad \textit{by (16).} \end{aligned}$$

Thus u is differentiable at x, and its gradient equals its weak gradient at x. 3. In case $p = \infty$, we note $W^{1,\infty}_{loc}(\Omega) \subset W^{1,p}_{loc}(\Omega)$ for all $1 \leq p < \infty$ and apply the reasoning above.

Theorem 5.7 (Rademacher's Theorem)**.** *Let u be locally Lipschitz continuous in Ω. Then u is differentiable almost everywhere in Ω.*

Part IV

Second Order Elliptic Equatons

Chapter VIII

Second Order Elliptic Equations

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§1 Elliptic Operator and Weak Solution

Suppoes U is an open, bounded subset of $\mathbb{R}^n$.

Definition 1.1. *We study the Drichlet problem*

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ u = 0 \text{ on } \partial U \end{cases}$$

where $u : \bar{U} \to \mathbb{R}$ is the unknown, $u = u(x)$. Here $f : U \to \mathbb{R}$ is given, and $\mathcal{L}$ denotes a second-order partial differential operator having either the form

$$\mathcal{L}u = -\sum_{i,j=1}^{n} \left(a^{ij}(x)u_{x_i}\right)_{x_j} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u$$

*we say that the PDE $\mathcal{L}u = f$ is in **divergence form**, or else*

$$\mathcal{L}u = -\sum_{i,j=1}^{n} a^{ij}(x)u_{x_i x_j} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u$$

*is in **nondivergence form**, for given coefficient functions a^{ij}, b^i, c $(i, j = 1, \ldots, n)$.*

Remark. *In this section, we always assume that*

1. *The partial differential operator $\mathcal{L}$ is **(uniformly) elliptic** if there exists a constant $\theta > 0$ such that*
$$\sum_{i,j=1}^{n} a^{ij}(x)\xi_i\xi_j \geq \theta|\xi|^2$$
for a.e. $x \in U$ and all $\xi \in \mathbb{R}^n$.

2. *a^{ij}, b^i, c are in $L^\infty(U)$ for $i, j = 1, \ldots, n$.*

3. *the symmetry condition $a^{ij} = a^{ji}$.*

Definition 1.2. *We study the Dirichlet's problem*

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ \quad u = 0 \text{ on } \partial U \end{cases} \tag{1}$$

where $\mathcal{L}$ is divergence form elliptic operator that

$$\mathcal{L}u = -\sum_{i,j=1}^{n} \left(a^{ij}(x)u_{x_j}\right)_{x_i} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u$$

and $f \in H^{-1}(U)$.

1. *The bilinear form $B[\ ,\ ]$ on $H_0^1(U)$ associated with the operator $\mathcal{L}$ is*
$$B[u, v] := \langle \mathcal{L}u, v\rangle_{H^{-1},H_0^1} = \int_U \sum_{i,j=1}^{n} a^{ij}u_{x_i}v_{x_j} + \sum_{i=1}^{n} b^i u_{x_i}v + cuv \,\mathrm{d}x$$
for $u, v \in H_0^1(U)$.

2. *We say that $u \in H_0^1(U)$ is a **weak solution** of (1) provided*
$$B[u, v] = \langle f, v\rangle$$
*for all $v \in H_0^1(U)$. The identity (3) is sometimes called the **variational formulation** of (1).*

Remark. *That is, we consider the bounded linear operator*

$$\mathcal{L} : H_0^1(U) \to H^{-1}(U)$$

and the derivatives are in the sense of distributions. The equation (1) is interpreted to equation $\mathcal{L}u = f$

We can give the definition of weak solution to nonhomogeneous boundary-value problems.

Definition 1.3. *Suppoes that* ∂U *is* C^1 *and* $u \in H^1(U)$ *is a weak solution of*

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ \quad u = g \text{ on } \partial U. \end{cases}$$

This means that $u = g$ *on* ∂U *in the trace sense and furthermore that the bilinear form identity*

$$B[u, v] = \langle f, v \rangle$$

holds for all $v \in H_0^1(U)$*. For this to be possible, it is necessary for* g *to be the trace of some* H^1 *function, say* w*. But then* $\tilde{u} := u - w$ *belongs to* $H_0^1(U)$ *and is a weak solution of the boundary-value problem*

$$\begin{cases} \mathcal{L}\tilde{u} = \tilde{f} \text{ in } U \\ \quad \tilde{u} = 0 \text{ on } \partial U \end{cases}$$

where $\tilde{f} := f - \mathcal{L}w \in H^{-1}(U)$.

§2 First Existence Theorem

Theorem 2.1 (Lax-Milgram Theorem)**.** *Assume that* H *is a real Hilbert space and* $B : H \times H \to \mathbb{R}$ *is a real bilinear mapping, for which there exist constants* $\alpha, \beta > 0$ *such that*

(i) $|B[u, v]| \leq \alpha \|u\| \|v\|$

(ii) $\beta \|u\|^2 \leq B[u, u]$

Then for each continuous linear functional $f \in H^*$*, there exists a unique element* $u_f \in H$ *such that*

$$B[u_f, v] = \langle f, v \rangle$$

for all $v \in H$ *and*

$$\|u_f\| \leq \frac{1}{\beta} \|f\|$$

Theorem 2.2 (Energy estimates)**.** *There exist constants* $\alpha, \beta > 0$ *and* $\gamma \geq 0$ *such that*

$$|B[u, v]| \leq \alpha \|u\|^2_{H_0^1(U)} \|v\|_{H_0^1(U)}$$

and

$$\beta \|u\|^2_{H_0^1(U)} \leq B[u, u] + \gamma \|u\|^2_{L^2(U)}$$

for all $u, v \in H_0^1(U)$.

Theorem 2.3 (First Existence Theorem)**.** *There is a number $\gamma \geq 0$ such that*

$$(-\infty, -\gamma] \subset \rho(\mathcal{L})$$

the set of all regular values of $\mathcal{L}$.

Corollary 2.4. *The invertible elliptic operator $\mathcal{L}_\mu$ induce an isomorphism*

$$\mathcal{L}_\mu^{-1} : H^{-1}(U) \to H_0^1(U)$$

given by $\langle f, v\rangle = B\left[\mathcal{L}_\mu^{-1} f, v\right]$ or $\langle \mathcal{L}_\mu u, v\rangle = B[u, v]$. Then we define an new inner product in $H^{-1}(U)$ by

$$(f, g)_{H^{-1}(U)} = \left(\mathcal{L}_\mu^{-1} f, \mathcal{L}_\mu^{-1} g\right)_{H_0^1(U)}$$

whence $\mathcal{L}_\mu$ is an isometry of Hilbert space.

(2) Especially operator $L = -\Delta + I$ induce an isomorphism

$$(-\Delta + I)^{-1} : H^{-1}(U) \to H_0^1(U)$$

given that $\langle f, v\rangle = B[L^{-1} f, v] = (L^{-1} f, v)_{H_0^1(U)}$ (the bilinear form associated with $L = -\Delta + I$ is $B[\ ,\ ] = (\ ,\ )_{H_0^1(U)}$). Furthermore, we have also

$$\begin{aligned}(f, g)_{H^{-1}(U)} &= \left(L^{-1} f, L^{-1} g\right)_{H_0^1(U)} \\ &= \left\langle f, L^{-1} g\right\rangle \\ &= \left\langle g, L^{-1} f\right\rangle\end{aligned}$$

and the norm induced by the inner product coincides with the dual norm

$$(f, f)_{H^{-1}(U)}^{\frac{1}{2}} = \sup \frac{|\langle f, u\rangle|}{\|u\|_{H_0^1(U)}}$$

§3 Second Existence Theorem

§3.1 Adjoint

We use the notation $\langle -, -\rangle_X$ for the duality pairing between a Banach space X and its dual X^*.

Theorem 3.1. *Suppoes that $(H_0^1)^* = H^{-1}$ and $(H^{-1})^* = H_0^1$ the adjoint operator*

$$\mathcal{L}^* : H_0^1 \to H^{-1}$$

of $\mathcal{L}$ is

$$\mathcal{L}^* v = -\sum \left(a^{ij} v_{x_i}\right)_{x_j} - \sum b^i v_{x_i} + \left(c - \sum b_{x_i}^i\right) v$$

Remark. $\langle \mathcal{L}u, v\rangle_{H^{-1}} = \langle u, \mathcal{L}^* v\rangle_{H_0^1}$

Proof. The adjoint operator $\mathcal{L}_\mu^* = (\mathcal{L}+\mu)^* = \mathcal{L}^* + \mu$ satisfies that for all $u, v \in H_0^1$

$$\langle v, \mathcal{L}_\mu u\rangle_{H_0^1, H^{-1}} = \left\langle \mathcal{L}_\mu^* v, u\right\rangle_{H^{-1}, H_0^1}$$

thus $\langle v, \mathcal{L}_\mu u\rangle_{H_0^1, H^{-1}} = \langle \mathcal{L}_\mu u, v\rangle_{H^{-1}, H_0^1} = \left\langle \mathcal{L}_\mu^* v, u\right\rangle_{H^{-1}, H_0^1}$. It is equivalent that

$$B_\mu[u, v] = B_\mu\left[\mathcal{L}_\mu^{-1}\mathcal{L}_\mu^* v, u\right]$$

Let $v^* = \mathcal{L}_\mu^{-1}\mathcal{L}_\mu^* v$, we have

$$\int \sum_{ij} a^{ij} v_{x_i}^* u_{x_j} - \sum_{ij} a^{ij} v_{x_i} u_{x_j} - \sum_i b^i v u_{x_i}\,\mathrm{d}x = \int \left[-\sum_i b^i v_{x_i}^* + (c+\mu)(v - v^*)\right] u\,\mathrm{d}x$$

integrating by parts

$$\int \left[-\sum_{ij}\left(a^{ij} v_{x_i}^*\right)_{x_j} + \sum_{ij}\left(a^{ij} v_{x_i}\right)_{x_j} + \sum_i \left(b^i v\right)_{x_i}\right] u\,\mathrm{d}x = \int \left[-\sum_i b^i v_{x_i}^* + (c+\mu)(v - v^*)\right] u\,\mathrm{d}x$$

$$-\sum_{ij}\left(a^{ij} v_{x_i}^*\right)_{x_j} + \sum_{ij}\left(a^{ij} v_{x_i}\right)_{x_j} + \sum_i \left(b^i v\right)_{x_i} = -\sum_i b^i v_{x_i}^* + (c+\mu)(v - v^*)$$

then

$$\begin{aligned}\mathcal{L}_\mu^* v = \mathcal{L}_\mu v^* &= -\sum_{ij}\left(a^{ij} v_{x_i}\right)_{x_j} - \sum_i \left(b^i v\right)_{x_i} + (c+\mu) v \\ &= -\sum\left(a^{ij} v_{x_i}\right)_{x_j} - \sum b^i v_{x_i} + \left(c + \mu - \sum b_{x_i}^i\right) v\end{aligned}$$

hence

$$\mathcal{L}^* v = -\sum\left(a^{ij} v_{x_i}\right)_{x_j} - \sum b^i v_{x_i} + \left(c - \sum b_{x_i}^i\right) v$$

which complete this proof. □

§3.2

Theorem 3.2 (Second existence theorem)**.** *$\mathcal{L}$ is a Fredholm operator of index zero, that is, the following three statements hold:*

1. *Precisely one of the following statements holds:*
   - $0 \in \rho(\mathcal{L})$
   - $\operatorname{Ker}(\mathcal{L}) \neq 0$
2. $\dim \operatorname{Ker}(\mathcal{L}) = \dim \operatorname{Ker}(\mathcal{L}^*) < \infty$
3. $\operatorname{Im}(\mathcal{L}) = \operatorname{Ker}(\mathcal{L}^*)^\perp$

Proof. Step 1. Choose $\mu > \gamma$ as in 2.4 and $\mathcal{L}_\mu^{-1} : H^{-1}(U) \to H_0^1(U)$ is a bounded linear operator. Then

$$K := \mu \mathrm{id} \cdot \mathcal{L}_\mu^{-1} : H^{-1}(U) \xrightarrow{\mathcal{L}_\mu^{-1}} H_0^1(U) \hookrightarrow\hookrightarrow L^2(U) \hookrightarrow H^{-1}(U)$$

is a bounded, linear, compact operator on $H^{-1}(U)$, since $H_0^1(U) \subset\subset L^2(U)$ according to 4.2.

Step 2. We rewrite this equality in $H^{-1}(U)$ to read

$$\mu^{-1}(\mathcal{L} + \mu) u - u = \mu^{-1} f$$

that is, the equality in $H_0^1(U)$,

$$(I - K) u = h,$$

where $h = \mu^{-1} K f = \mathcal{L}_\mu^{-1} f \in H_0^1(U)$.

Step 3. We may consequently apply the Fredholm alternative. Equation has solution if and only if

$$h \in N(I - K^*)^\perp$$

and the dimension of the space $N(I - K)$ of solution is finite and equals $\dim N(I - K^*)$.

Step 4. Finally, we recall that the solution of $(I - K) u = h$ is the weak solution of (1) and the solution of $(I - K^*) v = 0$ is weak solutions of (3). □

Remark. $\mathcal{L}_\mu^{-1} : L^2(U) \to H_0^1(U)$

$$K := \mu \mathrm{id} \cdot \mathcal{L}_\mu^{-1} : L^2(U) \xrightarrow{\mathcal{L}_\mu^{-1}} H_0^1(U) \hookrightarrow\hookrightarrow L^2(U)$$

$L^2(U)$

Theorem 3.3 (Third Existence Theorem)**.** *The spectrum $\sigma(L)$ of $\mathcal{L}$ is a discrete set of real eigenvalues.*

Remark.

$$\lambda_1 \le \lambda_2 \le \lambda_3 \le \dots, \quad \lim_{n\to\infty} \lambda_n = +\infty$$

§4 Regularity

§4.1 Interior regularity

Theorem 4.1 (Interior H^2-regularity)**.** *Assume*

$$a^{ij} \in C^1(U), b^i, c \in L^\infty(U) \quad (i, j = 1, \dots, n)$$

and

$$f \in L^2(U).$$

Suppose furthermore that $u \in H^1(U)$ is a weak solution of the elliptic PDE

$$\mathcal{L}u = f \quad \text{in } U \tag{1}$$

Then

$$u \in H^2_{\text{loc}}(U)$$

and for each open subset $V \subset\subset U$ we have the estimate

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right),$$

the constant C depending only on V, U, and the coefficients of L.

Proof. Step 1. Fix any open set $V \subset\subset U$, and choose an open set W such that $V \subset\subset W \subset\subset U$. Then select a smooth function ζ satisfying

$$\begin{cases} \zeta \equiv 1 \text{ on } V, \zeta \equiv 0 \text{ on } \mathbb{R}^n - W \\ 0 \leq \zeta \leq 1 \end{cases}$$

Step 2. Now since u is a weak solution of (1), we have $B[u, v] = (f, v)$ for all $v \in H^1_0(U)$. Consequently

$$\sum_{i,j=1}^n \int_U a^{ij} u_{x_i} v_{x_j} \, \mathrm{d}x = \int_U \tilde{f} v \, \mathrm{d}x$$

where $\tilde{f} := f - \sum_{i=1}^n b^i u_{x_i} - cu$

3. Now let $|h| > 0$ be small, choose $k \in \{1, \ldots, n\}$, and then substitute

$$v := -D_k^{-h} \left(\zeta^2 D_k^h u \right)$$

into (9), where the expression $D_k^h u$ denotes the difference quotient

$$D_k^h u(x) = \frac{u(x + he_k) - u(x)}{h} \quad (h \in \mathbb{R}, h \neq 0).$$

we have the formulas

$$\int_U v D_k^{-h} w \, \mathrm{d}x = -\int_U w D_k^h v \, \mathrm{d}x$$

and

$$D_k^h(vw) = v^h D_k^h w + w D_k^h v,$$

for $v^h(x) := v(x + he_k)$. We write the resulting expression as

$$A = B,$$

for

$$A := \sum_{i,j=1}^{n} \int_U a^{ij} u_{x_i} v_{x_j} \, \mathrm{d}x$$

and

$$B := \int_U \tilde{f} v \, \mathrm{d}x$$

Step 4. Estimate of A. We have

$$\begin{aligned}
A &= -\sum_{i,j=1}^{n} \int_U a^{ij} u_{x_i} \left[D_k^{-h} \left(\zeta^2 D_k^h u \right) \right]_{x_j} \, \mathrm{d}x \\
&= -\sum_{i,j=1}^{n} \int_U a^{ij} u_{x_i} D_k^{-h} \left(\zeta^2 D_k^h u \right)_{x_j} \, \mathrm{d}x \\
&= \sum_{i,j=1}^{n} \int_U D_k^h \left(a^{ij} u_{x_i} \right) \left(\zeta^2 D_k^h u \right)_{x_j} \, \mathrm{d}x \\
&= \sum_{i,j=1}^{n} \int_U a^{ij,h} \left(D_k^h u_{x_i} \right) \left(\zeta^2 D_k^h u \right)_{x_j} + \left(D_k^h a^{ij} \right) u_{x_i} \left(\zeta^2 D_k^h u \right)_{x_j} \, \mathrm{d}x
\end{aligned}$$

Returning now to (15), we find

$$\begin{aligned}
A = &\sum_{i,j=1}^{n} \int_U a^{ij,h} D_k^h u_{x_i} D_k^h u_{x_j} \zeta^2 \, \mathrm{d}x \\
&+ \sum_{i,j=1}^{n} \int_U \left[a^{ij,h} D_k^h u_{x_i} D_k^h u 2\zeta\zeta_{x_j} + \left(D_k^h a^{ij} \right) u_{x_i} D_k^h u_{x_j} \zeta^2 \right. \\
&\left. + \left(D_k^h a^{ij} \right) u_{x_i} D_k^h u 2\zeta\zeta_{x_j} \right] \, \mathrm{d}x \\
=: &A_1 + A_2.
\end{aligned}$$

The uniform ellipticity condition implies

$$A_1 \geq \theta \int_U \zeta^2 \left| D_k^h Du \right|^2 dx \tag{2}$$

Furthermore we see that

$$|A_2| \leq C \int_U \zeta \left| D_k^h Du \right| \left| D_k^h u \right| + \zeta \left| D_k^h Du \right| |Du| + \zeta \left| D_k^h u \right| |Du| dx,$$

for some appropriate constant C. But then Cauchy's inequality with ϵ yields the bound

$$|A_2| \leq \epsilon \int_U \zeta^2 \left| D_k^h Du \right|^2 dx + \frac{C}{\epsilon} \int_W \left| D_k^h u \right|^2 + |Du|^2 dx$$

We choose $\epsilon = \frac{\theta}{2}$ and further recall from Theorem 3(i) in §5.8.2 the estimate

$$\int_W \left|D_k^h u\right|^2 dx \leq C \int_U |Du|^2 dx,$$

thereby obtaining the inequality

$$|A_2| \leq \frac{\theta}{2} \int_U \zeta^2 \left|D_k^h Du\right|^2 dx + C \int_U |Du|^2 dx \tag{3}$$

This estimate, (19) and (18) imply finally

$$A \geq \frac{\theta}{2} \int_U \zeta^2 \left|D_k^h Du\right|^2 dx - C \int_U |Du|^2 dx$$

Step 5. Estimate of B. Recalling now (10), (11), and (14), we estimate

$$|B| \leq C \int_U (|f| + |Du| + |u|)|v| dx$$

Now Theorem 3(i) in §5.8.2 implies

$$\begin{aligned}
\int_U |v|^2 dx &\leq C \int_U \left|D\left(\zeta^2 D_k^h u\right)\right|^2 dx \\
&\leq C \int_W \left|D_k^h u\right|^2 + \zeta^2 \left|D_k^h Du\right|^2 dx \\
&\leq C \int_U |Du|^2 + \zeta^2 \left|D_k^h Du\right|^2 dx
\end{aligned}$$

Thus (21) and Cauchy's inequality with ϵ imply

$$|B| \leq \epsilon \int_U \zeta^2 \left|D_k^h Du\right|^2 dx + \frac{C}{\epsilon} \int_U f^2 + u^2 dx + \frac{C}{\epsilon} \int_U |Du|^2 dx$$

Select $\epsilon = \frac{\theta}{4}$, to obtain

$$|B| \leq \frac{\theta}{4} \int_U \zeta^2 \left|D_k^h Du\right|^2 dx + C \int_U f^2 + u^2 + |Du|^2 dx$$

Step 6. We finally combine (12), (20) and (22), to discover

$$\int_V \left|D_k^h Du\right|^2 dx \leq \int_U \zeta^2 \left|D_k^h Du\right|^2 dx \leq C \int_U f^2 + u^2 + |Du|^2 dx$$

for $k = 1, \ldots, n$ and all sufficiently small $|h| \neq 0$. In view of Theorem 3(ii) in §5.8.2, we deduce $Du \in H^1_{\text{loc}}\ (U; \mathbb{R}^n)$, and thus $u \in H^2_{\text{loc}}(U)$, with the estimate

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\right)$$

Step 7. We now refine estimate (23) by noting that if $V \subset\subset W \subset\subset U$, then the same argument shows

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(W)} + \|u\|_{H^1(W)} \right),$$

for an appropriate constant C depending on V, W, etc. Choose a new cutoff function ζ satisfying

$$\begin{cases} \zeta \equiv 1 \text{ on } W, \text{ spt } \zeta \subset U \\ 0 \leq \zeta \leq 1 \end{cases}$$

Now set $v = \zeta^2 u$ in identity (9) and perform elementary calculations, to discover

$$\int_U \zeta^2 |Du|^2 dx \leq C \int_U f^2 + u^2 dx$$

Thus

$$\|u\|_{H^1(W)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right)$$

This inequality and (24) yield (8). □

Theorem 4.2 (Higher interior regularity)**.** *Let m be a nonnegative integer, and assume*

$$a^{ij}, b^i, c \in C^{m+1}(U) \quad (i, j = 1, \ldots, n)$$

and

$$f \in H^m(U)$$

Suppose $u \in H^1(U)$ is a weak solution of the elliptic PDE

$$\mathcal{L}u = f \quad \text{in } U$$

Then

$$u \in H^{m+2}_{\text{loc}}(U)$$

and for each $V \subset\subset U$ we have the estimate

$$\|u\|_{H^{m+2}(V)} \leq C \left(\|f\|_{H^m(U)} + \|u\|_{L^2(U)} \right),$$

the constant C depending only on m, U, V and the coefficients of L.

Proof. 1. We will establish (27), (28) by induction on m, the case $m = 0$ being Theorem 1 above.

2. Assume now assertions (27) and (28) are valid for some nonnegative integer m and all open sets U, coefficients a^{ij}, b^i, c, etc., as above. Suppose then

$$a^{ij}, b^i, c \in C^{m+2}(U)$$

and $u \in H^1(U)$ is a weak solution of $\mathcal{L}u = f$ in U. By the induction hypotheses, we have

$$u \in H^{m+2}_{\text{loc}}(U),$$

with the estimate

$$\|u\|_{H^{m+2}(W)} \leq C\left(\|f\|_{H^m(U)} + \|u\|_{L^2(U)}\right)$$

for each $W \subset\subset U$ and an appropriate constant C, depending only on W, the coefficients of L, etc. Fix $V \subset\subset W \subset\subset U$.

3. Now let α be any multiindex with

$$|\alpha| = m + 1,$$

and choose any test function $\tilde{v} \in C_c^\infty(W)$. Insert

$$v := (-1)^{|\alpha|} D^\alpha \tilde{v}$$

into the identity $B[u, v] = (f, v)_{L^2(U)}$, and perform some integrations by parts, eventually to discover

$$B[\tilde{u}, \tilde{v}] = (\tilde{f}, \tilde{v})$$

for

$$\tilde{u} := D^\alpha u \in H^1(W)$$

and

$$\tilde{f} := D^\alpha f - \sum_{\substack{\beta \leq \alpha \\ \beta \neq \alpha}} \binom{\alpha}{\beta} \Big[-\sum_{i,j=1}^n \left(D^{\alpha-\beta} a^{ij} D^\beta u_{x_i}\right)_{x_j} + \sum_{i=1}^n D^{\alpha-\beta} b^i D^\beta u_{x_i} + D^{\alpha-\beta} c D^\beta u \Big]$$

Since the identity (34) holds for each $\tilde{v} \in C_c^\infty(W)$, we see that $\tilde{u}$ is a weak solution of

$$L\tilde{u} = \tilde{f} \quad \text{in } W.$$

In view of (29)-(32) and (36), we have $\tilde{f} \in L^2(W)$, with

$$\|\tilde{f}\|_{L^2(W)} \leq C\left(\|f\|_{H^{m+1}(U)} + \|u\|_{L^2(U)}\right)$$

Step 4. In light of 4.1 then, we see $\tilde{u} \in H^2(V)$, with the estimate

$$\begin{aligned} \|\tilde{u}\|_{H^2(V)} &\leq C\left(\|\tilde{f}\|_{L^2(W)} + \|\tilde{u}\|_{L^2(W)}\right) \\ &\leq C\left(\|f\|_{H^{m+1}(U)} + \|u\|_{L^2(U)}\right) \end{aligned}$$

This inequality holds for each multiindex α with $|\alpha| = m+1$ and $\tilde{u} = D^\alpha u$ as above. Consequently $u \in H^{m+3}(V)$, and

$$\|u\|_{H^{m+3}(V)} \leq C \left(\|f\|_{H^{m+1}(U)} + \|u\|_{L^2(U)} \right)$$

□

Corollary 4.3 (Infinite differentiability in the interior)**.** *Assume*

$$a^{ij}, b^i, c \in C^\infty(U) \quad (i, j = 1, \ldots, n)$$

and

$$f \in C^\infty(U)$$

Suppose $u \in H^1(U)$ is a weak solution of the elliptic PDE

$$\mathcal{L}u = f \quad \text{in } U$$

Then

$$u \in C^\infty(U)$$

Proof. According to Theorem 2, we have $u \in H^m_{\text{loc}}(U)$ for each integer $m = 1, 2, \ldots$. Hence Theorem 6 in §5.6.3 implies $u \in C^k(U)$ for each $k = 1, 2, \ldots$. □

§4.2 Boundary regularity

Theorem 4.4 (Boundary H^2-regularity)**.** *Assume*

$$a^{ij} \in C^1(\bar{U}), b^i, c \in L^\infty(U) \quad (i, j = 1, \ldots, n)$$

,

$$f \in L^2(U)$$

and

$$\partial U \text{ is } C^2.$$

Suppose that $u \in H^1_0(U)$ is a weak solution of the elliptic boundary-value problem

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ u = 0 \text{ on } \partial U. \end{cases}$$

Then

$$u \in H^2(U)$$

and we have the estimate

$$\|u\|_{H^2(U)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right),$$

the constant C depending only on U and the coefficients of L.

Theorem 4.5 (Higher boundary regularity)**.** *Let m be a nonnegative integer, and assume*

$$a^{ij}, b^i, c \in C^{m+1}(\bar{U}) \quad (i, j = 1, \ldots, n)$$

,

$$f \in H^m(U)$$

and

$$\partial U \text{ is } C^{m+2}$$

Suppose that $u \in H_0^1(U)$ is a weak solution of the boundary-value problem

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ u = 0 \text{ on } \partial U \end{cases}$$

Then

$$u \in H^{m+2}(U)$$

and we have the estimate

$$\|u\|_{H^{m+2}(U)} \leq C \left(\|f\|_{H^m(U)} + \|u\|_{L^2(U)} \right)$$

the constant C depending only on m, U and the coefficients of L.

Theorem 4.6 (Infinite differentiability up to the boundary)**.** *Assume*

$$a^{ij}, b^i, c \in C^\infty(\bar{U}) \quad (i, j = 1, \ldots, n)$$

and

$$f \in C^\infty(\bar{U})$$

Assume also that ∂U is C^∞. Suppose $u \in H_0^1(U)$ is a weak solution of the boundary-value problem

$$\begin{cases} \mathcal{L}u = f \text{ in } U \\ u = 0 \text{ on } \partial U \end{cases}$$

then

$$u \in C^\infty(\bar{U})$$

§5 Maximum principle

In this section, we always assume that

1. U is an open bounded subset of $\mathbb{R}^n$

2. $\mathcal{L}u = -\sum_{i,j=1}^n a^{ij} u_{x_i x_j} + \sum_{i=1}^n b^i u_{x_i} + cu$
3. uniform ellipticity condition
4. a^{ij}, b^i, c are continuous
5. the symmetry condition $a^{ij} = a^{ji}$.

§5.1 Weak maximum principle

Theorem 5.1. *Assume $u \in C^2(U) \cap C(\bar{U})$ and $c \equiv 0$ in U*

1. *If u is a **subsolution**, that is, $\mathcal{L}u \leq 0$ in U, then*

$$\max_{\bar{U}} u = \max_{\partial U} u$$

2. *If u is a **supersolution**, that is, $\mathcal{L}u \geq 0$ in U, then*

$$\min_{\bar{U}} u = \min_{\partial U} u$$

Theorem 5.2 (Weak maximum principle)**.** *Assume $u \in C^2(U) \cap C(\bar{U})$ and $c \geq 0$ in U.*

1. *If $\mathcal{L}u \leq 0$ in U, then*

$$\max_{\bar{U}} u \leq \max_{\partial U} u^+.$$

2. *If $\mathcal{L}u \geq 0$ in U, then*

$$\min_{\bar{U}} u \geq \min_{\partial U} (-u)^+ = -\max_{\partial U} u^-.$$

3. *So in particular, if $\mathcal{L}u = 0$ in U, then*

$$\max_{\bar{U}} |u| = \max_{\partial U} |u|.$$

Proof. 1. Let u be a subsolution and set $V := \{x \in U \mid u(x) > 0\} \neq \varnothing$ (if $V = \varnothing$, $u(x) \leq 0$ for all $x \in U$ the theorem is proved). Then

$$\begin{aligned} Ku &:= \mathcal{L}u - cu \\ &\leq -cu \leq 0 \quad \text{in } V. \end{aligned}$$

The operator $K = L - c$ has no zeroth-order term and consequently 5.1 implies

$$\max_{\bar{V}} u = \max_{\partial V} u = \max_{\partial U} u^+$$

This gives (11) in the case that $V \neq \varnothing$. Otherwise $u \leq 0$ everywhere in U, and (11) likewise follows.

2. Assertion (ii) follows from (i) applied to $-u$, once we observe that $(-u)^+ = u^-$. □

§5.2 Strong maximum principle

Theorem 5.3 (Hopf's Lemma). *Assume $u \in C^2(U) \cap C^1(\bar{U})$ and*

$$c \equiv 0 \quad \text{in } U.$$

Suppose further

$$\mathcal{L}u \leq 0 \quad \text{in } U$$

and there exists a point $x^0 \in \partial U$ such that

$$u\left(x^0\right) > u(x) \quad \text{for all } x \in U$$

Assume finally that U satisfies the interior ball condition at x^0; that is, there exists an open ball $B \subset U$ with $x^0 \in \partial B$. Then

$$\frac{\partial u}{\partial \nu}\left(x^0\right) > 0$$

where ν is the outer unit normal to B at x^0.

(2) If

$$c \geq 0 \quad \text{in } U,$$

the same conclusion holds provided

$$u\left(x^0\right) \geq 0.$$

Proof. Step 1. Assume $c \geq 0$. We may as well further assume $B = B^0(0,r)$ for some radius $r > 0$. Define

$$v(x) := e^{-\lambda|x|^2} - e^{-\lambda r^2} \quad (x \in B(0,r))$$

for $\lambda > 0$ as selected below. Then using the uniform ellipticity condition, we compute

$$\begin{aligned}
Lv =& -\sum_{i,j=1}^n a^{ij} v_{x_i x_j} + \sum_{i=1}^n b^i v_{x_i} + cv \\
=& e^{-\lambda|x|^2} \sum_{i,j=1}^n a^{ij}\left(-4\lambda^2 x_i x_j + 2\lambda\delta_{ij}\right) - e^{-\lambda|x|^2} \sum_{i=1}^n b^i 2\lambda x_i + c\left(e^{-\lambda|x|^2} - e^{-\lambda r^2}\right) \\
\leq& e^{-\lambda|x|^2}\left(-4\theta\lambda^2|x|^2 + 2\lambda \operatorname{tr} A + 2\lambda \|b\| \, |x| + c\right)
\end{aligned}$$

Consider next the open annular region $R := B^0(0,r) - B(0,r/2)$. We have

$$Lv \leq e^{-\lambda|x|^2}\left(-\theta\lambda^2 r^2 + 2\lambda \operatorname{tr} \mathbf{A} + 2\lambda|\mathbf{b}|r + c\right) \leq 0$$

in R, provided $\lambda > 0$ is fixed large enough.

2. In view of (14) there exists a constant $\epsilon > 0$ so small that

$$u\left(x^0\right) \geq u(x) + \epsilon v(x) \quad (x \in \partial B(0,r/2))$$

In addition note

$$u\left(x^0\right) \geq u(x) + \epsilon v(x) \quad (x \in \partial B(0,r))$$

since $v \equiv 0$ on $\partial B(0,r)$.

3. From (15) we see

$$L\left(u + \epsilon v - u\left(x^0\right)\right) \leq -cu\left(x^0\right) \leq 0 \quad \text{in } R$$

and from (16), (17) we observe

$$u + \epsilon v - u\left(x^0\right) \leq 0 \quad \text{on } \partial R$$

In view of the weak maximum principle, Theorem $2, u + \epsilon v - u\left(x^0\right) \leq 0$ in R. But $u\left(x^0\right) + \epsilon v\left(x^0\right) - u\left(x^0\right) = 0$, and so

$$\frac{\partial u}{\partial \nu}\left(x^0\right) + \epsilon \frac{\partial v}{\partial \nu}\left(x^0\right) \geq 0$$

Consequently

$$\frac{\partial u}{\partial \nu}\left(x^0\right) \geq -\epsilon \frac{\partial v}{\partial \nu}\left(x^0\right) = -\frac{\epsilon}{r} Dv\left(x^0\right) \cdot x^0 = 2\lambda \epsilon r e^{-\lambda r^2} > 0$$

as required. □

Theorem 5.4. *Assume $u \in C^2(U) \cap C(\bar{U})$ and $c \equiv 0$ in U. Suppose also U is connected, open and bounded.*

1. *If*

$$\mathcal{L}u \leq 0 \quad \text{in } U$$

and u attains its maximum over $\bar{U}$ at an interior point, then u is constant within U.

2. *if*

$$\mathcal{L}u \geq 0 \quad \text{in } U$$

and u attains its minimum over $\bar{U}$ at an interior point, then u is constant within U.

Chapter IX

Linear Evolution Equations

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For this chapter we assume domain U to be an open, bounded subset of $\mathbb{R}^n$ and set $U_T = U \times (0, T]$ for some fixed time $T > 0$.

§1 Second-order parabolic equations

§1.1 Bochner space

Definition 1.1.

$$L^2\left(0, T; H_0^1(U)\right) := \left\{ \mathbf{u} : [0, T] \to H_0^1(U) \mid \int_0^T \|\mathbf{u}(t)\|_{H_0^1(U)}^2 \, \mathrm{d}t < \infty \right\}$$

with norm $\|\mathbf{u}\|_{L^2(0,T;H_0^1(U))} = \left(\int_0^T \|\mathbf{u}(t)\|_{H_0^1(U)}^2 \, \mathrm{d}t \right)^{1/2}$

The derivatives of $\mathbf{u}$ *are defined in the distributional sense.*

§1.2 Weak solution

Definition 1.2. *We will first study the initial Dirichlet problem*

$$\begin{cases} u_t + \mathcal{L}u = f \text{ in } U_T \\ \qquad u = 0 \text{ on } \partial U \times [0,T] \\ \qquad u = g \text{ on } U \times \{t = 0\} \end{cases}$$

where $f : U_T \to \mathbb{R}$ and $g : U \to \mathbb{R}$ are given and $u : \bar{U}_T \to \mathbb{R}$ is the unknown, $u = u(x,t)$. The letter $\mathcal{L}$ denotes for each time t a second-order partial differential operator, having either the divergence form

$$\mathcal{L}u = -\sum_{i,j=1}^{n} \left(a^{ij}(x,t)u_{x_i}\right) x_j + \sum_{i=1}^{n} b^i(x,t)u_{x_i} + c(x,t)u$$

or else the nondivergence form

$$\mathcal{L}u = -\sum_{i,j=1}^{n} a^{ij}(x,t)u_{x_i x_j} + \sum_{i=1}^{n} b^i(x,t)u_{x_i} + c(x,t)u$$

for given coefficients $a^{ij}, b^i, c (i,j = 1,\dots,n)$.

Definition 1.3. *We say that the partial differential operator $\frac{\partial}{\partial t} + L$ is **(uniformly) parabolic** if there exists a constant $\theta > 0$ such that*

$$\sum_{i,j=1}^{n} a^{ij}(x,t)\xi_i\xi_j \geq \theta|\xi|^2$$

for all $(x,t) \in U_T, \xi \in \mathbb{R}^n$.

Definition 1.4. *We consider first the case that L has the divergence form and try to find an appropriate notion of weak solution for the initial/boundaryvalue problem. We assume for now that*

$$a^{ij}, b^i, c \in L^\infty(U_T)$$

$$f \in L^2(U_T)$$

$$g \in L^2(U)$$

We will also always suppose $a^{ij} = a^{ji} (i,j = 1,\dots,n)$. Let us now define the time-dependent bilinear form

$$B[u,v;t] := \int_U \sum_{i,j=1}^{n} a^{ij}(\cdot,t)u_{x_i}v_{x_j} + \sum_{i=1}^{n} b^i(\cdot,t)u_{x_i}v + c(\cdot,t)uv \,\mathrm{d}x$$

for $u(\cdot,t), v(\cdot,t) \in H_0^1(U)$ and a.e. $0 \leq t \leq T$.

We say a function

$$\mathbf{u} \in L^2\left(0,T;H_0^1(U)\right), \text{ with } \mathbf{u}' \in L^2\left(0,T;H^{-1}(U)\right),$$

is a ***weak solution*** *of the parabolic initial/boundary-value problem (1) provided*

1. $\langle \mathbf{u}', v\rangle + B[\mathbf{u}, v; t] = (\mathbf{f}, v)$ *for each* $v \in H_0^1(U)$ *and a.e.* $0 \le t \le T$.
2. $\mathbf{u}(0) = g$.

§1.3 Existence of weak solution

Galerkin approximations

Definition 1.5. *We intend to build a weak solution of the parabolic problem*

$$\begin{cases} u_t + \mathcal{L}u = f \text{ in } U_T \\ \qquad u = 0 \text{ on } \partial U \times [0,T] \\ \qquad u = u_0 \text{ on } U \times \{t = 0\} \end{cases} \tag{1}$$

by first constructing solutions of certain finite-dimensional approximations to (1) and then passing to limits. This is called ***Galerkin's method****.*

Assume the functions $w_k = w_k(x)(k = 1, \ldots)$ *are smooth, and*

(i) $\{w_k\}_{k=1}^{\infty}$ *is an orthogonal basis of* $H_0^1(U)$*,*

(ii) $\{w_k\}_{k=1}^{\infty}$ *is an orthonormal basis of* $L^2(U)$

Fix now a positive integer m*. We will look for a function* $\mathbf{u}_m : [0,T] \to H_0^1(U)$ *of the form*

$$\mathbf{u}_m(t) := \sum_{k=1}^{m} d_m^k(t) w_k \tag{2}$$

where we hope to select the coefficients $d_m^k(t)(0 \le t \le T, k = 1, \ldots, m)$ *so that*

$$\begin{cases} d_m^k(0) = (g, w_k) & (k = 1, \ldots, m) \\ (\mathbf{u}_m', w_k) + B\left[\mathbf{u}_m, w_k; t\right] = (\mathbf{f}, w_k) & (0 \le t \le T, k = 1, \ldots, m) \end{cases} \tag{3}$$

Thus we seek a function $\mathbf{u}_m$ *of the form (2) that satisfies the "projection" (3) of problem (1) onto the finite-dimensional subspace spanned by* $\{w_k\}_{k=1}^{m}$.

Theorem 1.6 (Construction of approximate solutions)**.** *For each integer* $m = 1, 2, \ldots$ *there exists a unique function* $\mathbf{u}_m$ *of the form (2) satisfying (3).*

Proof. Assuming $\mathbf{u}_m$ has the structure (2), we first note that

$$(\mathbf{u}_m'(t), w_k) = d_m^{k'}(t)$$

Furthermore

$$B\left[\mathbf{u}_m, w_k; t\right] = \sum_{l=1}^{m} e^{kl}(t) d_m^l(t)$$

for $e^{kl}(t) := B\left[w_l, w_k; t\right]$ $(k, l = 1, \ldots, m)$. Let us further write $f^k(t) := (\mathbf{f}(t), w_k)$ $(k = 1, \ldots, m)$. Then (3) becomes the linear system of ODE

$$\begin{cases} d_m^{k'}(t) + \displaystyle\sum_{l=1}^{m} e^{kl}(t) d_m^l(t) = f^k(t) (k = 1, \ldots, m) \\ \qquad\qquad\qquad\qquad d_m^k(0) = (g, w_k) (k = 1, \ldots, m) \end{cases} \tag{4}$$

According to standard existence theory for ordinary differential equations, there exists a unique absolutely continuous function $\mathbf{d}_m(t) = (d_m^1(t), \ldots, d_m^m(t))$ satisfying (4) for a.e. $0 \le t \le T$. And then $\mathbf{u}_m$ defined by (2) solves (3) for a.e. $0 \le t \le T$. □

Energey estimates

Theorem 1.7 (Energy estimates)**.** *There exists a constant C, depending only on U, T and the coefficients of L, such that*

$$\max_{0 \le l \le T} \|\mathbf{u}_m(t)\|_{L^2(U)} + \|\mathbf{u}_m\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\mathbf{u}_m'\|_{L^2(0,T;H^{-1}(U))} \\ \le C\left(\|\mathbf{f}\|_{L^2(0,T;L^2(U))} + \|g\|_{L^2(U)}\right)$$

for $m = 1, 2, \ldots$.

Existence and uniqueness

Theorem 1.8 (Existence of weak solution)**.** *There exists a weak solution of (11).*

Theorem 1.9 (Uniqueness of weak solutions)**.** *A weak solution of (11) is unique.*

§1.4 Regularity